TRANSNORDESTINA Logística S.A.	**FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS**

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	**NUMERAÇÃO DA TLSA**	**PÁGINA**
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)	FTN-MVP10-NV4D-0643-001	1 / 5
MVP 10 - KM 450+000 A KM 500+789.330	**CONTRATO**	**REV.**
CAPA - PLANILHA DE CUBAÇÃO	-	00

REVISÕES						
REV.	**DESCRIÇÃO**	**POR**	**VER.**	**APR.**	**AUT.**	**DATA**
00	EMISSÃO INICIAL - REVISÃO DE PROJETOS	INS	LPB	LPB	UAS	2/18/2025
01	REVISÃO DE GREIDE E TALUDE	INS	LPB	LPB	UAS	7/24/2025

TRANSNORDESTINA Logística S.A.	**FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS**	
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS **TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)** **MVP 10 - KM 450+000 A KM 500+789.330** **CAPA - PLANILHA DE CUBAÇÃO**	**NUMERAÇÃO DA TLSA** FTN-MVP10-NV4D-0643-001 **CONTRATO** -	**PÁGINA** 1 / 5 **REV.** 00

REVISÕES						
REV.	**DESCRIÇÃO**	**POR**	**VER.**	**APR.**	**AUT.**	**DATA**

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
450+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.04	8.97	33.07	0.00	0.00	0.00	0.00	1.36	0.00	0.00
450+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.72	8.97	57.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	94.94	8.97	85.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	95.53	8.97	86.56	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	96.02	8.97	87.05	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.97	8.97	93.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	103.73	8.97	94.76	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.39	8.97	93.42	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.46	8.97	92.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.75	8.97	97.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.16	8.97	101.19	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	167.22	8.97	158.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	94.80	8.97	85.83	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.71	8.97	77.74	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.67	8.97	72.70	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.08	8.97	55.11	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.09	8.97	45.12	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.66	8.97	39.69	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.82	8.97	35.85	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.45	8.97	29.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.83	8.97	23.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.88	8.97	18.91	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.30	8.97	14.33	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.73	8.97	12.76	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.76	8.97	10.79	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.95	8.97	5.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.05	8.97	3.08	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
450+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.75	8.97	2.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.64	8.97	1.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.40	8.97	2.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.50	8.97	2.53	0.00	0.00	0.00	0.00	1.36	20.00	0.00
450+620.000	0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.86	8.97	0.88	0.00	0.00	0.00	0.00	1.36	20.00	1.98
450+640.000	1.74	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	19.35
450+660.000	3.70	4.84	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	54.37
450+680.000	0.78	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	44.80
450+700.000	2.67	0.29	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	34.50
450+720.000	3.16	0.24	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	58.30
450+740.000	2.72	0.07	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	58.80
450+760.000	4.26	0.05	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	69.80
450+780.000	6.34	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	106.00
450+800.000	4.89	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	112.30
450+820.000	2.92	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.10
450+840.000	1.24	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	41.60
450+860.000	8.88	3.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	101.24
450+880.000	9.54	1.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	184.25
450+900.000	9.95	0.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	194.91
450+920.000	8.40	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	183.51
450+940.000	7.21	0.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	156.08
450+960.000	8.08	0.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	152.90
450+980.000	7.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	158.66
451+000.000	7.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	150.77
451+020.000	6.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	135.67
451+040.000	5.62	0.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	118.88
451+060.000	4.21	1.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	98.23
451+080.000	5.15	2.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	93.61

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
451+100.000	2.62	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	77.73
451+120.000	1.36	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	39.80
451+140.000	1.24	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	26.00
451+160.000	3.20	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	44.40
451+180.000	5.25	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	84.50
451+200.000	7.42	0.04	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	126.70
451+220.000	8.26	1.51	0.00	43.18	29.37	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	156.80
451+240.000	8.81	3.35	0.00	44.06	37.37	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	170.70
451+260.000	9.36	5.16	0.00	45.27	45.96	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	181.70
451+280.000	10.08	6.96	0.00	47.06	59.81	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	194.40
451+300.000	10.62	8.82	0.00	47.97	70.21	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	207.00
451+320.000	11.02	10.70	0.00	48.23	79.20	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	216.40
451+340.000	10.35	12.65	0.00	44.33	88.59	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	213.70
451+360.000	9.79	15.18	0.00	40.75	101.09	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	201.40
451+380.000	9.98	19.55	0.00	39.15	118.02	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	197.70
451+400.000	10.41	24.32	0.00	38.62	138.29	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	203.90
451+420.000	11.70	29.20	0.00	41.23	156.96	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	221.10
451+440.000	12.43	35.55	0.00	41.26	174.20	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	241.30
451+460.000	23.53	69.18	8.08	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	359.60
451+480.000	23.61	59.59	18.79	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	471.40
451+500.000	22.53	52.44	30.88	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	461.40
451+520.000	21.27	47.42	44.32	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	438.00
451+540.000	17.85	45.44	58.95	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	391.20
451+560.000	29.61	44.42	75.19	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	474.60
451+580.000	55.79	45.91	94.83	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	854.00
451+600.000	61.78	49.56	119.30	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,175.70
451+620.000	52.31	59.59	135.50	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,140.90
451+640.000	53.38	59.66	131.65	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,056.90
451+660.000	38.80	59.90	135.46	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	921.80
451+680.000	58.59	61.50	140.62	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	973.90
451+700.000	106.29	61.58	147.64	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,648.80
451+720.000	9.29	60.55	155.62	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,155.80
451+740.000	52.45	50.50	121.63	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	617.40
451+760.000	32.89	50.54	96.98	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	853.40
451+780.000	13.15	74.27	28.90	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	460.40
451+800.000	6.25	11.73	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	194.00
451+820.000	25.70	50.64	5.81	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	319.50
451+840.000	19.89	76.98	24.59	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	455.90
451+860.000	15.72	64.56	30.67	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	356.10
451+880.000	12.75	55.05	24.70	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	284.70
451+900.000	4.48	44.17	13.69	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	172.30
451+920.000	6.16	9.52	0.13	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	106.40
451+940.000	9.65	17.23	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	158.10
451+960.000	9.63	15.52	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	192.80
451+980.000	9.62	4.68	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	192.50

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
452+000.000	4.15	4.39	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	137.70
452+020.000	7.47	9.79	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	116.20
452+040.000	3.34	8.85	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	108.10
452+060.000	0.00	2.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	33.40
452+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.47	8.97	0.00	6.50	0.00	0.00	0.00	1.36	20.00	0.00
452+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	31.21	8.97	0.00	22.24	0.00	0.00	0.00	1.36	20.00	0.00
452+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.23	8.97	0.00	21.26	0.00	0.00	0.00	1.36	20.00	0.00
452+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.54	8.97	0.00	8.57	0.00	0.00	0.00	1.36	20.00	0.00
452+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.11	8.97	0.00	18.14	0.00	0.00	0.00	1.36	20.00	0.00
452+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.81	8.97	0.00	26.20	2.64	0.00	0.00	1.36	20.00	0.00
452+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.52	8.97	0.00	26.20	7.35	0.00	0.00	1.36	20.00	0.00
452+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.74	8.97	0.00	26.20	7.57	0.00	0.00	1.36	20.00	0.00
452+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.38	8.97	0.00	26.20	7.21	0.00	0.00	1.36	20.00	0.00
452+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.22	8.97	0.00	26.20	10.05	0.00	0.00	1.36	20.00	0.00
452+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.63	8.97	0.00	26.20	13.46	0.00	0.00	1.36	20.00	0.00
452+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.57	8.97	0.00	26.20	4.40	0.00	0.00	1.36	20.00	0.00
452+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.17	8.97	0.00	26.20	11.00	0.00	0.00	1.36	20.00	0.00
452+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.80	8.97	0.00	26.20	24.63	0.00	0.00	1.36	20.00	0.00
452+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.07	8.97	0.00	26.20	33.90	0.00	0.00	1.36	20.00	0.00
452+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	67.68	8.97	0.00	26.20	32.51	0.00	0.00	1.36	20.00	0.00
452+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.66	8.97	0.00	26.20	31.49	0.00	0.00	1.36	20.00	0.00
452+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	65.71	8.97	0.00	26.20	30.54	0.00	0.00	1.36	20.00	0.00
452+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.97	8.97	0.00	26.20	20.80	0.00	0.00	1.36	20.00	0.00
452+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.40	8.97	0.00	26.20	13.23	0.00	0.00	1.36	20.00	0.00
452+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.17	8.97	0.00	23.20	0.00	0.00	0.00	1.36	20.00	0.00
452+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.10	8.97	0.00	26.20	11.93	0.00	0.00	1.36	20.00	0.00
452+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.26	8.97	0.00	26.20	13.09	0.00	0.00	1.36	20.00	0.00
452+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.46	8.97	0.00	26.20	9.29	0.00	0.00	1.36	20.00	0.00
452+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.99	8.97	0.00	26.20	0.82	0.00	0.00	1.36	20.00	0.00
452+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.38	8.97	0.00	26.20	1.21	0.00	0.00	1.36	20.00	0.00
452+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.67	8.97	0.00	21.70	0.00	0.00	0.00	1.36	20.00	0.00
452+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.41	8.97	0.00	15.44	0.00	0.00	0.00	1.36	20.00	0.00
452+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.40	8.97	0.00	9.43	0.00	0.00	0.00	1.36	20.00	0.00
452+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.46	8.97	0.00	4.49	0.00	0.00	0.00	1.36	20.00	0.00
452+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.37	8.97	0.00	0.40	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
452+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.52	8.97	0.00	2.55	0.00	0.00	0.00	1.36	20.00	0.00
452+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+840.000	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	0.00
452+860.000	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	0.00
452+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.06	8.97	0.00	0.08	0.00	0.00	0.00	1.36	20.00	0.00
452+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
452+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.52	8.97	0.00	3.55	0.00	0.00	0.00	1.36	20.00	0.00
452+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.34	8.97	0.00	8.37	0.00	0.00	0.00	1.36	20.00	0.00
452+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.44	8.97	0.00	14.47	0.00	0.00	0.00	1.36	20.00	0.00
453+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.95	8.97	0.00	20.98	0.00	0.00	0.00	1.36	20.00	0.00
453+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.39	8.97	0.00	26.20	1.22	0.00	0.00	1.36	20.00	0.00
453+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.24	8.97	0.00	26.20	9.07	0.00	0.00	1.36	20.00	0.00
453+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.09	8.97	0.00	26.20	17.92	0.00	0.00	1.36	20.00	0.00
453+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.92	8.97	0.00	26.20	11.75	0.00	0.00	1.36	20.00	0.00
453+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.25	8.97	0.00	26.20	8.08	0.00	0.00	1.36	20.00	0.00
453+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.12	8.97	0.00	26.20	9.95	0.00	0.00	1.36	20.00	0.00
453+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.79	8.97	0.00	26.20	13.62	0.00	0.00	1.36	20.00	0.00
453+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.15	8.97	0.00	26.20	14.98	0.00	0.00	1.36	20.00	0.00
453+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.85	8.97	0.00	26.20	7.68	0.00	0.00	1.36	20.00	0.00
453+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.35	8.97	0.00	21.38	0.00	0.00	0.00	1.36	20.00	0.00
453+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.83	8.97	0.00	21.86	0.00	0.00	0.00	1.36	20.00	0.00
453+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.92	8.97	0.00	25.95	0.00	0.00	0.00	1.36	20.00	0.00
453+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.73	8.97	0.00	21.76	0.00	0.00	0.00	1.36	20.00	0.00
453+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.25	8.97	0.00	5.28	0.00	0.00	0.00	1.36	20.00	0.00
453+300.000	4.41	3.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	44.09
453+320.000	3.38	6.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	77.89
453+340.000	0.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	41.59
453+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.79	8.97	0.00	15.82	0.00	0.00	0.00	1.36	20.00	7.79
453+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.70	8.97	0.00	21.73	0.00	0.00	0.00	1.36	20.00	0.00
453+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.22	8.97	0.00	26.20	12.05	0.00	0.00	1.36	20.00	0.00
453+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.38	8.97	0.00	26.20	7.21	0.00	0.00	1.36	20.00	0.00
453+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.56	8.97	0.00	26.20	1.39	0.00	0.00	1.36	20.00	0.00
453+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.38	8.97	0.00	26.20	1.21	0.00	0.00	1.36	20.00	0.00
453+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	67.06	8.97	0.00	26.20	31.89	0.00	0.00	1.36	20.00	0.00
453+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.85	8.97	0.00	26.20	40.68	0.00	0.00	1.36	20.00	0.00
453+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.08	8.97	0.00	26.20	41.91	0.00	0.00	1.36	20.00	0.00
453+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.18	8.97	0.00	26.20	42.01	0.00	0.00	1.36	20.00	0.00
453+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.26	8.97	0.00	26.20	42.09	0.00	0.00	1.36	20.00	0.00
453+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.15	8.97	0.00	26.20	41.98	0.00	0.00	1.36	20.00	0.00
453+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.62	8.97	0.00	26.20	42.45	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
453+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.24	8.97	0.00	26.20	42.07	0.00	0.00	1.36	20.00	0.00
453+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.13	8.97	0.00	26.20	41.96	0.00	0.00	1.36	20.00	0.00
453+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.29	8.97	0.00	26.20	42.12	0.00	0.00	1.36	20.00	0.00
453+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.08	8.97	0.00	26.20	41.91	0.00	0.00	1.36	20.00	0.00
453+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	80.82	8.97	0.00	26.20	45.65	0.00	0.00	1.36	20.00	0.00
453+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.28	8.97	0.00	26.20	53.11	0.00	0.00	1.36	20.00	0.00
453+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	98.59	8.97	0.00	26.20	63.42	0.00	0.00	1.36	20.00	0.00
453+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.39	8.97	0.00	26.20	66.22	0.00	0.00	1.36	20.00	0.00
453+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.99	8.97	0.00	26.20	66.82	0.00	0.00	1.36	20.00	0.00
453+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.50	8.97	0.00	26.20	51.33	0.00	0.00	1.36	20.00	0.00
453+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	72.66	8.97	0.00	26.20	37.49	0.00	0.00	1.36	20.00	0.00
453+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.41	8.97	0.00	26.20	23.24	0.00	0.00	1.36	20.00	0.00
453+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.82	8.97	0.00	26.20	14.65	0.00	0.00	1.36	20.00	0.00
453+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.72	8.97	0.00	26.20	14.55	0.00	0.00	1.36	20.00	0.00
453+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.55	8.97	0.00	26.20	21.38	0.00	0.00	1.36	20.00	0.00
453+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	67.13	8.97	0.00	26.20	31.96	0.00	0.00	1.36	20.00	0.00
453+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.24	8.97	0.00	26.20	34.07	0.00	0.00	1.36	20.00	0.00
453+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	72.10	8.97	0.00	26.20	36.93	0.00	0.00	1.36	20.00	0.00
453+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.42	8.97	0.00	26.20	34.25	0.00	0.00	1.36	20.00	0.00
454+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.20	8.97	0.00	26.20	22.03	0.00	0.00	1.36	20.00	0.00
454+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.17	8.97	0.00	26.20	13.00	0.00	0.00	1.36	20.00	0.00
454+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.96	8.97	0.00	26.20	5.79	0.00	0.00	1.36	20.00	0.00
454+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.27	8.97	0.00	26.20	9.10	0.00	0.00	1.36	20.00	0.00
454+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.03	8.97	0.00	26.20	15.86	0.00	0.00	1.36	20.00	0.00
454+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.41	8.97	0.00	26.20	16.24	0.00	0.00	1.36	20.00	0.00
454+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.34	8.97	0.00	26.20	17.17	0.00	0.00	1.36	20.00	0.00
454+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.74	8.97	0.00	26.20	25.57	0.00	0.00	1.36	20.00	0.00
454+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.27	8.97	0.00	26.20	46.10	0.00	0.00	1.36	20.00	0.00
454+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.71	8.97	0.00	26.20	51.54	0.00	0.00	1.36	20.00	0.00
454+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	72.78	8.97	0.00	26.20	37.61	0.00	0.00	1.36	20.00	0.00
454+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.12	8.97	0.00	26.20	33.95	0.00	0.00	1.36	20.00	0.00
454+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.39	8.97	0.00	26.20	27.22	0.00	0.00	1.36	20.00	0.00
454+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.79	8.97	0.00	26.20	20.62	0.00	0.00	1.36	20.00	0.00
454+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.01	8.97	0.00	26.20	8.84	0.00	0.00	1.36	20.00	0.00
454+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.17	8.97	0.00	26.20	3.00	0.00	0.00	1.36	20.00	0.00
454+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.64	8.97	0.00	26.20	1.47	0.00	0.00	1.36	20.00	0.00
454+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.21	8.97	0.00	17.24	0.00	0.00	0.00	1.36	20.00	0.00
454+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.29	8.97	0.00	1.32	0.00	0.00	0.00	1.36	20.00	0.00
454+380.000	3.96	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	39.60
454+400.000	3.66	16.71	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	76.20
454+420.000	3.82	16.08	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	74.80
454+440.000	2.84	0.07	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	66.60
454+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.85	8.97	10.88	0.00	0.00	0.00	0.00	1.36	20.00	28.40
454+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.13	8.97	37.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
454+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.65	8.97	46.68	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	61.72	8.97	52.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.21	8.97	62.24	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.93	8.97	74.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.51	8.97	92.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	118.27	8.97	109.30	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	127.94	8.97	118.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	135.41	8.97	126.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	128.98	8.97	120.01	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	124.62	8.97	115.65	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	119.76	8.97	110.79	0.00	0.00	0.00	0.00	1.36	20.00	0.00
454+715.565	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	112.04	8.97	103.07	0.00	0.00	0.00	0.00	1.36	15.57	0.00
454+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.43	0.00
454+734.740	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.96	8.97	0.00	26.20	87.78	0.00	0.00	1.36	14.74	0.00
454+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	136.07	8.97	0.00	26.20	100.90	0.00	0.00	1.36	5.26	0.00
454+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	159.63	8.97	0.00	26.20	124.46	0.00	0.00	1.36	20.00	0.00
454+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	161.39	8.97	0.00	26.20	126.22	0.00	0.00	1.36	20.00	0.00
454+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	126.61	8.97	0.00	26.20	91.44	0.00	0.00	1.36	20.00	0.00
454+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	112.82	8.97	0.00	26.20	77.65	0.00	0.00	1.36	20.00	0.00
454+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	115.32	8.97	0.00	26.20	80.15	0.00	0.00	1.36	20.00	0.00
454+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	127.79	8.97	0.00	26.20	92.62	0.00	0.00	1.36	20.00	0.00
454+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	132.67	8.97	0.00	26.20	97.50	0.00	0.00	1.36	20.00	0.00
454+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	123.09	8.97	0.00	26.20	87.92	0.00	0.00	1.36	20.00	0.00
454+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.95	8.97	0.00	26.20	75.78	0.00	0.00	1.36	20.00	0.00
454+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	109.42	8.97	0.00	26.20	74.25	0.00	0.00	1.36	20.00	0.00
454+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	95.05	8.97	0.00	26.20	59.88	0.00	0.00	1.36	20.00	0.00
454+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.92	8.97	0.00	26.20	38.75	0.00	0.00	1.36	20.00	0.00
455+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.40	8.97	0.00	26.20	18.23	0.00	0.00	1.36	20.00	0.00
455+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.96	8.97	0.00	18.99	0.00	0.00	0.00	1.36	20.00	0.00
455+040.000	2.17	3.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	21.71
455+060.000	5.26	10.11	0.00	23.31	55.92	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	74.31
455+080.000	9.26	27.52	0.00	32.25	116.89	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	145.20
455+100.000	9.11	32.25	0.00	30.10	139.16	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	183.70
455+120.000	7.24	28.34	0.00	24.52	104.25	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	163.50
455+140.000	1.03	13.64	0.00	2.17	39.48	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	82.70
455+160.000	0.08	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.15	8.97	1.17	0.00	0.00	0.00	0.00	1.36	20.00	11.10
455+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.57	8.97	46.60	0.00	0.00	0.00	0.00	1.36	20.00	0.80
455+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	132.88	8.97	123.91	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	227.79	8.97	218.82	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	182.19	8.97	173.22	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	168.87	8.97	159.90	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	157.83	8.97	148.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	155.22	8.97	146.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	148.68	8.97	139.71	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	139.77	8.97	130.80	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	132.32	8.97	123.35	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
455+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.28	8.97	113.31	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	112.04	8.97	103.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.75	8.97	98.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.73	8.97	96.76	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.03	8.97	91.06	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	90.47	8.97	81.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.25	8.97	57.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.56	8.97	31.59	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.67	8.97	23.70	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.66	8.97	28.69	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.92	8.97	35.95	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.38	8.97	33.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.70	8.97	13.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+640.000	3.57	4.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	35.65
455+660.000	2.08	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	56.45
455+680.000	6.96	4.87	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	90.40
455+700.000	6.06	3.44	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	130.20
455+720.000	0.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.37	8.97	2.40	0.00	0.00	0.00	0.00	1.36	20.00	67.85
455+740.000	1.15	0.29	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	18.75
455+760.000	2.13	1.47	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	32.80
455+780.000	2.03	1.63	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	41.60
455+800.000	1.73	0.32	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	37.60
455+820.000	2.78	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.02	8.97	0.05	0.00	0.00	0.00	0.00	1.36	20.00	45.12
455+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.46	8.97	21.49	0.00	0.00	0.00	0.00	1.36	20.00	27.82
455+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.70	8.97	27.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.62	8.97	4.65	0.00	0.00	0.00	0.00	1.36	20.00	0.00
455+900.000	4.65	0.00	0.00	10.56	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	46.50
455+920.000	5.09	15.29	0.00	6.48	29.64	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	97.40
455+940.000	2.79	25.36	0.03	3.26	41.22	11.17	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.80
455+960.000	8.78	39.66	15.24	5.26	26.09	41.31	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	115.70
455+980.000	16.47	38.09	32.46	9.18	22.72	60.66	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	252.50
456+000.000	15.64	52.80	26.30	8.94	32.61	53.11	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	321.10
456+020.000	17.40	8.77	0.00	26.75	33.56	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	330.40
456+040.000	8.47	14.93	0.00	12.41	38.75	19.91	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	258.70
456+060.000	6.67	58.95	6.40	4.13	41.60	27.42	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	151.40

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
456+080.000	18.38	69.19	2.79	10.57	45.97	10.88	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	250.50
456+100.000	6.59	25.33	0.00	7.36	40.17	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	249.70
456+120.000	6.71	0.92	0.00	10.88	6.61	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	133.00
456+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.19	8.97	0.00	0.22	0.00	0.00	0.00	1.36	20.00	67.10
456+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.65	8.97	0.00	23.68	0.00	0.00	0.00	1.36	20.00	0.00
456+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.59	8.97	0.00	26.20	25.42	0.00	0.00	1.36	20.00	0.00
456+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	82.87	8.97	0.00	26.20	47.70	0.00	0.00	1.36	20.00	0.00
456+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.42	8.97	0.00	26.20	64.25	0.00	0.00	1.36	20.00	0.00
456+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	116.11	8.97	0.00	26.20	80.94	0.00	0.00	1.36	20.00	0.00
456+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	126.06	8.97	0.00	26.20	90.89	0.00	0.00	1.36	20.00	0.00
456+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	137.27	8.97	0.00	26.20	102.10	0.00	0.00	1.36	20.00	0.00
456+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	135.61	8.97	0.00	26.20	100.44	0.00	0.00	1.36	20.00	0.00
456+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	127.58	8.97	0.00	26.20	92.41	0.00	0.00	1.36	20.00	0.00
456+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	143.53	8.97	0.00	26.20	108.36	0.00	0.00	1.36	20.00	0.00
456+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	149.09	8.97	0.00	26.20	113.92	0.00	0.00	1.36	20.00	0.00
456+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	145.47	8.97	0.00	26.20	110.30	0.00	0.00	1.36	20.00	0.00
456+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	141.16	8.97	0.00	26.20	105.99	0.00	0.00	1.36	20.00	0.00
456+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	132.88	8.97	0.00	26.20	97.71	0.00	0.00	1.36	20.00	0.00
456+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.35	8.97	0.00	26.20	87.18	0.00	0.00	1.36	20.00	0.00
456+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	104.02	8.97	0.00	26.20	68.85	0.00	0.00	1.36	20.00	0.00
456+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.81	8.97	0.00	26.20	53.64	0.00	0.00	1.36	20.00	0.00
456+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.09	8.97	0.00	26.20	37.92	0.00	0.00	1.36	20.00	0.00
456+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.00	8.97	0.00	26.20	19.83	0.00	0.00	1.36	20.00	0.00
456+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.40	8.97	0.00	26.20	14.23	0.00	0.00	1.36	20.00	0.00
456+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.17	8.97	0.00	26.20	10.00	0.00	0.00	1.36	20.00	0.00
456+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.13	8.97	0.00	26.20	30.96	0.00	0.00	1.36	20.00	0.00
456+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	93.24	8.97	0.00	26.20	58.07	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
456+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	93.84	8.97	0.00	26.20	58.67	0.00	0.00	1.36	20.00	0.00
456+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.70	8.97	0.00	26.20	42.53	0.00	0.00	1.36	20.00	0.00
456+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	82.02	8.97	0.00	26.20	46.85	0.00	0.00	1.36	20.00	0.00
456+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	94.12	8.97	0.00	26.20	58.95	0.00	0.00	1.36	20.00	0.00
456+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	103.96	8.97	0.00	26.20	68.79	0.00	0.00	1.36	20.00	0.00
456+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	111.08	8.97	0.00	26.20	75.91	0.00	0.00	1.36	20.00	0.00
456+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	114.12	8.97	0.00	26.20	78.95	0.00	0.00	1.36	20.00	0.00
456+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	116.77	8.97	0.00	26.20	81.60	0.00	0.00	1.36	20.00	0.00
456+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	118.33	8.97	0.00	26.20	83.16	0.00	0.00	1.36	20.00	0.00
456+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	119.58	8.97	0.00	26.20	84.41	0.00	0.00	1.36	20.00	0.00
456+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	118.41	8.97	0.00	26.20	83.24	0.00	0.00	1.36	20.00	0.00
456+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	115.32	8.97	0.00	26.20	80.15	0.00	0.00	1.36	20.00	0.00
456+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.02	8.97	0.00	26.20	70.85	0.00	0.00	1.36	20.00	0.00
456+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.70	8.97	0.00	26.20	51.53	0.00	0.00	1.36	20.00	0.00
456+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.42	8.97	0.00	26.20	29.25	0.00	0.00	1.36	20.00	0.00
456+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.18	8.97	0.00	26.20	3.01	0.00	0.00	1.36	20.00	0.00
456+940.000	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.97	8.97	0.00	3.00	0.00	0.00	0.00	1.36	20.00	0.30
456+960.000	7.05	0.09	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	70.80
456+980.000	23.47	18.89	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	305.20
457+000.000	13.58	26.90	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	370.50
457+020.000	8.29	6.04	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	218.70
457+040.000	5.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	133.95
457+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.14	8.97	18.17	0.00	0.00	0.00	0.00	1.36	20.00	51.05
457+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.63	8.97	43.66	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.36	8.97	54.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.72	8.97	9.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+140.000	3.24	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00		0.00	5.10	0.00	1.64	20.00	32.40
457+160.000	2.31	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	55.50
457+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.19	8.97	6.22	0.00	0.00	0.00	0.00	1.36	20.00	23.10
457+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.81	8.97	39.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	79.62	8.97	70.65	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.17	8.97	74.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.33	8.97	65.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.02	8.97	53.05	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.42	8.97	24.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+320.000	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	1.30
457+340.000	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.65	8.97	0.68	0.00	0.00	0.00	0.00	1.36	20.00	2.00
457+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.85	8.97	21.88	0.00	0.00	0.00	0.00	1.36	20.00	0.70
457+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.90	8.97	39.93	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.36	8.97	41.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.93	8.97	25.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.93	8.97	12.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.38	8.97	19.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.10	8.97	39.13	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
457+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.66	8.97	55.69	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.16	8.97	54.19	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.49	8.97	54.52	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.28	8.97	42.31	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.07	8.97	36.10	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.15	8.97	25.18	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.43	8.97	10.46	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+640.000	2.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.00	8.97	0.03	0.00	0.00	0.00	0.00	1.36	20.00	23.44
457+660.000	9.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	123.36
457+680.000	0.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.68	8.97	1.70	0.00	0.00	0.00	0.00	1.36	20.00	108.16
457+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.22	8.97	13.25	0.00	0.00	0.00	0.00	1.36	20.00	8.24
457+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.36	8.97	25.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+740.000	0.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.99	8.97	0.02	0.00	0.00	0.00	0.00	1.36	20.00	5.58
457+760.000	7.73	0.34	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	82.88
457+780.000	4.02	21.42	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	117.50
457+800.000	6.41	1.78	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	104.30
457+820.000	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.71	8.97	2.74	0.00	0.00	0.00	0.00	1.36	20.00	64.30
457+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.74	8.97	30.77	0.00	0.00	0.00	0.00	1.36	20.00	0.20
457+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.58	8.97	23.61	0.00	0.00	0.00	0.00	1.36	20.00	0.00
457+880.000	2.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	20.76
457+900.000	9.49	6.19	0.00	35.67	43.93	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	115.66
457+920.000	13.84	36.77	0.00	35.55	128.34	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	233.30
457+940.000	10.57	55.90	0.20	23.45	163.69	16.84	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	244.10
457+960.000	12.81	41.36	0.00	31.86	151.89	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	233.80
457+980.000	10.33	10.34	0.00	36.44	59.19	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	231.40
458+000.000	4.72	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.08	8.97	0.00	0.11	0.00	0.00	0.00	1.36	20.00	150.46
458+020.000	14.82	21.74	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	195.36
458+040.000	3.24	68.70	18.80	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	180.60
458+060.000	11.83	2.94	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	150.70
458+080.000	0.71	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	125.40
458+100.000	11.23	20.95	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	119.40
458+120.000	7.48	31.84	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	187.10
458+140.000	4.23	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	117.13
458+160.000	9.40	30.68	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	136.33
458+180.000	11.85	48.31	12.38	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	212.50
458+200.000	4.35	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	162.00
458+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.63	8.97	0.00	9.66	0.00	0.00	0.00	1.36	20.00	43.50
458+240.000	5.25	1.46	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	52.50
458+260.000	5.56	5.71	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	108.10
458+280.000	4.97	1.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	105.25
458+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.28	8.97	8.31	0.00	0.00	0.00	0.00	1.36	20.00	49.65
458+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.42	8.97	24.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.22	8.97	61.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.81	8.97	31.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
458+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.68	8.97	10.71	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+400.000	1.42	1.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	14.20
458+420.000	0.91	1.37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	23.30
458+440.000	0.51	0.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	14.20
458+460.000	1.46	2.38	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	19.70
458+480.000	0.00	15.02	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	14.60
458+500.000	1.10	2.34	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	11.00
458+520.000	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.66	8.97	1.69	0.00	0.00	0.00	0.00	1.36	20.00	12.30
458+540.000	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.76	8.97	2.78	0.00	0.00	0.00	0.00	1.36	20.00	1.40
458+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.70	8.97	11.73	0.00	0.00	0.00	0.00	1.36	20.00	0.10
458+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.87	8.97	26.90	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.43	8.97	43.46	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.46	8.97	62.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.27	8.97	92.30	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	127.22	8.97	118.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.31	8.97	97.34	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.56	8.97	80.59	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.70	8.97	59.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.21	8.97	57.24	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.64	8.97	61.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.40	8.97	62.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	78.95	8.97	69.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	90.96	8.97	81.99	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.62	8.97	66.65	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	61.34	8.97	52.37	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.95	8.97	33.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.14	8.97	25.17	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.91	8.97	18.94	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.60	8.97	18.63	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.24	8.97	24.27	0.00	0.00	0.00	0.00	1.36	20.00	0.00
458+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.83	8.97	47.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.12	8.97	80.15	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.26	8.97	65.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.24	8.97	16.27	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+060.000	1.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	17.77
459+080.000	0.80	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	25.77
459+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.65	8.97	0.00	16.68	0.00	0.00	0.00	1.36	20.00	8.00
459+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.91	8.97	0.00	12.94	0.00	0.00	0.00	1.36	20.00	0.00
459+140.000	3.38	5.12	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	33.80
459+160.000	6.97	37.16	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	103.50
459+180.000	8.83	84.35	25.78	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	158.00
459+200.000	14.49	106.06	53.03	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	233.20
459+220.000	27.88	167.68	71.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	423.70
459+240.000	35.58	269.61	89.48	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	634.60
459+260.000	28.80	309.65	94.92	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	643.80
459+280.000	34.15	228.69	95.99	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	629.50
459+300.000	15.62	150.98	87.97	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	497.70
459+320.000	17.14	84.77	79.07	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	327.60
459+340.000	0.00	117.97	124.17	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	171.40
459+360.000	16.79	113.03	167.55	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	167.90
459+380.000	25.70	107.50	74.35	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	424.90
459+400.000	7.42	58.51	37.81	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	331.20
459+420.000	6.05	87.92	16.55	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	134.70
459+440.000	8.95	29.03	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	150.00
459+460.000	8.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	170.14
459+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.27	8.97	34.30	0.00	0.00	0.00	0.00	1.36	20.00	80.64

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
459+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.36	8.97	44.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.08	8.97	72.11	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	78.13	8.97	69.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.51	8.97	61.53	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.83	8.97	8.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+600.000	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.03	8.97	4.05	0.00	0.00	0.00	0.00	1.36	20.00	10.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
459+620.000	1.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.68	8.97	2.71	0.00	0.00	0.00	0.00	1.36	20.00	29.19
459+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.33	8.97	25.36	0.00	0.00	0.00	0.00	1.36	20.00	19.19
459+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.65	8.97	20.68	0.00	0.00	0.00	0.00	1.36	20.00	0.00
459+680.000	0.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.74	8.97	0.76	0.00	0.00	0.00	0.00	1.36	20.00	7.93
459+700.000	9.07	11.63	0.00	7.54	8.40	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	98.63
459+720.000	5.89	54.60	21.12	2.01	20.20	21.43	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	149.60
459+740.000	16.19	36.51	6.09	8.66	22.49	13.71	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	220.80
459+760.000	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	163.40
459+780.000	9.64	6.29	0.00	10.78	23.66	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	97.90
459+800.000	14.94	7.43	0.00	30.29	42.86	16.02	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	245.80
459+820.000	14.21	7.59	53.28	10.54	6.49	74.84	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	291.50
459+840.000	17.11	42.74	85.29	9.17	25.47	103.47	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	313.20
459+860.000	20.91	56.73	87.18	9.97	30.00	93.24	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	380.20
459+880.000	16.87	113.85	92.15	7.41	54.80	79.90	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	377.80
459+900.000	15.24	9.56	70.51	10.00	6.69	99.69	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	321.10
459+920.000	16.02	13.68	80.48	9.93	9.24	98.46	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	312.60
459+940.000	6.97	14.37	29.96	5.78	13.16	65.14	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	229.90
459+960.000	2.46	3.68	0.06	4.84	14.58	18.50	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	94.30
459+980.000	1.44	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.97	8.97	0.00	4.00	0.00	0.00	0.00	1.36	20.00	38.98
460+000.000	7.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	87.03

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
460+020.000	5.89	20.84	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	131.55
460+040.000	7.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	133.40
460+060.000	10.56	2.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	180.08
460+080.000	5.60	0.55	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	161.58
460+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.67	15.65	11.02	0.00	0.00	0.00	0.00	2.36	20.00	56.00
460+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	92.61	15.65	76.96	0.00	0.00	0.00	0.00	2.36	20.00	0.00
460+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.58	15.65	86.93	0.00	0.00	0.00	0.00	2.36	20.00	0.00
460+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.19	15.65	32.54	0.00	0.00	0.00	0.00	2.36	20.00	0.00
460+180.000	10.71	3.77	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	107.10
460+200.000	16.95	24.67	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	276.60
460+220.000	20.44	33.59	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	373.90
460+240.000	17.77	79.16	0.84	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	382.10
460+260.000	11.32	119.86	17.10	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	290.90
460+280.000	25.41	353.54	41.67	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	367.30
460+300.000	15.50	189.09	57.72	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	409.10
460+320.000	7.71	107.59	49.98	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	232.10
460+340.000	7.92	78.20	48.53	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	156.30
460+360.000	14.49	78.47	35.26	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	224.10
460+380.000	17.71	79.32	14.03	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	322.00
460+400.000	14.95	27.34	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	326.60
460+420.000	6.42	4.70	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	213.70
460+440.000	15.30	15.07	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	217.20
460+460.000	14.91	68.65	1.47	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	302.10
460+480.000	11.96	110.96	5.87	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	268.70
460+500.000	2.81	38.54	1.41	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	147.70
460+520.000	4.69	11.31	0.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	75.00
460+540.000	3.51	3.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	82.03

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
460+560.000	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.61	15.65	0.96	0.00	0.00	0.00	0.00	2.36	20.00	40.13
460+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.64	15.65	44.99	0.00	0.00	0.00	0.00	2.36	20.00	5.00
460+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.34	15.65	35.69	0.00	0.00	0.00	0.00	2.36	20.00	0.00
460+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.71	15.65	20.06	0.00	0.00	0.00	0.00	2.36	20.00	0.00
460+640.000	19.77	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	197.70
460+660.000	14.20	35.88	46.87	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	339.70
460+680.000	16.73	30.65	30.20	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	309.30
460+700.000	5.15	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	218.80
460+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.35	15.65	0.00	5.70	0.00	0.00	0.00	2.36	20.00	51.50
460+740.000	9.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	99.02
460+760.000	9.31	13.88	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	192.12
460+780.000	17.22	35.52	43.53	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	265.30
460+800.000	16.72	62.67	63.92	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	339.40
460+820.000	36.33	63.36	28.25	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	530.50
460+840.000	15.77	2.26	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	521.00
460+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.30	15.65	0.00	9.65	0.00	0.00	0.00	2.36	20.00	157.70
460+880.000	7.47	1.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.22	15.65	0.00	1.57	0.00	0.00	0.00	2.36	20.00	74.70
460+900.000	8.87	0.82	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	163.40
460+920.000	3.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.17	15.65	0.00	10.52	0.00	0.00	0.00	2.36	20.00	120.18
460+940.000	0.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.14	15.65	0.00	3.49	0.00	0.00	0.00	2.36	20.00	33.06
460+960.000	6.02	22.23	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	61.78
460+980.000	2.38	56.61	38.74	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	84.00
461+000.000	14.23	69.37	90.72	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	166.10
461+020.000	36.62	71.98	118.94	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	508.50
461+040.000	27.40	83.87	87.37	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	640.20
461+060.000	14.68	48.43	71.26	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	420.80
461+080.000	46.21	110.64	74.59	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	608.90
461+100.000	50.60	95.91	92.78	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	968.10

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
461+120.000	21.17	78.85	109.67	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	717.70
461+140.000	32.72	84.02	112.95	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	538.90
461+160.000	23.46	89.67	106.53	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	561.80
461+180.000	11.18	79.40	92.21	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	346.40
461+200.000	3.86	74.17	58.46	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	150.40
461+220.000	16.47	49.85	7.73	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	203.30
461+240.000	3.28	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	197.50
461+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.67	15.65	14.02	0.00	0.00	0.00	0.00	2.36	20.00	32.80
461+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.23	15.65	65.58	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	116.45	15.65	100.80	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	126.33	15.65	110.68	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	148.03	15.65	132.38	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	145.44	15.65	129.79	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	146.20	15.65	130.55	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	152.35	15.65	136.70	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	132.09	15.65	116.44	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	134.24	15.65	118.59	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	139.53	15.65	123.88	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	167.59	15.65	151.94	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	187.11	15.65	171.46	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	188.50	15.65	172.85	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	191.62	15.65	175.97	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	186.48	15.65	170.83	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.29	15.65	163.64	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	156.26	15.65	140.61	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	130.72	15.65	115.07	0.00	0.00	0.00	0.00	2.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
461+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	131.96	15.65	116.31	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	145.28	15.65	129.63	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	166.59	15.65	150.94	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	155.59	15.65	139.94	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.92	15.65	84.27	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.62	15.65	38.97	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.03	15.65	21.38	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.03	15.65	40.38	0.00	0.00	0.00	0.00	2.36	20.00	0.00
461+800.000	5.80	0.36	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	58.00
461+820.000	7.56	6.24	0.00	14.72	6.36	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	133.60
461+840.000	14.05	25.96	0.00	23.45	52.19	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	216.10
461+860.000	25.82	82.65	16.58	22.31	78.98	35.18	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	398.70
461+880.000	5.57	114.51	21.61	4.54	103.09	44.86	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	313.90
461+900.000	10.38	101.42	4.88	9.12	99.52	10.04	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	159.50
461+920.000	10.85	21.36	0.00	19.00	43.60	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	212.30
461+940.000	6.57	0.51	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	174.20
461+960.000	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	66.60
461+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.09	15.65	30.44	0.00	0.00	0.00	0.00	2.36	20.00	0.90
462+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.55	15.65	19.90	0.00	0.00	0.00	0.00	2.36	20.00	0.00
462+020.000	1.03	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	10.30
462+040.000	9.73	28.88	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	107.60
462+060.000	11.51	75.20	20.83	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	212.40
462+080.000	11.80	67.77	16.20	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	233.10
462+100.000	7.69	14.57	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	194.90
462+120.000	20.07	1.08	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	277.60
462+140.000	20.35	70.39	24.79	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	404.20
462+160.000	27.60	75.19	63.07	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	479.50
462+180.000	43.28	98.13	72.88	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	708.80
462+200.000	50.58	102.87	75.78	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	938.60
462+220.000	40.89	88.79	78.22	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	914.70
462+240.000	36.37	63.54	79.83	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	772.60
462+260.000	24.19	55.51	80.07	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	605.60
462+280.000	38.37	61.55	81.16	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	625.60
462+300.000	44.70	75.56	71.92	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	830.70
462+320.000	47.60	95.40	50.15	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	923.00
462+340.000	14.33	83.11	35.65	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	619.30
462+360.000	11.42	48.71	18.55	0.00	0.00	0.00	0.00	0.00	5.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.40	2.64	20.00	257.50
462+380.000	2.62	5.68	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	140.40
462+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.94	15.65	0.00	11.29	0.00	0.00	0.00	2.36	20.00	26.20
462+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	79.17	15.65	0.00	63.52	0.00	0.00	0.00	2.36	20.00	0.00
462+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	109.22	15.65	0.00	93.57	0.00	0.00	0.00	2.36	20.00	0.00
462+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	131.60	15.65	0.00	115.95	0.00	0.00	0.00	2.36	20.00	0.00
462+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.27	8.97	0.00	26.20	33.10	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
462+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.28	8.97	0.00	12.31	0.00	0.00	0.00	1.36	20.00	0.00
462+520.000	1.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.64	8.97	0.00	0.67	0.00	0.00	0.00	1.36	20.00	14.13
462+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.19	8.97	0.00	4.22	0.00	0.00	0.00	1.36	20.00	14.13
462+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.21	8.97	0.00	25.24	0.00	0.00	0.00	1.36	20.00	0.00
462+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.21	8.97	0.00	5.24	0.00	0.00	0.00	1.36	20.00	0.00
462+600.000	4.21	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	42.10
462+620.000	8.97	9.54	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	131.80
462+640.000	8.51	13.82	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	174.80
462+660.000	8.25	3.70	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	167.60
462+680.000	6.91	0.40	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	151.60
462+700.000	5.25	5.14	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	121.60
462+720.000	7.63	24.14	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	128.80
462+740.000	23.62	61.36	6.80	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	312.50
462+760.000	19.56	11.64	2.23	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	431.80
462+780.000	5.44	0.41	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	250.00
462+800.000	6.91	7.55	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	123.50
462+820.000	11.84	11.36	36.93	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	187.50
462+840.000	13.61	9.70	36.64	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	254.50
462+860.000	15.81	14.85	20.26	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	294.20
462+880.000	8.45	0.75	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	242.60
462+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.13	8.97	0.00	26.20	7.96	0.00	0.00	1.36	20.00	84.50
462+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.20	8.97	0.00	26.20	8.03	0.00	0.00	1.36	20.00	0.00
462+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.31	8.97	0.00	21.34	0.00	0.00	0.00	1.36	20.00	0.00
462+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.44	8.97	0.00	26.20	10.27	0.00	0.00	1.36	20.00	0.00
462+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.80	8.97	0.00	26.20	24.63	0.00	0.00	1.36	20.00	0.00
463+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.64	8.97	0.00	15.67	0.00	0.00	0.00	1.36	20.00	0.00
463+020.000	3.17	1.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.47	8.97	0.00	2.49	0.00	0.00	0.00	1.36	20.00	31.69
463+040.000	1.90	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	50.69
463+060.000	8.56	0.98	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	104.60
463+080.000	11.91	0.06	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	204.70
463+100.000	1.14	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	130.50
463+120.000	13.66	0.64	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	148.00
463+140.000	10.48	11.12	0.90	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	241.40
463+160.000	14.81	20.45	7.79	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	252.90
463+180.000	30.88	26.02	14.55	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	456.90
463+200.000	26.68	27.52	22.01	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	575.60
463+220.000	21.34	26.74	15.94	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	480.20
463+240.000	8.57	7.36	0.59	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	299.10
463+260.000	5.89	3.06	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	144.60
463+280.000	7.58	2.55	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	134.69
463+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.20	8.97	0.00	26.20	1.03	0.00	0.00	1.36	20.00	75.79
463+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	79.71	8.97	0.00	26.20	44.54	0.00	0.00	1.36	20.00	0.00
463+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	79.22	8.97	0.00	26.20	44.05	0.00	0.00	1.36	20.00	0.00
463+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.27	8.97	0.00	26.20	23.10	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
463+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.03	8.97	0.00	26.20	19.86	0.00	0.00	1.36	20.00	0.00
463+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.95	8.97	0.00	17.98	0.00	0.00	0.00	1.36	20.00	0.00
463+420.000	13.15	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	131.50
463+440.000	10.23	0.28	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	233.80
463+460.000	5.16	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	153.90
463+480.000	5.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.35	8.97	0.00	1.38	0.00	0.00	0.00	1.36	20.00	105.98
463+500.000	9.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	150.45
463+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.46	8.97	0.00	19.49	0.00	0.00	0.00	1.36	20.00	96.07
463+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.59	8.97	0.00	26.20	42.42	0.00	0.00	1.36	20.00	0.00
463+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.67	8.97	0.00	26.20	35.50	0.00	0.00	1.36	20.00	0.00
463+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.87	8.97	0.00	26.20	16.70	0.00	0.00	1.36	20.00	0.00
463+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	61.42	8.97	0.00	26.20	26.25	0.00	0.00	1.36	20.00	0.00
463+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.71	8.97	0.00	26.20	3.54	0.00	0.00	1.36	20.00	0.00
463+640.000	3.90	0.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	39.00
463+660.000	4.68	14.59	1.40	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	85.80
463+680.000	8.18	35.15	11.81	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	128.60
463+700.000	10.81	36.13	14.64	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	189.90
463+720.000	9.17	38.19	22.20	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	199.80
463+740.000	8.95	37.18	21.46	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	181.20
463+760.000	13.41	29.14	1.40	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	223.60
463+780.000	7.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	207.43
463+800.000	6.44	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	137.73
463+820.000	7.05	41.36	41.46	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	134.90
463+840.000	5.14	45.93	60.85	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	121.90
463+860.000	9.82	39.13	55.49	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	149.60
463+880.000	11.72	47.36	42.57	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	215.40
463+900.000	12.09	26.74	30.21	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	238.10
463+920.000	20.83	44.55	60.66	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	329.20
463+940.000	10.12	52.82	95.15	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	309.50
463+960.000	18.96	54.30	105.51	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	290.80
463+980.000	15.19	51.04	89.41	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	341.50
464+000.000	11.82	45.78	54.29	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	270.10
464+020.000	10.84	46.55	51.08	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	226.60
464+040.000	20.47	49.49	76.58	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	313.10
464+060.000	15.56	52.17	89.33	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	360.30
464+080.000	17.38	49.84	78.27	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	329.40
464+100.000	24.64	46.13	57.39	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	420.20
464+120.000	35.94	40.42	30.12	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	605.80
464+140.000	15.58	37.72	13.36	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	515.20
464+160.000	44.36	41.28	32.29	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	599.40
464+180.000	48.08	46.57	57.89	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	924.40
464+200.000	47.58	49.75	75.48	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	956.60
464+220.000	49.77	49.38	73.99	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	973.50
464+240.000	40.12	47.50	62.47	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	898.90
464+260.000	28.08	40.65	36.57	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	682.00
464+280.000	21.89	38.42	20.24	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	499.70
464+300.000	28.33	41.41	34.07	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	502.20
464+320.000	27.43	47.46	62.83	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	557.60
464+340.000	40.84	51.01	88.79	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	682.70
464+360.000	46.30	54.76	109.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	871.40
464+380.000	53.42	63.95	117.11	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	997.20
464+400.000	52.56	50.38	105.75	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,059.80
464+420.000	60.28	54.21	113.12	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,128.40
464+440.000	50.25	76.59	165.56	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,105.30
464+460.000	35.02	81.73	219.32	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	852.70
464+480.000	22.08	87.42	284.68	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	571.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
464+500.000	33.75	80.76	359.79	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	558.30
464+520.000	21.12	80.52	457.68	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	548.70
464+540.000	20.87	63.22	563.29	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	419.90
464+560.000	25.98	61.22	593.46	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	468.50
464+580.000	36.38	59.64	598.67	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	623.60
464+600.000	45.07	67.94	579.47	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	814.50

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
464+620.000	41.83	90.06	488.05	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	869.00
464+640.000	80.38	76.75	362.47	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,222.10
464+660.000	118.79	64.83	186.28	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,991.70
464+680.000	37.33	41.51	83.28	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,561.20
464+700.000	79.33	54.77	141.42	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,166.60
464+720.000	69.17	67.56	294.73	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,485.00
464+740.000	73.90	53.92	429.96	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,430.70
464+760.000	80.27	41.93	476.38	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,541.70
464+780.000	40.28	30.62	476.61	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,205.50
464+800.000	53.71	43.40	406.50	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	939.90
464+820.000	71.39	65.00	291.88	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,251.00
464+840.000	80.51	62.52	163.66	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,519.00
464+860.000	5.90	55.48	107.90	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	864.10
464+880.000	76.46	52.61	123.95	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	823.60
464+900.000	68.10	66.49	191.95	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,445.60
464+920.000	73.08	65.43	285.10	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,411.80
464+940.000	36.67	38.33	406.00	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,097.50
464+960.000	38.94	25.84	509.08	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	756.10
464+980.000	61.50	33.47	494.37	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,004.40
465+000.000	47.71	42.09	376.49	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,092.10
465+020.000	56.89	55.92	236.86	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,046.00
465+040.000	55.72	48.01	167.15	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,126.10
465+060.000	53.75	72.18	275.33	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,094.70
465+080.000	43.29	85.41	385.56	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	970.40
465+100.000	22.12	91.43	497.39	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	654.10
465+120.000	12.99	93.86	529.40	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	351.10
465+140.000	26.41	87.80	496.05	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	394.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
465+160.000	23.68	58.89	411.38	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	500.90
465+180.000	27.84	40.77	306.11	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	515.20
465+200.000	25.08	51.92	201.28	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	529.20
465+220.000	34.09	50.40	123.08	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	591.70
465+240.000	9.63	44.00	76.59	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	437.20
465+260.000	37.10	42.12	76.96	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	467.30
465+280.000	28.06	41.62	75.43	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	651.60
465+300.000	20.32	40.78	69.58	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	483.80
465+320.000	17.77	38.17	54.17	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	380.90
465+340.000	20.16	34.83	34.53	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	379.30
465+360.000	15.28	31.38	17.63	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	354.40
465+380.000	16.20	27.07	2.13	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	314.80
465+400.000	5.16	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	213.60
465+420.000	0.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.04	8.97	0.00	2.07	0.00	0.00	0.00	1.36	20.00	59.51
465+440.000	1.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	18.58
465+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.38	8.97	0.00	7.41	0.00	0.00	0.00	1.36	20.00	10.67
465+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.48	8.97	0.00	18.51	0.00	0.00	0.00	1.36	20.00	0.00
465+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.41	8.97	0.00	26.20	5.24	0.00	0.00	1.36	20.00	0.00
465+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.31	8.97	0.00	26.20	17.14	0.00	0.00	1.36	20.00	0.00
465+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	65.28	8.97	0.00	26.20	30.11	0.00	0.00	1.36	20.00	0.00
465+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.35	8.97	0.00	26.20	39.18	0.00	0.00	1.36	20.00	0.00
465+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.52	8.97	0.00	26.20	54.35	0.00	0.00	1.36	20.00	0.00
465+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	95.93	8.97	0.00	26.20	60.76	0.00	0.00	1.36	20.00	0.00
465+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.87	8.97	0.00	26.20	71.70	0.00	0.00	1.36	20.00	0.00
465+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.10	8.97	0.00	26.20	86.93	0.00	0.00	1.36	20.00	0.00
465+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	133.93	8.97	0.00	26.20	98.76	0.00	0.00	1.36	20.00	0.00
465+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	139.46	8.97	0.00	26.20	104.29	0.00	0.00	1.36	20.00	0.00
465+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	144.21	8.97	0.00	26.20	109.04	0.00	0.00	1.36	20.00	0.00
465+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	136.99	8.97	0.00	26.20	101.82	0.00	0.00	1.36	20.00	0.00
465+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	143.30	8.97	0.00	26.20	108.13	0.00	0.00	1.36	20.00	0.00
465+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	162.78	8.97	0.00	26.20	127.61	0.00	0.00	1.36	20.00	0.00
465+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	171.51	8.97	0.00	26.20	136.34	0.00	0.00	1.36	20.00	0.00
465+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	172.27	8.97	0.00	26.20	137.10	0.00	0.00	1.36	20.00	0.00
465+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	138.98	8.97	0.00	26.20	103.81	0.00	0.00	1.36	20.00	0.00
465+826.635	9.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.07	8.97	0.00	26.20	86.90	0.00	0.00	1.36	6.64	30.45
465+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.36	61.35
465+847.399	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	94.21	8.97	0.00	26.20	59.04	0.00	0.00	1.36	7.40	0.00
465+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	82.05	8.97	0.00	26.20	46.88	0.00	0.00	1.36	12.60	0.00
465+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.76	8.97	0.00	26.20	16.59	0.00	0.00	1.36	20.00	0.00
465+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.08	8.97	0.00	18.11	0.00	0.00	0.00	1.36	20.00	0.00
465+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.64	8.97	0.00	5.67	0.00	0.00	0.00	1.36	20.00	0.00
465+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.54	8.97	0.00	6.57	0.00	0.00	0.00	1.36	20.00	0.00
465+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.35	8.97	0.00	9.38	0.00	0.00	0.00	1.36	20.00	0.00
465+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.93	8.97	0.00	15.96	0.00	0.00	0.00	1.36	20.00	0.00
466+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.56	8.97	0.00	24.59	0.00	0.00	0.00	1.36	20.00	0.00
466+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.64	8.97	0.00	26.20	10.47	0.00	0.00	1.36	20.00	0.00
466+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.59	8.97	0.00	26.20	14.42	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
466+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.00	8.97	0.00	26.20	20.83	0.00	0.00	1.36	20.00	0.00
466+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.25	8.97	0.00	26.20	29.08	0.00	0.00	1.36	20.00	0.00
466+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.93	8.97	0.00	26.20	38.76	0.00	0.00	1.36	20.00	0.00
466+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.72	8.97	0.00	26.20	42.55	0.00	0.00	1.36	20.00	0.00
466+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.71	8.97	0.00	26.20	38.54	0.00	0.00	1.36	20.00	0.00
466+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.37	8.97	0.00	26.20	42.20	0.00	0.00	1.36	20.00	0.00
466+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.40	8.97	0.00	26.20	52.23	0.00	0.00	1.36	20.00	0.00
466+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	98.35	8.97	0.00	26.20	63.18	0.00	0.00	1.36	20.00	0.00
466+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	103.75	8.97	0.00	26.20	68.58	0.00	0.00	1.36	20.00	0.00
466+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.17	8.97	0.00	26.20	52.00	0.00	0.00	1.36	20.00	0.00
466+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.41	8.97	0.00	26.20	39.24	0.00	0.00	1.36	20.00	0.00
466+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	72.97	8.97	0.00	26.20	37.80	0.00	0.00	1.36	20.00	0.00
466+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.17	8.97	0.00	26.20	53.00	0.00	0.00	1.36	20.00	0.00
466+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	95.78	8.97	0.00	26.20	60.61	0.00	0.00	1.36	20.00	0.00
466+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.43	8.97	0.00	26.20	56.26	0.00	0.00	1.36	20.00	0.00
466+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.91	8.97	0.00	26.20	52.74	0.00	0.00	1.36	20.00	0.00
466+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.20	8.97	0.00	26.20	87.03	0.00	0.00	1.36	20.00	0.00
466+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.95	8.97	0.00	26.20	87.78	0.00	0.00	1.36	20.00	0.00
466+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	126.73	8.97	0.00	26.20	91.56	0.00	0.00	1.36	20.00	0.00
466+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	127.98	8.97	0.00	26.20	92.81	0.00	0.00	1.36	20.00	0.00
466+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	132.73	8.97	0.00	26.20	97.56	0.00	0.00	1.36	20.00	0.00
466+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	138.31	8.97	0.00	26.20	103.14	0.00	0.00	1.36	20.00	0.00
466+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	148.24	8.97	0.00	26.20	113.07	0.00	0.00	1.36	20.00	0.00
466+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	157.31	8.97	0.00	26.20	122.14	0.00	0.00	1.36	20.00	0.00
466+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	149.33	8.97	0.00	26.20	114.16	0.00	0.00	1.36	20.00	0.00
466+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	142.74	8.97	0.00	26.20	107.57	0.00	0.00	1.36	20.00	0.00
466+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	133.63	8.97	0.00	26.20	98.46	0.00	0.00	1.36	20.00	0.00
466+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	125.19	8.97	0.00	26.20	90.02	0.00	0.00	1.36	20.00	0.00
466+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.14	8.97	0.00	26.20	86.97	0.00	0.00	1.36	20.00	0.00
466+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	124.03	8.97	0.00	26.20	88.86	0.00	0.00	1.36	20.00	0.00
466+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	128.24	8.97	0.00	26.20	93.07	0.00	0.00	1.36	20.00	0.00
466+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	133.29	8.97	0.00	26.20	98.12	0.00	0.00	1.36	20.00	0.00
466+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	134.97	8.97	0.00	26.20	99.80	0.00	0.00	1.36	20.00	0.00
466+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	136.60	8.97	0.00	26.20	101.43	0.00	0.00	1.36	20.00	0.00
466+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	140.76	8.97	0.00	26.20	105.59	0.00	0.00	1.36	20.00	0.00
466+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	139.27	8.97	0.00	26.20	104.10	0.00	0.00	1.36	20.00	0.00
466+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.78	8.97	0.00	26.20	87.61	0.00	0.00	1.36	20.00	0.00
466+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.72	8.97	0.00	26.20	72.55	0.00	0.00	1.36	20.00	0.00
466+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.36	8.97	0.00	26.20	52.19	0.00	0.00	1.36	20.00	0.00
466+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	65.55	8.97	0.00	26.20	30.38	0.00	0.00	1.36	20.00	0.00
466+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.69	8.97	0.00	26.20	5.52	0.00	0.00	1.36	20.00	0.00
466+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.81	8.97	0.00	8.84	0.00	0.00	0.00	1.36	20.00	0.00
466+900.000	3.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	37.31
466+920.000	2.69	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	64.21

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
466+940.000	13.72	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	164.10
466+960.000	23.24	0.05	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	369.60
466+980.000	23.91	1.14	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	471.50
467+000.000	17.53	0.08	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	414.40
467+020.000	0.78	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	183.10
467+040.000	36.73	11.99	3.03	25.75	9.19	6.39	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	375.10
467+060.000	44.58	10.36	19.91	27.32	6.87	33.02	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	813.10
467+080.000	57.42	15.25	34.59	31.58	9.07	51.25	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,020.00
467+100.000	63.85	24.94	53.77	32.00	13.59	72.05	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,212.70
467+120.000	62.43	32.90	66.52	31.87	17.43	83.18	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,262.80
467+140.000	50.83	24.96	68.86	26.15	13.14	85.24	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,132.60
467+160.000	51.91	20.01	52.09	27.07	11.23	68.50	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,027.40
467+180.000	41.89	25.19	33.82	25.42	16.24	50.16	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	938.00
467+200.000	48.56	14.08	17.99	29.67	9.50	27.12	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	904.50
467+220.000	47.91	15.35	16.98	28.65	9.56	24.53	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	964.70
467+240.000	37.76	17.01	15.71	23.03	11.31	25.58	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	856.70
467+260.000	36.24	14.13	10.07	23.45	10.04	16.40	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	740.00
467+280.000	35.15	10.53	2.51	24.09	8.05	3.43	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	713.90
467+300.000	8.96	0.00	0.00	10.68	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	441.10
467+320.000	3.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	127.80
467+340.000	18.89	11.54	0.00	25.37	19.55	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	227.10
467+360.000	39.31	23.91	10.77	23.87	16.37	15.46	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	582.00
467+380.000	34.23	15.28	17.40	20.85	10.08	25.63	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	735.40
467+400.000	30.20	14.23	22.44	18.19	9.23	32.62	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	644.30
467+420.000	28.20	13.74	29.61	16.41	8.38	42.06	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	584.00
467+440.000	27.44	10.80	28.33	16.27	7.33	39.23	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	556.40
467+460.000	21.46	18.76	11.18	14.15	13.06	16.72	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	489.00
467+480.000	10.88	0.60	0.00	15.34	3.28	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	323.40
467+500.000	17.00	0.00	0.00	28.46	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	278.80
467+520.000	39.97	17.87	9.94	24.64	12.12	15.23	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	569.70
467+540.000	11.83	12.33	0.48	16.15	18.57	2.92	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	518.00
467+560.000	8.02	0.00	0.00	10.04	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	198.50
467+580.000	22.28	3.30	0.00	31.36	3.95	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	303.00
467+600.000	50.18	25.39	9.15	29.45	15.96	13.15	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	724.60
467+620.000	51.04	20.05	16.83	30.74	13.36	25.32	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,012.20
467+640.000	43.85	21.70	23.07	25.44	13.78	35.84	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	948.90
467+660.000	37.94	27.22	19.11	22.17	17.44	30.56	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	817.90
467+680.000	43.04	28.20	9.77	26.04	19.09	17.62	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	809.80
467+700.000	49.22	24.19	2.42	31.56	17.62	4.90	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	922.60
467+720.000	44.10	18.21	2.11	28.66	13.29	2.73	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	933.20
467+740.000	16.74	1.54	0.00	25.68	3.38	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	608.40
467+760.000	3.03	0.00	0.00	5.16	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	197.70
467+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.17	8.97	13.20	0.00	0.00	0.00	0.00	1.36	20.00	30.30
467+800.000	15.91	0.00	0.00	24.95	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	159.10
467+820.000	8.92	18.88	14.24	5.67	12.11	29.39	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	248.30
467+840.000	54.08	24.51	15.83	32.14	16.69	26.72	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	630.00
467+860.000	40.10	36.09	38.62	21.21	21.18	55.84	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	941.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
467+880.000	47.05	28.23	62.14	23.45	15.30	80.78	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	871.50
467+900.000	52.72	28.69	86.21	24.31	14.09	102.46	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	997.70
467+920.000	66.49	34.44	98.80	29.16	16.47	112.17	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,192.10
467+940.000	81.12	45.44	100.58	48.80	29.06	114.28	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,476.10
467+960.000	109.45	62.73	65.18	51.44	32.87	79.75	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,905.70
467+980.000	119.02	60.51	19.57	61.49	35.89	31.32	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	2,284.70
468+000.000	27.41	0.00	0.00	39.25	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	1,464.30
468+020.000	5.13	0.00	0.00	6.99	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	325.40
468+040.000	72.00	44.42	6.65	24.64	17.92	7.15	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	771.30
468+060.000	38.43	23.08	7.21	27.14	10.79	9.50	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,104.30
468+080.000	10.19	0.00	0.00	12.76	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	486.20
468+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.59	8.97	20.62	0.00	0.00	0.00	0.00	1.36	20.00	101.90
468+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.93	8.97	16.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
468+140.000	9.94	0.79	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	99.40
468+160.000	34.07	12.05	34.02	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	440.10
468+180.000	43.60	21.07	33.33	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	776.70
468+200.000	15.92	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	595.20
468+220.000	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.81	8.97	9.84	0.00	0.00	0.00	0.00	1.36	20.00	160.25
468+240.000	1.71	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.29	8.97	0.32	0.00	0.00	0.00	0.00	1.36	20.00	18.12
468+260.000	50.93	25.33	5.20	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	526.37
468+280.000	32.86	3.16	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	837.90
468+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.95	8.97	0.00	26.20	13.78	0.00	0.00	1.36	20.00	328.60
468+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	170.98	8.97	0.00	26.20	135.81	0.00	0.00	1.36	20.00	0.00
468+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	339.28	8.97	0.00	26.20	304.11	0.00	0.00	1.36	20.00	0.00
468+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	450.90	8.97	0.00	26.20	415.73	0.00	0.00	1.36	20.00	0.00
468+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	282.56	8.97	0.00	26.20	247.39	0.00	0.00	1.36	20.00	0.00
468+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	72.42	8.97	0.00	26.20	37.25	0.00	0.00	1.36	20.00	0.00
468+420.000	7.59	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	75.90
468+440.000	10.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	180.34
468+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.67	8.97	0.00	26.20	22.50	0.00	0.00	1.36	20.00	104.44
468+480.000	9.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.49	8.97	0.00	0.52	0.00	0.00	0.00	1.36	20.00	98.45
468+500.000	7.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	168.45
468+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.52	8.97	0.00	26.20	23.35	0.00	0.00	1.36	20.00	70.00
468+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	201.58	8.97	0.00	26.20	166.41	0.00	0.00	1.36	20.00	0.00
468+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	368.78	8.97	0.00	26.20	333.61	0.00	0.00	1.36	20.00	0.00
468+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	488.21	8.97	0.00	26.20	453.04	0.00	0.00	1.36	20.00	0.00
468+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	508.90	8.97	0.00	26.20	473.73	0.00	0.00	1.36	20.00	0.00
468+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	474.62	8.97	0.00	26.20	439.45	0.00	0.00	1.36	20.00	0.00
468+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	448.04	8.97	0.00	26.20	412.87	0.00	0.00	1.36	20.00	0.00
468+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	338.08	8.97	0.00	26.20	302.91	0.00	0.00	1.36	20.00	0.00
468+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	190.18	8.97	0.00	26.20	155.01	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
468+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	133.51	8.97	0.00	26.20	98.34	0.00	0.00	1.36	20.00	0.00
468+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.14	8.97	0.00	26.20	40.97	0.00	0.00	1.36	20.00	0.00
468+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.92	8.97	0.00	26.20	5.75	0.00	0.00	1.36	20.00	0.00
468+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.21	8.97	0.00	26.20	19.04	0.00	0.00	1.36	20.00	0.00
468+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.29	8.97	0.00	26.20	75.12	0.00	0.00	1.36	20.00	0.00
468+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	239.38	8.97	0.00	26.20	204.21	0.00	0.00	1.36	20.00	0.00
468+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	236.50	8.97	0.00	26.20	201.33	0.00	0.00	1.36	20.00	0.00
468+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	117.30	8.97	0.00	26.20	82.13	0.00	0.00	1.36	20.00	0.00
468+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.47	8.97	0.00	26.20	48.30	0.00	0.00	1.36	20.00	0.00
468+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.72	8.97	0.00	26.20	49.55	0.00	0.00	1.36	20.00	0.00
468+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	168.14	8.97	0.00	26.20	132.97	0.00	0.00	1.36	20.00	0.00
468+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	207.30	8.97	0.00	26.20	172.13	0.00	0.00	1.36	20.00	0.00
468+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	187.48	8.97	0.00	26.20	152.31	0.00	0.00	1.36	20.00	0.00
468+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	203.19	8.97	0.00	26.20	168.02	0.00	0.00	1.36	20.00	0.00
468+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	316.82	8.97	0.00	26.20	281.65	0.00	0.00	1.36	20.00	0.00
469+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	372.97	8.97	0.00	26.20	337.80	0.00	0.00	1.36	20.00	0.00
469+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	332.26	8.97	0.00	26.20	297.09	0.00	0.00	1.36	20.00	0.00
469+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	388.07	8.97	0.00	26.20	352.90	0.00	0.00	1.36	20.00	0.00
469+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	518.39	8.97	0.00	26.20	483.22	0.00	0.00	1.36	20.00	0.00
469+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	628.82	8.97	0.00	26.20	593.65	0.00	0.00	1.36	20.00	0.00
469+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	625.25	8.97	0.00	26.20	590.08	0.00	0.00	1.36	20.00	0.00
469+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	491.83	8.97	0.00	26.20	456.66	0.00	0.00	1.36	20.00	0.00
469+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	256.16	8.97	0.00	26.20	220.99	0.00	0.00	1.36	20.00	0.00
469+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	195.04	8.97	0.00	26.20	159.87	0.00	0.00	1.36	20.00	0.00
469+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	122.82	8.97	0.00	26.20	87.65	0.00	0.00	1.36	20.00	0.00
469+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	123.34	8.97	0.00	26.20	88.17	0.00	0.00	1.36	20.00	0.00
469+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	116.36	8.97	0.00	26.20	81.19	0.00	0.00	1.36	20.00	0.00
469+240.000	0.57	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.40	8.97	0.00	26.20	16.23	0.00	0.00	1.36	20.00	5.71
469+260.000	1.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.78	8.97	0.00	26.20	8.60	0.00	0.00	1.36	20.00	20.43
469+280.000	10.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.19	8.97	0.00	7.21	0.00	0.00	0.00	1.36	20.00	122.59
469+300.000	25.94	12.28	0.04	17.88	35.16	22.51	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	367.27
469+320.000	49.98	30.88	41.92	22.90	18.05	63.08	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	759.20
469+340.000	54.09	46.45	26.02	26.65	27.73	42.99	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,040.70
469+360.000	10.03	0.00	0.00	36.91	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	641.20
469+380.000	109.03	53.77	37.49	57.40	30.94	63.11	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,190.60
469+400.000	68.76	52.61	80.42	38.79	31.93	114.07	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,777.90
469+420.000	55.51	34.16	45.20	36.22	24.56	86.90	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,242.70
469+440.000	52.11	15.95	0.00	70.35	31.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	1,076.20
469+460.000	43.69	7.50	0.00	63.83	13.89	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	958.00
469+480.000	137.57	38.73	30.35	90.81	29.62	49.96	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,812.60
469+500.000	105.76	43.17	50.28	71.76	31.91	84.58	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	2,433.30
469+520.000	15.17	29.08	34.25	13.03	30.45	47.87	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,209.30
469+540.000	0.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.84	8.97	0.00	8.87	0.00	0.00	0.00	1.36	20.00	156.64
469+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.13	8.97	0.00	26.20	55.96	0.00	0.00	1.36	20.00	4.94

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
469+580.000	0.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.60	8.97	0.00	0.63	0.00	0.00	0.00	1.36	20.00	4.94
469+600.000	0.00	6.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	4.94
469+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	207.44	8.97	0.00	26.20	172.27	0.00	0.00	1.36	20.00	0.00
469+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	261.18	8.97	0.00	26.20	226.01	0.00	0.00	1.36	20.00	0.00
469+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	212.53	8.97	0.00	26.20	177.36	0.00	0.00	1.36	20.00	0.00
469+680.000	12.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	82.04	8.97	0.00	26.20	46.87	0.00	0.00	1.36	20.00	128.17
469+700.000	16.41	4.45	5.39	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	292.27
469+720.000	34.00	13.83	22.77	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	504.10
469+740.000	11.30	3.95	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	453.00
469+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.93	8.97	0.00	26.20	27.76	0.00	0.00	1.36	20.00	113.00
469+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	202.87	8.97	0.00	26.20	167.70	0.00	0.00	1.36	20.00	0.00
469+800.000	0.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.07	8.97	0.00	20.10	0.00	0.00	0.00	1.36	20.00	7.45
469+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.02	8.97	0.00	26.20	9.85	0.00	0.00	1.36	20.00	7.45
469+840.000	3.51	14.03	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.07	8.97	0.00	1.10	0.00	0.00	0.00	1.36	20.00	35.09
469+860.000	86.18	155.64	22.06	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	896.89
469+880.000	21.59	390.93	35.80	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,077.70
469+900.000	28.81	345.12	44.68	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	504.00
469+920.000	59.49	132.14	55.91	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	883.00
469+940.000	19.65	52.88	54.30	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	791.40
469+960.000	13.27	32.40	49.99	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	329.20
469+980.000	18.75	27.52	29.22	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	320.20
470+000.000	9.99	7.46	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	287.40
470+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.25	8.97	0.00	26.20	36.08	0.00	0.00	1.36	20.00	99.90
470+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.55	8.97	0.00	26.20	27.38	0.00	0.00	1.36	20.00	0.00
470+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.10	8.97	0.00	26.20	143.93	0.00	0.00	1.36	20.00	0.00
470+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	78.71	8.97	0.00	26.20	43.54	0.00	0.00	1.36	20.00	0.00
470+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.94	8.97	0.00	26.20	33.77	0.00	0.00	1.36	20.00	0.00
470+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.96	8.97	0.00	26.20	75.79	0.00	0.00	1.36	20.00	0.00
470+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	200.83	8.97	0.00	26.20	165.66	0.00	0.00	1.36	20.00	0.00
470+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	320.65	8.97	0.00	26.20	285.48	0.00	0.00	1.36	20.00	0.00
470+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	427.47	8.97	0.00	26.20	392.30	0.00	0.00	1.36	20.00	0.00
470+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	497.55	8.97	0.00	26.20	462.38	0.00	0.00	1.36	20.00	0.00
470+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	536.02	8.97	0.00	26.20	500.85	0.00	0.00	1.36	20.00	0.00
470+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	537.87	8.97	0.00	26.20	502.70	0.00	0.00	1.36	20.00	0.00
470+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	555.05	8.97	0.00	26.20	519.88	0.00	0.00	1.36	20.00	0.00
470+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	504.70	8.97	0.00	26.20	469.53	0.00	0.00	1.36	20.00	0.00
470+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	422.28	8.97	0.00	26.20	387.11	0.00	0.00	1.36	20.00	0.00
470+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	474.08	8.97	0.00	26.20	438.91	0.00	0.00	1.36	20.00	0.00
470+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	618.61	8.97	0.00	26.20	583.44	0.00	0.00	1.36	20.00	0.00
470+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	807.12	8.97	0.00	26.20	771.95	0.00	0.00	1.36	20.00	0.00
470+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	846.38	8.97	0.00	26.20	811.21	0.00	0.00	1.36	20.00	0.00
470+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	850.51	8.97	0.00	26.20	815.34	0.00	0.00	1.36	20.00	0.00
470+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	842.13	8.97	0.00	26.20	806.96	0.00	0.00	1.36	20.00	0.00
470+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	621.54	8.97	0.00	26.20	586.37	0.00	0.00	1.36	20.00	0.00
470+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	542.77	8.97	0.00	26.20	507.60	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
470+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	431.14	8.97	0.00	26.20	395.97	0.00	0.00	1.36	20.00	0.00
470+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	327.49	8.97	0.00	26.20	292.32	0.00	0.00	1.36	20.00	0.00
470+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	316.50	8.97	0.00	26.20	281.33	0.00	0.00	1.36	20.00	0.00
470+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	198.13	8.97	0.00	26.20	162.96	0.00	0.00	1.36	20.00	0.00
470+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	163.17	8.97	0.00	26.20	128.00	0.00	0.00	1.36	20.00	0.00
470+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	196.19	8.97	0.00	26.20	161.02	0.00	0.00	1.36	20.00	0.00
470+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	257.04	8.97	0.00	26.20	221.87	0.00	0.00	1.36	20.00	0.00
470+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	346.28	8.97	0.00	26.20	311.11	0.00	0.00	1.36	20.00	0.00
470+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	428.53	8.97	0.00	26.20	393.36	0.00	0.00	1.36	20.00	0.00
470+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	349.65	8.97	0.00	26.20	314.48	0.00	0.00	1.36	20.00	0.00
470+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	125.03	8.97	0.00	26.20	89.86	0.00	0.00	1.36	20.00	0.00
470+700.000	12.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.72	8.97	0.00	1.74	0.00	0.00	0.00	1.36	20.00	122.50
470+720.000	7.90	15.07	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	201.50
470+740.000	5.18	4.95	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	130.80
470+760.000	3.46	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	86.40
470+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.91	8.97	13.94	0.00	0.00	0.00	0.00	1.36	20.00	34.60
470+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.26	8.97	46.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
470+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.49	8.97	18.52	0.00	0.00	0.00	0.00	1.36	20.00	0.00
470+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.04	8.97	20.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
470+860.000	8.32	8.76	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	83.20
470+880.000	8.00	3.80	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	163.20
470+900.000	3.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.21	8.97	0.00	10.24	0.00	0.00	0.00	1.36	20.00	117.40
470+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.29	8.97	0.00	26.20	46.12	0.00	0.00	1.36	20.00	37.40
470+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	189.05	8.97	0.00	26.20	153.88	0.00	0.00	1.36	20.00	0.00
470+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	169.54	8.97	0.00	26.20	134.37	0.00	0.00	1.36	20.00	0.00
470+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.29	8.97	0.00	26.20	144.12	0.00	0.00	1.36	20.00	0.00
471+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	155.63	8.97	0.00	26.20	120.46	0.00	0.00	1.36	20.00	0.00
471+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.88	8.97	0.00	12.91	0.00	0.00	0.00	1.36	20.00	0.00
471+040.000	6.59	7.87	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	65.90
471+060.000	10.83	12.51	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	174.20
471+080.000	5.27	10.31	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	161.00
471+100.000	4.84	5.06	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	101.10
471+120.000	2.87	8.57	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.10	8.97	0.00	0.13	0.00	0.00	0.00	1.36	20.00	77.05
471+140.000	3.03	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	58.99
471+160.000	4.36	29.78	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	73.94
471+180.000	11.22	76.39	15.07	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	155.80
471+200.000	7.25	46.10	55.58	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	184.70
471+220.000	21.27	66.85	92.60	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	285.20
471+240.000	15.14	103.36	130.35	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	364.10
471+260.000	19.06	110.34	148.02	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	342.00
471+280.000	15.26	133.22	146.25	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	343.20
471+300.000	23.64	119.88	131.15	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	389.00
471+320.000	25.20	91.36	106.64	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	488.40
471+340.000	26.38	60.92	94.60	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	515.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
471+360.000	19.41	58.97	106.75	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	457.90
471+380.000	11.85	70.41	121.05	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	312.60
471+400.000	25.89	40.07	112.14	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	377.40
471+420.000	17.31	52.78	30.45	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	432.00
471+440.000	5.61	22.15	12.23	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	229.20
471+460.000	3.36	22.29	10.76	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	89.70
471+480.000	13.33	17.07	6.40	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	166.90
471+500.000	35.09	53.33	6.37	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	484.20
471+520.000	18.65	31.02	49.03	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	537.40
471+540.000	13.50	23.54	60.93	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	321.50
471+560.000	14.71	80.53	82.53	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	282.10
471+580.000	98.73	96.06	126.94	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,134.40
471+600.000	45.68	140.13	310.42	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,444.10
471+620.000	36.21	127.57	358.21	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	818.90
471+640.000	39.92	106.32	216.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	761.30
471+660.000	32.84	62.75	106.28	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	727.60
471+680.000	17.07	56.12	47.91	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	499.10
471+700.000	30.72	38.06	4.64	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	477.90
471+720.000	10.28	0.43	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	410.00
471+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.83	8.97	0.00	1.86	0.00	0.00	0.00	1.36	20.00	102.80
471+760.000	1.29	0.03	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	12.90
471+780.000	7.42	0.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	87.13
471+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.48	8.97	0.00	17.51	0.00	0.00	0.00	1.36	20.00	74.23
471+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.49	8.97	0.00	26.20	31.32	0.00	0.00	1.36	20.00	0.00
471+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.80	8.97	0.00	26.20	22.63	0.00	0.00	1.36	20.00	0.00
471+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.62	8.97	0.00	26.20	24.45	0.00	0.00	1.36	20.00	0.00
471+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.59	8.97	0.00	18.62	0.00	0.00	0.00	1.36	20.00	0.00
471+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.94	8.97	0.00	11.97	0.00	0.00	0.00	1.36	20.00	0.00
471+920.000	2.32	0.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	23.21
471+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.49	8.97	0.00	9.52	0.00	0.00	0.00	1.36	20.00	23.21
471+960.000	2.00	4.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	19.96
471+980.000	2.52	3.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	45.16
472+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.96	8.97	0.00	3.99	0.00	0.00	0.00	1.36	20.00	25.20
472+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.87	8.97	0.00	26.20	2.70	0.00	0.00	1.36	20.00	0.00
472+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.70	8.97	0.00	26.20	13.53	0.00	0.00	1.36	20.00	0.00
472+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.11	8.97	0.00	26.20	10.94	0.00	0.00	1.36	20.00	0.00
472+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.28	8.97	0.00	26.20	5.11	0.00	0.00	1.36	20.00	0.00
472+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.70	8.97	0.00	26.20	5.53	0.00	0.00	1.36	20.00	0.00
472+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.28	8.97	0.00	26.20	7.11	0.00	0.00	1.36	20.00	0.00
472+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.77	8.97	0.00	26.20	25.60	0.00	0.00	1.36	20.00	0.00
472+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.72	8.97	0.00	26.20	25.55	0.00	0.00	1.36	20.00	0.00
472+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.16	8.97	0.00	26.20	45.99	0.00	0.00	1.36	20.00	0.00
472+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	98.38	8.97	0.00	26.20	63.21	0.00	0.00	1.36	20.00	0.00
472+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.77	8.97	0.00	26.20	67.60	0.00	0.00	1.36	20.00	0.00
472+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	98.31	8.97	0.00	26.20	63.14	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
472+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.10	8.97	0.00	26.20	71.93	0.00	0.00	1.36	20.00	0.00
472+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	138.13	8.97	0.00	26.20	102.96	0.00	0.00	1.36	20.00	0.00
472+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	136.77	8.97	0.00	26.20	101.60	0.00	0.00	1.36	20.00	0.00
472+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.12	8.97	0.00	26.20	21.95	0.00	0.00	1.36	20.00	0.00
472+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.18	8.97	0.00	26.20	18.01	0.00	0.00	1.36	20.00	0.00
472+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	96.35	8.97	0.00	26.20	61.18	0.00	0.00	1.36	20.00	0.00
472+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	130.08	8.97	0.00	26.20	94.91	0.00	0.00	1.36	20.00	0.00
472+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	97.43	8.97	0.00	26.20	62.26	0.00	0.00	1.36	20.00	0.00
472+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.84	8.97	0.00	26.20	39.67	0.00	0.00	1.36	20.00	0.00
472+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.23	8.97	0.00	26.20	20.06	0.00	0.00	1.36	20.00	0.00
472+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.27	8.97	0.00	26.20	21.10	0.00	0.00	1.36	20.00	0.00
472+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.32	8.97	0.00	17.35	0.00	0.00	0.00	1.36	20.00	0.00
472+500.000	1.33	1.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	13.30
472+520.000	2.69	2.20	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	40.20
472+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.87	8.97	0.00	2.90	0.00	0.00	0.00	1.36	20.00	26.90
472+560.000	1.16	1.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.57	8.97	0.00	0.60	0.00	0.00	0.00	1.36	20.00	11.63
472+580.000	1.59	0.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.13	8.97	0.00	0.16	0.00	0.00	0.00	1.36	20.00	27.57
472+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.79	8.97	0.00	8.82	0.00	0.00	0.00	1.36	20.00	15.94
472+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.65	8.97	0.00	26.20	7.48	0.00	0.00	1.36	20.00	0.00
472+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.37	8.97	0.00	26.20	27.20	0.00	0.00	1.36	20.00	0.00
472+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.30	8.97	0.00	26.20	1.13	0.00	0.00	1.36	20.00	0.00
472+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.16	8.97	0.00	8.19	0.00	0.00	0.00	1.36	20.00	0.00
472+700.000	5.68	3.89	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	56.80
472+720.000	2.11	15.93	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	77.90
472+740.000	2.85	1.29	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	49.60
472+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.23	8.97	9.26	0.00	0.00	0.00	0.00	1.36	20.00	28.50
472+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.39	8.97	72.42	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	188.89	8.97	179.92	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	288.54	8.97	279.57	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	242.70	8.97	233.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	240.99	8.97	232.02	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	260.40	8.97	251.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	140.30	8.97	131.33	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.98	8.97	76.01	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.70	8.97	62.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	146.36	8.97	137.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
472+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	261.99	8.97	253.02	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	251.13	8.97	242.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	248.25	8.97	239.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	248.85	8.97	239.88	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	237.53	8.97	228.56	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	211.94	8.97	202.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	164.75	8.97	155.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.35	8.97	101.38	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.09	8.97	40.12	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.73	8.97	15.76	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.02	8.97	8.05	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.64	8.97	6.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
473+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.28	8.97	4.31	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.12	8.97	2.15	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.50	8.97	0.53	0.00	0.00	0.00	0.00	1.36	20.00	0.00
473+280.000	1.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	12.47
473+300.000	0.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.09	8.97	0.00	0.12	0.00	0.00	0.00	1.36	20.00	21.44
473+320.000	4.95	0.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	58.47
473+340.000	7.22	4.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	121.72
473+360.000	2.65	0.00	0.00	27.10	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	98.72
473+380.000	4.85	2.40	0.00	30.93	52.49	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	75.00
473+400.000	5.73	7.41	0.00	32.97	79.76	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	105.80
473+420.000	4.39	12.74	0.00	23.42	113.54	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	101.20
473+440.000	2.98	16.85	0.00	15.22	139.10	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	73.70
473+460.000	2.26	15.16	0.00	6.06	142.38	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	52.40
473+480.000	5.54	7.89	0.00	31.80	93.61	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.00
473+500.000	1.67	0.26	0.00	12.21	43.31	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	72.10
473+520.000	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.79	8.97	0.00	0.82	0.00	0.00	0.00	1.36	20.00	17.60
473+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.44	8.97	0.00	23.47	0.00	0.00	0.00	1.36	20.00	0.90
473+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.61	8.97	0.00	26.20	25.44	0.00	0.00	1.36	20.00	0.00
473+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.99	8.97	0.00	26.20	25.82	0.00	0.00	1.36	20.00	0.00
473+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.26	8.97	0.00	15.29	0.00	0.00	0.00	1.36	20.00	0.00
473+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.36	8.97	0.00	21.39	0.00	0.00	0.00	1.36	20.00	0.00
473+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.01	8.97	0.00	26.20	5.84	0.00	0.00	1.36	20.00	0.00
473+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.22	8.97	0.00	26.20	6.05	0.00	0.00	1.36	20.00	0.00
473+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.72	8.97	0.00	25.75	0.00	0.00	0.00	1.36	20.00	0.00
473+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.79	8.97	0.00	26.20	4.62	0.00	0.00	1.36	20.00	0.00
473+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.01	8.97	0.00	26.20	10.84	0.00	0.00	1.36	20.00	0.00
473+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.65	8.97	0.00	26.20	24.48	0.00	0.00	1.36	20.00	0.00
473+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.31	8.97	0.00	26.20	66.14	0.00	0.00	1.36	20.00	0.00
473+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	158.33	8.97	0.00	26.20	123.16	0.00	0.00	1.36	20.00	0.00
473+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	222.87	8.97	0.00	26.20	187.70	0.00	0.00	1.36	20.00	0.00
473+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	245.79	8.97	0.00	26.20	210.62	0.00	0.00	1.36	20.00	0.00
473+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	183.45	8.97	0.00	26.20	148.28	0.00	0.00	1.36	20.00	0.00
473+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.27	8.97	0.00	26.20	49.10	0.00	0.00	1.36	20.00	0.00
473+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.32	8.97	0.00	26.20	14.15	0.00	0.00	1.36	20.00	0.00
473+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.61	8.97	0.00	10.64	0.00	0.00	0.00	1.36	20.00	0.00
473+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.50	8.97	0.00	12.53	0.00	0.00	0.00	1.36	20.00	0.00
473+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.73	8.97	0.00	26.20	19.56	0.00	0.00	1.36	20.00	0.00
473+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	94.92	8.97	0.00	26.20	59.75	0.00	0.00	1.36	20.00	0.00
473+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.91	8.97	0.00	26.20	25.74	0.00	0.00	1.36	20.00	0.00
474+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.10	8.97	0.00	26.20	28.93	0.00	0.00	1.36	20.00	0.00
474+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.05	8.97	0.00	26.20	27.88	0.00	0.00	1.36	20.00	0.00
474+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.50	8.97	0.00	26.20	3.33	0.00	0.00	1.36	20.00	0.00
474+060.000	1.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.80	8.97	0.00	5.83	0.00	0.00	0.00	1.36	20.00	10.48
474+080.000	2.61	23.91	0.00	4.41	50.75	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	36.58

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
474+100.000	4.60	70.08	22.89	3.18	53.85	45.33	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	72.10
474+120.000	22.46	35.10	62.46	13.72	22.42	95.25	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	270.60
474+140.000	18.72	8.05	86.39	11.20	5.50	119.12	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	411.80
474+160.000	16.12	0.69	95.25	13.27	0.64	127.84	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	348.40
474+180.000	19.33	6.60	92.12	14.51	5.53	133.78	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	354.50
474+200.000	21.62	8.17	75.96	13.30	5.85	108.47	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	409.50
474+220.000	18.29	9.88	52.35	5.62	7.47	81.31	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	399.10
474+240.000	19.44	18.95	23.22	14.35	14.45	44.65	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	377.30
474+260.000	11.99	11.06	0.00	21.27	25.89	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	314.30
474+280.000	1.89	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	138.80
474+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.72	8.97	0.00	8.75	0.00	0.00	0.00	1.36	20.00	18.90
474+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.50	8.97	0.00	26.20	12.33	0.00	0.00	1.36	20.00	0.00
474+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.89	8.97	0.00	24.92	0.00	0.00	0.00	1.36	20.00	0.00
474+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.29	8.97	0.00	19.32	0.00	0.00	0.00	1.36	20.00	0.00
474+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.23	8.97	0.00	26.20	11.06	0.00	0.00	1.36	20.00	0.00
474+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.33	8.97	0.00	26.20	12.16	0.00	0.00	1.36	20.00	0.00
474+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.50	8.97	0.00	16.53	0.00	0.00	0.00	1.36	20.00	0.00
474+440.000	7.94	1.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	79.37
474+460.000	7.89	2.06	0.00	29.40	48.62	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	158.27
474+480.000	6.92	9.57	0.00	29.26	81.17	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	148.10
474+500.000	3.62	11.63	0.00	26.24	89.74	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	105.40
474+520.000	6.28	10.42	0.00	40.63	79.42	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	99.00
474+540.000	5.07	4.61	0.00	42.41	53.05	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	113.50
474+560.000	4.94	1.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	100.09
474+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.34	8.97	20.37	0.00	0.00	0.00	0.00	1.36	20.00	49.39
474+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.93	8.97	38.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
474+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.35	8.97	11.38	0.00	0.00	0.00	0.00	1.36	20.00	0.00
474+640.000	5.85	1.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	58.51
474+660.000	5.99	10.29	0.00	17.40	68.42	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	118.41
474+680.000	7.80	35.91	0.00	15.72	111.18	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	137.90
474+700.000	6.18	31.76	0.00	13.51	95.15	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	139.80
474+720.000	6.99	30.86	0.00	14.54	90.37	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	131.70
474+740.000	10.14	32.09	0.00	20.29	91.55	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	171.30
474+760.000	8.12	19.28	0.00	19.11	61.98	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	182.60
474+780.000	4.11	6.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	122.28
474+800.000	6.22	2.77	0.00	19.29	11.02	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	103.28
474+820.000	3.43	26.68	0.00	7.55	70.91	5.11	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	96.50
474+840.000	10.42	33.60	21.15	11.22	38.27	59.89	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	138.50
474+860.000	9.35	34.36	23.69	9.92	37.94	68.17	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	197.70
474+880.000	9.80	29.71	19.48	10.92	34.65	60.26	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	191.50
474+900.000	9.80	27.48	14.27	11.50	34.36	46.94	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	196.00
474+920.000	11.47	28.17	9.97	13.28	34.26	34.18	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	212.70
474+940.000	10.53	35.45	5.76	12.21	43.14	21.35	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	220.00
474+960.000	14.70	38.22	1.12	14.10	39.79	3.01	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	252.30
474+980.000	3.22	14.19	0.00	8.27	44.57	2.81	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	179.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
475+000.000	7.22	8.06	0.00	19.96	27.14	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	104.40
475+020.000	4.73	2.30	0.00	14.44	16.36	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	119.50
475+040.000	2.86	0.33	0.00	8.27	10.22	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	75.90
475+060.000	1.06	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	39.20
475+080.000	5.15	6.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	62.14
475+100.000	4.22	4.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	93.78
475+120.000	4.03	4.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	82.52
475+140.000	5.23	4.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	92.59
475+160.000	5.27	3.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	105.04
475+180.000	1.81	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	70.83
475+200.000	3.17	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	49.80
475+220.000	6.37	5.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	95.42
475+240.000	4.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	109.33
475+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.06	8.97	2.09	0.00	0.00	0.00	0.00	1.36	20.00	45.61
475+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.19	8.97	11.22	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.28	8.97	13.31	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.04	8.97	12.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.80	8.97	8.83	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.51	8.97	8.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.45	8.97	14.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.12	8.97	13.15	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.54	8.97	11.57	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.94	8.97	9.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.70	8.97	9.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.99	8.97	11.02	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.58	8.97	12.61	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.31	8.97	12.34	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.47	8.97	12.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.41	8.97	16.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.98	8.97	22.01	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.90	8.97	21.93	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.83	8.97	30.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.12	8.97	32.15	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.17	8.97	29.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.35	8.97	31.38	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.32	8.97	34.35	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.17	8.97	37.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.71	8.97	36.74	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.80	8.97	31.83	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.32	8.97	19.35	0.00	0.00	0.00	0.00	1.36	20.00	0.00
475+800.000	0.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.78	8.97	1.80	0.00	0.00	0.00	0.00	1.36	20.00	5.55
475+820.000	2.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.99	8.97	0.01	0.00	0.00	0.00	0.00	1.36	20.00	34.47
475+840.000	2.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.02	8.97	0.05	0.00	0.00	0.00	0.00	1.36	20.00	50.58
475+860.000	2.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	49.26
475+880.000	7.21	2.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	99.67

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
475+900.000	4.43	1.31	0.00	19.25	18.93	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	116.37
475+920.000	3.69	7.96	0.00	14.20	46.85	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	81.20
475+940.000	2.50	14.16	0.00	8.78	71.08	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	61.90
475+960.000	3.35	17.61	0.00	11.15	80.56	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	58.50
475+980.000	3.30	16.44	0.00	11.17	79.06	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	66.50
476+000.000	4.91	11.90	0.00	17.52	65.29	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	82.10
476+020.000	4.02	7.50	0.00	15.59	47.57	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	89.30
476+040.000	3.78	2.62	0.00	14.98	26.96	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.00
476+060.000	0.80	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	45.80
476+080.000	3.63	3.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	44.33
476+100.000	2.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	57.96
476+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.81	8.97	1.84	0.00	0.00	0.00	0.00	1.36	20.00	21.63
476+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.68	8.97	6.71	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.03	8.97	12.06	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.91	8.97	17.94	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.41	8.97	21.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.40	8.97	27.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.24	8.97	35.27	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.16	8.97	42.19	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.61	8.97	49.64	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.35	8.97	53.38	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.48	8.97	45.51	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.66	8.97	38.69	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.29	8.97	33.32	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.28	8.97	33.31	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.88	8.97	34.91	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.14	8.97	38.17	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.26	8.97	42.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.38	8.97	42.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.13	8.97	44.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.69	8.97	48.72	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.46	8.97	57.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.27	8.97	65.30	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.42	8.97	66.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	78.59	8.97	69.62	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	79.46	8.97	70.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.90	8.97	66.93	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	72.51	8.97	63.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.22	8.97	60.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.44	8.97	57.47	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.59	8.97	49.62	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.20	8.97	23.23	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.33	8.97	15.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.47	8.97	15.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.17	8.97	11.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
476+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.78	8.97	5.81	0.00	0.00	0.00	0.00	1.36	20.00	0.00
476+820.000	0.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.15	8.97	1.18	0.00	0.00	0.00	0.00	1.36	20.00	3.40
476+840.000	3.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	41.01
476+860.000	5.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	90.45
476+880.000	7.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	125.03
476+900.000	8.74	1.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	159.57
476+920.000	9.32	2.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	180.62
476+940.000	7.53	2.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	168.55
476+960.000	4.87	0.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	123.96
476+980.000	3.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	84.89
477+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.87	8.97	5.90	0.00	0.00	0.00	0.00	1.36	20.00	36.24
477+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.47	8.97	13.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.63	8.97	14.66	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.92	8.97	14.95	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.58	8.97	12.61	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.70	8.97	18.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.78	8.97	3.81	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+140.000	5.50	6.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	55.01
477+160.000	4.41	9.40	0.00	12.04	33.30	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	99.11
477+180.000	4.64	21.07	0.00	10.80	62.76	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	90.50
477+200.000	2.93	26.47	0.00	6.61	83.36	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	75.70
477+220.000	9.98	66.61	5.26	10.96	82.50	22.48	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	129.10
477+240.000	8.46	77.41	12.39	8.57	87.75	40.82	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	184.40
477+260.000	0.00	73.94	21.19	0.00	78.46	65.15	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	84.60
477+280.000	2.06	70.90	23.35	2.02	75.16	68.28	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	20.60
477+300.000	6.67	61.90	19.41	6.80	68.43	57.47	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	87.30
477+320.000	15.26	46.97	12.83	16.48	54.79	41.26	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	219.30
477+340.000	15.38	44.50	6.83	17.40	55.28	24.39	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	306.40
477+360.000	14.71	49.76	1.61	13.83	52.30	8.64	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	300.90
477+380.000	7.22	18.45	0.00	17.37	62.97	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	219.30
477+400.000	5.95	8.70	0.00	16.95	40.62	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	131.70
477+420.000	3.85	3.41	0.00	12.24	25.36	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	98.00
477+440.000	2.58	0.00	0.00	8.52	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	64.30
477+460.000	6.14	2.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	87.15
477+480.000	4.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	107.94
477+500.000	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.95	8.97	0.98	0.00	0.00	0.00	0.00	1.36	20.00	47.92
477+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.09	8.97	4.12	0.00	0.00	0.00	0.00	1.36	20.00	1.33
477+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.42	8.97	1.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+560.000	5.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	50.81
477+580.000	9.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	150.39
477+600.000	4.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	141.30
477+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.59	8.97	4.62	0.00	0.00	0.00	0.00	1.36	20.00	41.72
477+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.66	8.97	15.69	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	31.25	8.97	22.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
477+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.72	8.97	21.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.34	8.97	18.37	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.20	8.97	10.23	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.63	8.97	10.66	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.93	8.97	10.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.81	8.97	16.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.57	8.97	25.60	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.06	8.97	35.09	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.04	8.97	42.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.97	8.97	36.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.93	8.97	35.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.90	8.97	35.93	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.82	8.97	35.85	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.44	8.97	36.47	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.78	8.97	38.81	0.00	0.00	0.00	0.00	1.36	20.00	0.00
477+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.94	8.97	41.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.50	8.97	44.53	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.99	8.97	45.02	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.93	8.97	45.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.75	8.97	48.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.14	8.97	53.17	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.70	8.97	57.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.84	8.97	61.87	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.76	8.97	66.79	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	80.06	8.97	71.09	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	82.54	8.97	73.57	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.06	8.97	75.09	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	85.58	8.97	76.61	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.09	8.97	77.12	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.52	8.97	75.55	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.41	8.97	74.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.19	8.97	67.22	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.74	8.97	57.77	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.08	8.97	48.11	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.42	8.97	38.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.17	8.97	37.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.33	8.97	37.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.04	8.97	43.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.07	8.97	55.10	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.72	8.97	59.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.72	8.97	62.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.88	8.97	64.91	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.76	8.97	67.79	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	79.54	8.97	70.57	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
478+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.96	8.97	72.99	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.94	8.97	72.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.42	8.97	72.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.02	8.97	72.05	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.90	8.97	77.93	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	90.62	8.97	81.65	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	98.14	8.97	89.17	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	93.79	8.97	84.82	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.26	8.97	80.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.84	8.97	75.87	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	80.47	8.97	71.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.83	8.97	68.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	78.45	8.97	69.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	78.88	8.97	69.91	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.26	8.97	68.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.54	8.97	64.57	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.95	8.97	51.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.48	8.97	62.51	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.98	8.97	56.01	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.29	8.97	49.32	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.25	8.97	49.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
478+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.25	8.97	49.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.25	8.97	49.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.22	8.97	49.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.63	8.97	42.66	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.70	8.97	37.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.21	8.97	33.24	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.81	8.97	30.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.81	8.97	30.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.81	8.97	30.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.81	8.97	30.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.81	8.97	30.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.11	8.97	25.14	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.68	8.97	15.71	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.55	8.97	13.58	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.74	8.97	7.77	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.30	8.97	2.33	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+300.000	1.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	17.92
479+320.000	7.43	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	92.18
479+340.000	1.72	0.00	0.00	37.20	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	91.46
479+360.000	6.33	0.00	0.00	57.43	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	80.50
479+380.000	7.49	0.00	0.00	56.46	20.55	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	138.20
479+400.000	7.29	0.00	0.00	55.29	19.71	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	147.80
479+420.000	5.25	0.00	0.00	44.68	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	125.40

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
479+440.000	2.60	0.00	0.00	32.86	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.50
479+460.000	8.05	3.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	106.48
479+480.000	7.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	158.27
479+500.000	3.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	110.76
479+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.07	8.97	5.10	0.00	0.00	0.00	0.00	1.36	20.00	32.97
479+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.34	8.97	13.37	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.37	8.97	15.40	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.37	8.97	15.40	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.75	8.97	12.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.51	8.97	5.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
479+640.000	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.65	8.97	0.67	0.00	0.00	0.00	0.00	1.36	20.00	1.17
479+660.000	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.52	8.97	0.55	0.00	0.00	0.00	0.00	1.36	20.00	5.20
479+680.000	4.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	45.34
479+700.000	5.92	2.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	100.55
479+720.000	6.18	4.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	121.03
479+740.000	7.15	3.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	133.32
479+760.000	7.26	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	144.09
479+780.000	2.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	99.85
479+800.000	9.76	1.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	124.85
479+820.000	8.88	2.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	186.32
479+840.000	5.73	2.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	146.07
479+860.000	5.74	2.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	114.73
479+880.000	5.75	2.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	114.87
479+900.000	5.82	2.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	115.65
479+920.000	6.44	2.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	122.62
479+940.000	2.64	8.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	90.83
479+960.000	7.60	6.31	0.00	43.63	73.36	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	102.41
479+980.000	10.53	12.17	0.00	53.98	99.99	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	181.30
480+000.000	4.62	13.05	0.00	24.50	130.68	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	151.50
480+020.000	4.94	12.32	0.00	26.58	108.65	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	95.60
480+040.000	4.72	10.06	0.00	26.31	97.18	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	96.60
480+060.000	4.44	7.60	0.00	25.80	81.96	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	91.60
480+080.000	4.72	4.95	0.00	28.67	65.07	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	91.60
480+100.000	5.08	2.22	0.00	32.84	53.62	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	98.00
480+120.000	5.43	1.85	0.00	37.50	66.23	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	105.10
480+140.000	7.58	14.77	0.00	38.39	117.97	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	130.10
480+160.000	2.01	26.50	0.00	19.00	158.01	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	95.90
480+180.000	4.90	19.63	0.00	23.25	122.53	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	69.10
480+200.000	4.54	10.93	0.00	24.56	87.15	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	94.40
480+220.000	1.79	4.29	0.00	11.24	58.77	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	63.30
480+240.000	1.74	0.02	0.00	15.31	20.10	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	35.30
480+260.000	5.60	2.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	73.36
480+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.93	8.97	2.96	0.00	0.00	0.00	0.00	1.36	20.00	55.96
480+300.000	2.93	4.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	29.33

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
480+320.000	3.93	1.29	0.00	25.87	36.16	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	68.63
480+340.000	3.70	8.15	0.00	21.33	81.72	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	76.30
480+360.000	3.91	13.37	0.00	20.73	113.70	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	76.10
480+380.000	3.32	13.96	0.00	17.66	122.35	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	72.30
480+400.000	3.71	13.57	0.00	19.82	122.10	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	70.30
480+420.000	6.13	9.26	0.00	34.31	99.53	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	98.40
480+440.000	3.04	7.85	0.00	17.78	86.36	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	91.70
480+460.000	4.82	11.36	0.00	25.68	76.84	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.60
480+480.000	1.98	15.30	0.00	10.15	106.75	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	68.00
480+500.000	0.14	17.86	0.00	0.69	120.66	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	21.20
480+520.000	0.27	17.44	0.00	1.39	129.05	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	4.10
480+540.000	0.00	14.19	0.00	0.00	118.67	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	2.70
480+560.000	0.32	8.65	0.00	1.90	99.49	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	3.20
480+580.000	2.38	3.92	0.00	15.19	61.87	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	27.00
480+600.000	1.03	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	34.10
480+620.000	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.52	8.97	1.55	0.00	0.00	0.00	0.00	1.36	20.00	10.40
480+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.22	8.97	6.25	0.00	0.00	0.00	0.00	1.36	20.00	0.10
480+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.82	8.97	14.85	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+680.000	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.82	8.97	1.85	0.00	0.00	0.00	0.00	1.36	20.00	1.01
480+700.000	3.16	3.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	32.62
480+720.000	3.42	1.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	65.78
480+740.000	2.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	56.12
480+760.000	0.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.51	8.97	0.54	0.00	0.00	0.00	0.00	1.36	20.00	23.50
480+780.000	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	6.80
480+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.54	8.97	1.57	0.00	0.00	0.00	0.00	1.36	20.00	5.25
480+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.93	8.97	2.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.25	8.97	5.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.73	8.97	8.76	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.32	8.97	12.35	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.17	8.97	16.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.66	8.97	20.69	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.51	8.97	28.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.83	8.97	37.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
480+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.00	8.97	50.03	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.21	8.97	61.24	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.22	8.97	65.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.10	8.97	72.13	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.09	8.97	82.12	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	93.42	8.97	84.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.21	8.97	92.24	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	104.45	8.97	95.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.46	8.97	97.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.53	8.97	96.56	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.33	8.97	104.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	123.70	8.97	114.73	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	124.14	8.97	115.17	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	134.35	8.97	125.38	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	138.17	8.97	129.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	140.58	8.97	131.61	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	150.63	8.97	141.66	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	169.61	8.97	160.64	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	147.81	8.97	138.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	124.38	8.97	115.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.10	8.97	82.13	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.46	8.97	54.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.64	8.97	62.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
481+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.26	8.97	67.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.10	8.97	74.13	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.45	8.97	75.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.56	8.97	67.59	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	67.60	8.97	58.63	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.55	8.97	50.58	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.67	8.97	46.70	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.04	8.97	41.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.82	8.97	40.85	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.26	8.97	42.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.51	8.97	53.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.91	8.97	64.94	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.77	8.97	72.80	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.74	8.97	77.77	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.95	8.97	74.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.24	8.97	67.27	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.76	8.97	57.79	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.60	8.97	53.63	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	57.83	8.97	48.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.29	8.97	42.32	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.27	8.97	36.30	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.53	8.97	28.56	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.96	8.97	19.99	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.15	8.97	12.18	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.84	8.97	10.87	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.69	8.97	14.72	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.83	8.97	21.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
481+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.60	8.97	27.63	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.98	8.97	32.01	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.17	8.97	31.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.95	8.97	31.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.45	8.97	34.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.91	8.97	33.94	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.04	8.97	33.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.65	8.97	32.68	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.31	8.97	34.34	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.76	8.97	35.79	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.98	8.97	38.01	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.20	8.97	43.23	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.02	8.97	46.05	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.46	8.97	49.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.42	8.97	51.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.15	8.97	53.18	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	67.50	8.97	58.53	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.82	8.97	62.85	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.73	8.97	65.76	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.69	8.97	64.72	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.40	8.97	77.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	92.04	8.97	83.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.86	8.97	61.89	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.01	8.97	47.04	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.35	8.97	28.38	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.40	8.97	9.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+500.000	4.14	1.31	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	41.40
482+520.000	14.79	40.90	6.61	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	189.30
482+540.000	13.16	27.66	20.66	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	279.50

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
482+560.000	13.88	20.13	17.94	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	270.40
482+580.000	10.26	19.48	2.16	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	241.40
482+600.000	2.40	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	126.60
482+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.33	8.97	9.36	0.00	0.00	0.00	0.00	1.36	20.00	24.00
482+640.000	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.42	8.97	2.45	0.00	0.00	0.00	0.00	1.36	20.00	0.68
482+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.70	8.97	9.73	0.00	0.00	0.00	0.00	1.36	20.00	0.68
482+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.27	8.97	23.30	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	31.63	8.97	22.66	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+720.000	2.94	0.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	29.44
482+740.000	4.98	6.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	79.28
482+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.75	8.97	12.78	0.00	0.00	0.00	0.00	1.36	20.00	49.84
482+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.93	8.97	42.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.39	8.97	20.42	0.00	0.00	0.00	0.00	1.36	20.00	0.00
482+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.57	8.97	7.60	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
482+840.000	4.51	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	45.10
482+860.000	3.66	0.33	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	81.70
482+880.000	9.47	11.90	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	131.30
482+900.000	9.19	30.61	0.83	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	186.60
482+920.000	15.89	62.90	16.18	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	250.80
482+940.000	19.28	58.75	30.66	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	351.70
482+960.000	15.64	47.61	49.29	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	349.20
482+980.000	8.84	37.67	66.58	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	244.80
483+000.000	9.42	34.43	78.62	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	182.60
483+020.000	7.96	37.30	86.87	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	173.80
483+040.000	22.02	104.10	82.63	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	299.80
483+060.000	14.73	165.85	89.00	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	367.50
483+080.000	15.43	183.39	103.33	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	301.60
483+100.000	8.66	190.28	115.88	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	240.90
483+120.000	22.30	115.58	116.93	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	309.60
483+140.000	15.18	52.80	116.15	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	374.80
483+160.000	18.63	80.13	119.42	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	338.10
483+180.000	17.52	66.08	111.06	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	361.50
483+200.000	15.40	51.40	102.49	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	329.20
483+220.000	16.80	35.43	84.61	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	322.00
483+240.000	16.09	32.83	56.04	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	328.90
483+260.000	17.13	23.57	27.31	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	332.20
483+280.000	11.00	17.50	19.85	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	281.30
483+300.000	15.77	27.68	23.39	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	267.70
483+320.000	16.28	31.13	23.04	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	320.50
483+340.000	8.63	31.86	15.38	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	249.10
483+360.000	9.74	30.28	5.83	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	183.70
483+380.000	4.78	9.78	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	145.20
483+400.000	5.29	2.82	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	100.70
483+420.000	2.60	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.90
483+440.000	4.83	4.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	74.26
483+460.000	4.48	1.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	93.07
483+480.000	1.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	57.96
483+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.79	8.97	2.82	0.00	0.00	0.00	0.00	1.36	20.00	13.15
483+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.95	8.97	1.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.13	8.97	1.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.94	8.97	6.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.28	8.97	21.31	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.51	8.97	40.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.34	8.97	60.37	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.19	8.97	74.22	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.33	8.97	79.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.47	8.97	74.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	65.54	8.97	56.57	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.47	8.97	33.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.72	8.97	13.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.24	8.97	2.27	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+780.000	0.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	9.15
483+800.000	2.14	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	30.55
483+820.000	3.86	2.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	59.99
483+840.000	6.06	4.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	99.17
483+860.000	3.29	0.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	93.46
483+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.50	8.97	2.53	0.00	0.00	0.00	0.00	1.36	20.00	32.88

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
483+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.79	8.97	7.82	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.05	8.97	8.08	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.61	8.97	8.64	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.10	8.97	10.13	0.00	0.00	0.00	0.00	1.36	20.00	0.00
483+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.96	8.97	9.99	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.92	8.97	10.95	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.79	8.97	12.82	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.89	8.97	15.92	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.34	8.97	20.37	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.81	8.97	24.84	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.39	8.97	31.42	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.47	8.97	37.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.00	8.97	39.03	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.65	8.97	37.68	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.36	8.97	38.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.84	8.97	38.87	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.41	8.97	39.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.60	8.97	39.63	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.87	8.97	41.90	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.59	8.97	46.62	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.15	8.97	62.18	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.47	8.97	66.50	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.74	8.97	64.77	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.17	8.97	68.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	82.84	8.97	73.87	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.29	8.97	72.32	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.20	8.97	64.23	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.65	8.97	54.68	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.49	8.97	39.52	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.36	8.97	31.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.94	8.97	20.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.05	8.97	11.08	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.86	8.97	6.89	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.25	8.97	3.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.75	8.97	0.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
484+600.000	3.82	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	38.23
484+620.000	0.91	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	47.33
484+640.000	4.59	2.06	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	55.00
484+660.000	4.16	7.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	87.50
484+680.000	3.67	7.47	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.30
484+700.000	4.20	4.96	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.70
484+720.000	4.81	2.74	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	90.10
484+740.000	4.04	1.47	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	88.50
484+760.000	4.97	0.24	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	90.10
484+780.000	7.03	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	120.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
484+800.000	3.34	0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	103.72
484+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.43	8.97	0.00	6.46	0.00	0.00	0.00	1.36	20.00	33.42
484+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.66	8.97	0.00	12.69	0.00	0.00	0.00	1.36	20.00	0.00
484+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.54	8.97	0.00	9.57	0.00	0.00	0.00	1.36	20.00	0.00
484+880.000	1.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	10.40
484+900.000	5.86	4.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	69.03
484+920.000	6.07	0.50	0.00	38.61	33.45	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	119.33
484+940.000	5.63	1.95	0.00	36.31	49.65	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	117.00
484+960.000	3.05	0.00	0.00	24.63	41.80	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	86.80
484+980.000	3.71	5.36	0.00	22.73	73.25	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	67.60
485+000.000	3.14	11.94	0.00	17.08	103.30	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	68.50
485+020.000	2.87	10.10	0.00	16.01	90.95	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	60.10
485+040.000	3.43	5.61	0.00	20.70	62.19	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	63.00
485+060.000	4.54	2.43	0.00	28.78	44.56	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	79.70
485+080.000	2.22	0.00	0.00	51.50	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	67.60
485+100.000	4.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	67.79

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
485+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.25	8.97	0.00	7.28	0.00	0.00	0.00	1.36	20.00	45.59
485+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.81	8.97	0.00	20.84	0.00	0.00	0.00	1.36	20.00	0.00
485+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.91	8.97	0.00	26.20	1.74	0.00	0.00	1.36	20.00	0.00
485+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.89	8.97	0.00	26.20	8.72	0.00	0.00	1.36	20.00	0.00
485+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.48	8.97	0.00	26.20	17.31	0.00	0.00	1.36	20.00	0.00
485+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.07	8.97	0.00	26.20	23.90	0.00	0.00	1.36	20.00	0.00
485+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.84	8.97	0.00	26.20	28.67	0.00	0.00	1.36	20.00	0.00
485+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.77	8.97	0.00	26.20	34.60	0.00	0.00	1.36	20.00	0.00
485+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.36	8.97	0.00	26.20	41.19	0.00	0.00	1.36	20.00	0.00
485+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.06	8.97	0.00	26.20	47.89	0.00	0.00	1.36	20.00	0.00
485+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.37	8.97	0.00	26.20	53.20	0.00	0.00	1.36	20.00	0.00
485+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	93.74	8.97	0.00	26.20	58.57	0.00	0.00	1.36	20.00	0.00
485+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	98.72	8.97	0.00	26.20	63.55	0.00	0.00	1.36	20.00	0.00
485+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	97.58	8.97	0.00	26.20	62.41	0.00	0.00	1.36	20.00	0.00
485+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.77	8.97	0.00	26.20	65.60	0.00	0.00	1.36	20.00	0.00
485+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	104.33	8.97	0.00	26.20	69.16	0.00	0.00	1.36	20.00	0.00
485+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.95	8.97	0.00	26.20	72.78	0.00	0.00	1.36	20.00	0.00
485+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	111.84	8.97	0.00	26.20	76.67	0.00	0.00	1.36	20.00	0.00
485+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.22	8.97	0.00	26.20	78.05	0.00	0.00	1.36	20.00	0.00
485+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	111.66	8.97	0.00	26.20	76.49	0.00	0.00	1.36	20.00	0.00
485+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.24	8.97	0.00	26.20	75.07	0.00	0.00	1.36	20.00	0.00
485+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.84	8.97	0.00	26.20	72.67	0.00	0.00	1.36	20.00	0.00
485+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.82	8.97	0.00	26.20	72.65	0.00	0.00	1.36	20.00	0.00
485+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.23	8.97	0.00	26.20	72.06	0.00	0.00	1.36	20.00	0.00
485+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.57	8.97	0.00	26.20	71.40	0.00	0.00	1.36	20.00	0.00
485+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.08	8.97	0.00	26.20	70.91	0.00	0.00	1.36	20.00	0.00
485+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.77	8.97	0.00	26.20	70.60	0.00	0.00	1.36	20.00	0.00
485+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.63	8.97	0.00	26.20	70.46	0.00	0.00	1.36	20.00	0.00
485+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.66	8.97	0.00	26.20	70.49	0.00	0.00	1.36	20.00	0.00
485+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.75	8.97	0.00	26.20	70.58	0.00	0.00	1.36	20.00	0.00
485+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.07	8.97	0.00	26.20	69.90	0.00	0.00	1.36	20.00	0.00
485+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	104.09	8.97	0.00	26.20	68.92	0.00	0.00	1.36	20.00	0.00
485+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	103.39	8.97	0.00	26.20	68.22	0.00	0.00	1.36	20.00	0.00
485+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	103.47	8.97	0.00	26.20	68.30	0.00	0.00	1.36	20.00	0.00
485+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	105.58	8.97	0.00	26.20	70.41	0.00	0.00	1.36	20.00	0.00
485+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	108.42	8.97	0.00	26.20	73.25	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
485+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	111.03	8.97	0.00	26.20	75.86	0.00	0.00	1.36	20.00	0.00
485+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.42	8.97	0.00	26.20	78.25	0.00	0.00	1.36	20.00	0.00
485+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	115.61	8.97	0.00	26.20	80.44	0.00	0.00	1.36	20.00	0.00
485+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	117.65	8.97	0.00	26.20	82.48	0.00	0.00	1.36	20.00	0.00
485+910.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	118.71	8.97	0.00	26.20	83.54	0.00	0.00	1.36	10.00	0.00
485+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.00	0.00
485+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.00	0.00
485+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.00	0.00
485+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.00	0.00
485+999.600	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	118.81	8.97	0.00	26.20	83.63	0.00	0.00	1.36	19.60	0.00
486+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	118.69	8.97	0.00	26.20	83.52	0.00	0.00	1.36	0.40	0.00
486+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	112.91	8.97	0.00	26.20	77.74	0.00	0.00	1.36	20.00	0.00
486+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	107.23	8.97	0.00	26.20	72.06	0.00	0.00	1.36	20.00	0.00
486+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.83	8.97	0.00	26.20	67.66	0.00	0.00	1.36	20.00	0.00
486+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	94.77	8.97	0.00	26.20	59.60	0.00	0.00	1.36	20.00	0.00
486+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.77	8.97	0.00	26.20	42.60	0.00	0.00	1.36	20.00	0.00
486+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.26	8.97	0.00	26.20	20.09	0.00	0.00	1.36	20.00	0.00
486+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.60	8.97	0.00	26.20	1.43	0.00	0.00	1.36	20.00	0.00
486+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.37	8.97	0.00	11.40	0.00	0.00	0.00	1.36	20.00	0.00
486+180.000	1.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	18.20
486+200.000	6.53	0.77	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	83.50
486+220.000	11.45	17.14	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	179.80
486+240.000	19.83	45.21	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	312.80
486+260.000	40.49	93.91	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	603.20
486+280.000	54.62	221.44	13.17	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	951.10
486+300.000	51.50	249.71	38.46	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,061.20
486+320.000	59.39	261.68	64.67	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,108.90

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
486+340.000	63.76	263.40	80.56	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	1,231.50
486+360.000	30.17	246.14	89.98	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	939.30
486+380.000	20.99	245.09	93.12	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	511.60
486+400.000	31.19	255.66	91.99	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	521.80
486+420.000	30.13	265.50	84.62	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	613.20
486+440.000	39.85	251.04	69.91	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	699.80
486+460.000	23.07	245.31	50.79	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	629.20
486+480.000	23.57	215.11	33.30	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	466.40
486+500.000	27.21	180.34	16.30	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	507.80
486+520.000	17.84	90.25	0.31	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	450.50
486+540.000	15.26	65.11	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	331.00
486+560.000	9.49	46.86	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	247.50
486+580.000	4.93	34.15	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	144.20
486+600.000	3.04	22.70	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	79.70
486+620.000	2.95	11.73	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	59.90
486+640.000	3.31	1.89	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	62.60
486+660.000	4.07	7.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	73.84
486+680.000	4.03	1.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	81.01
486+700.000	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	50.27
486+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.60	8.97	4.63	0.00	0.00	0.00	0.00	1.36	20.00	10.00
486+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.36	8.97	11.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
486+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.84	8.97	17.87	0.00	0.00	0.00	0.00	1.36	20.00	0.00
486+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.21	8.97	16.24	0.00	0.00	0.00	0.00	1.36	20.00	0.00
486+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.61	8.97	13.64	0.00	0.00	0.00	0.00	1.36	20.00	0.00
486+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.58	8.97	10.61	0.00	0.00	0.00	0.00	1.36	20.00	0.00
486+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.36	8.97	7.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
486+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.05	8.97	3.08	0.00	0.00	0.00	0.00	1.36	20.00	0.00
486+880.000	5.79	0.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	57.85
486+900.000	2.49	0.05	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	82.75
486+920.000	3.93	5.93	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	64.20
486+940.000	4.00	10.89	0.03	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	79.30
486+960.000	9.77	28.10	1.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	137.70
486+980.000	5.19	13.05	0.16	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	149.60
487+000.000	3.06	9.10	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	82.50
487+020.000	5.78	0.95	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	88.40
487+040.000	3.30	6.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	90.78
487+060.000	0.79	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	40.88
487+080.000	3.85	8.55	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	46.41
487+100.000	1.11	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	49.61
487+120.000	1.18	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	22.90
487+140.000	5.69	3.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	68.68
487+160.000	4.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	104.06
487+180.000	1.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	60.10
487+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.26	8.97	1.29	0.00	0.00	0.00	0.00	1.36	20.00	12.92

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
487+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.45	8.97	7.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.09	8.97	3.12	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.13	8.97	2.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.74	8.97	4.77	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.34	8.97	24.37	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.99	8.97	29.02	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.02	8.97	10.05	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.71	8.97	2.74	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+380.000	4.31	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	43.10
487+400.000	1.59	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	59.00
487+420.000	6.50	4.07	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	80.90
487+440.000	3.61	6.47	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	101.10
487+460.000	4.03	4.80	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	76.40
487+480.000	5.73	1.34	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	97.60
487+500.000	1.58	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	73.10
487+520.000	5.32	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	68.97
487+540.000	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.96	8.97	0.99	0.00	0.00	0.00	0.00	1.36	20.00	57.90
487+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.67	8.97	3.70	0.00	0.00	0.00	0.00	1.36	20.00	4.73
487+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.97	8.97	10.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.06	8.97	16.09	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.63	8.97	14.66	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.61	8.97	3.64	0.00	0.00	0.00	0.00	1.36	20.00	0.00
487+660.000	6.13	0.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	61.28
487+680.000	3.30	0.34	0.00	19.62	29.69	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	94.28
487+700.000	6.79	5.90	0.00	33.55	62.70	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	100.90
487+720.000	8.39	13.10	0.00	36.52	100.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	151.80
487+740.000	7.62	22.37	0.00	29.42	128.68	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	160.10
487+760.000	7.59	28.76	0.00	26.97	134.42	18.44	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	152.10
487+780.000	10.27	47.22	2.95	21.49	112.29	32.72	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	178.60
487+800.000	14.47	64.72	4.43	20.45	103.10	34.05	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	247.40
487+820.000	14.14	71.91	6.48	19.23	111.81	44.02	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	286.10
487+840.000	14.47	76.52	8.23	19.17	115.77	54.45	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	286.10
487+860.000	13.47	75.84	8.46	18.00	115.40	58.80	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	279.40
487+880.000	21.83	73.90	9.05	28.45	110.66	63.28	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	353.00
487+900.000	14.48	67.91	8.96	19.89	104.37	65.99	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	363.10
487+920.000	11.42	55.81	6.85	16.60	87.22	53.17	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	259.00
487+940.000	6.57	24.90	0.35	24.50	89.83	38.67	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	179.90
487+960.000	3.79	14.56	0.00	14.07	71.84	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	103.60
487+980.000	3.56	0.62	0.00	13.47	36.38	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	73.50
488+000.000	0.74	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.74	8.97	0.77	0.00	0.00	0.00	0.00	1.36	20.00	42.96
488+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.95	8.97	7.98	0.00	0.00	0.00	0.00	1.36	20.00	7.36
488+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.12	8.97	18.15	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.61	8.97	29.64	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.97	8.97	46.00	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.40	8.97	62.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.18	8.97	82.21	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
488+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	115.69	8.97	106.72	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	144.45	8.97	135.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	173.13	8.97	164.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	192.01	8.97	183.04	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	207.51	8.97	198.54	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	219.11	8.97	210.14	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	220.14	8.97	211.17	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	226.34	8.97	217.37	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	254.80	8.97	245.83	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	238.00	8.97	229.03	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	232.76	8.97	223.79	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	196.13	8.97	187.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	177.18	8.97	168.21	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	158.45	8.97	149.48	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	149.64	8.97	140.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	118.38	8.97	109.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.62	8.97	77.65	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.22	8.97	55.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.00	8.97	36.03	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.06	8.97	15.09	0.00	0.00	0.00	0.00	1.36	20.00	0.00
488+540.000	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.43	8.97	0.46	0.00	0.00	0.00	0.00	1.36	20.00	0.60
488+560.000	2.93	0.02	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	29.90
488+580.000	3.70	8.05	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	66.30
488+600.000	3.75	11.25	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	74.50
488+620.000	3.43	15.86	0.21	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	71.80
488+640.000	11.40	21.58	15.42	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	148.30
488+660.000	10.50	10.23	34.20	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	219.00
488+680.000	12.57	8.39	40.21	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	230.70
488+700.000	9.57	7.77	39.44	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	221.40
488+720.000	8.96	12.89	38.04	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	185.30
488+740.000	9.46	13.26	37.73	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	184.20
488+760.000	12.24	15.24	38.07	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	217.00
488+780.000	11.16	17.28	38.33	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	234.00
488+800.000	16.47	27.36	39.80	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	276.30
488+820.000	22.60	37.39	55.27	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	390.70
488+840.000	18.18	62.62	75.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	407.80
488+860.000	20.57	71.78	97.32	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	387.50
488+880.000	21.83	72.79	118.06	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	424.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
488+900.000	20.83	79.70	134.41	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	426.60
488+920.000	26.42	84.21	137.67	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	472.50
488+940.000	18.08	83.89	131.98	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	445.00
488+960.000	21.67	79.99	109.58	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	397.50
488+980.000	23.45	95.05	81.87	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	451.20
489+000.000	18.00	98.55	47.50	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	414.50
489+020.000	27.70	78.64	24.12	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	457.00
489+040.000	26.15	60.47	5.78	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	538.50
489+060.000	9.74	13.47	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	358.90
489+080.000	6.38	0.71	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	161.20
489+100.000	6.97	2.11	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	133.50
489+120.000	5.99	1.54	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	129.60
489+140.000	6.53	4.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	125.15
489+160.000	1.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	84.10
489+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.35	8.97	10.38	0.00	0.00	0.00	0.00	1.36	20.00	18.85
489+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	53.58	8.97	44.61	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.46	8.97	30.49	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.78	8.97	23.81	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.40	8.97	18.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.30	8.97	13.33	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.83	8.97	6.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+320.000	1.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	17.82
489+340.000	1.08	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	28.62
489+360.000	6.36	0.16	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	74.40
489+380.000	6.34	0.29	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	127.00
489+400.000	10.33	0.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	166.65
489+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.26	8.97	11.29	0.00	0.00	0.00	0.00	1.36	20.00	103.25
489+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.42	8.97	25.45	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.88	8.97	8.91	0.00	0.00	0.00	0.00	1.36	20.00	0.00
489+480.000	4.14	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	41.40
489+500.000	3.44	0.45	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	75.80
489+520.000	7.14	9.43	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	105.80
489+540.000	9.12	23.93	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	162.60
489+560.000	8.16	33.73	6.58	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	172.80
489+580.000	10.49	36.10	21.30	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	186.50
489+600.000	12.05	40.61	39.15	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	225.40
489+620.000	8.53	50.82	60.48	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	205.80
489+640.000	17.26	67.23	83.57	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	257.90
489+660.000	28.68	80.03	108.34	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	459.40
489+680.000	28.15	84.02	140.26	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	568.30
489+700.000	32.24	82.65	162.01	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	603.90
489+720.000	41.45	81.25	165.18	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	736.90
489+740.000	43.26	75.37	155.45	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	847.10

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
489+760.000	46.75	60.91	146.10	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	900.10
489+780.000	47.45	38.03	137.55	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	942.00
489+800.000	35.50	24.54	130.05	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	829.50
489+820.000	30.95	23.69	127.39	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	664.50
489+840.000	29.31	29.38	133.43	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	602.60
489+860.000	27.34	42.44	147.12	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	566.50
489+880.000	30.97	52.52	163.61	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	583.10
489+900.000	27.67	62.17	176.72	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	586.40
489+920.000	20.91	53.43	185.55	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	485.80
489+940.000	14.88	50.58	181.82	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	357.90
489+960.000	15.29	52.66	173.98	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	301.70
489+980.000	17.73	44.42	167.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	330.20
490+000.000	17.86	36.68	170.17	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	355.90
490+020.000	12.91	43.25	196.97	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	307.70
490+040.000	12.22	32.63	249.94	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	251.30
490+060.000	17.19	41.26	297.52	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	294.10
490+080.000	18.67	46.86	338.29	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	358.60
490+100.000	12.99	61.76	377.72	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	316.60
490+120.000	14.20	65.11	406.48	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	271.90
490+140.000	12.56	49.08	408.29	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	267.60
490+160.000	6.71	32.49	396.65	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	192.70
490+180.000	3.06	23.37	381.24	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	97.70
490+200.000	0.00	21.59	359.54	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	30.60
490+220.000	5.26	21.94	327.82	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	52.60
490+240.000	10.67	22.67	296.94	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	159.30
490+260.000	9.89	24.06	270.74	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	205.60
490+280.000	7.78	27.14	245.06	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	176.70
490+300.000	7.47	33.77	223.37	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	152.50
490+320.000	10.33	40.34	206.09	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	178.00
490+340.000	13.01	37.89	192.83	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	233.40
490+360.000	11.29	28.21	181.16	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	243.00
490+380.000	8.66	23.83	171.44	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	199.50
490+400.000	13.94	23.67	159.32	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	226.00
490+420.000	14.84	23.90	145.52	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	287.80
490+440.000	19.60	21.66	123.31	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	344.40
490+460.000	16.10	17.82	102.06	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	357.00
490+480.000	10.59	25.10	78.50	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	266.90
490+500.000	9.84	28.80	59.67	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	204.30
490+520.000	7.63	29.20	46.91	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	174.70
490+540.000	7.30	27.53	37.63	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	149.30
490+560.000	7.19	25.27	29.01	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	144.90
490+580.000	5.77	19.98	23.23	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	129.60
490+600.000	6.54	12.27	18.72	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	123.10
490+620.000	5.83	7.09	14.39	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	123.70
490+640.000	3.85	4.56	11.42	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	96.80
490+660.000	5.18	4.09	8.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	90.30

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
490+680.000	3.78	3.62	5.84	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	89.60
490+700.000	5.69	3.49	2.97	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	94.70
490+720.000	3.34	3.24	0.62	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	90.30
490+740.000	2.80	1.76	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	61.40
490+760.000	1.65	0.05	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	44.50
490+780.000	3.47	5.04	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	51.15
490+800.000	1.33	1.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	47.95
490+820.000	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.16	8.97	0.00	0.19	0.00	0.00	0.00	1.36	20.00	20.08
490+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.01	8.97	0.00	2.04	0.00	0.00	0.00	1.36	20.00	6.78
490+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.19	8.97	0.00	3.22	0.00	0.00	0.00	1.36	20.00	0.00
490+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.58	8.97	0.00	5.61	0.00	0.00	0.00	1.36	20.00	0.00
490+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	8.97	0.00	6.68	0.00	0.00	0.00	1.36	20.00	0.00
490+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.70	8.97	0.00	11.73	0.00	0.00	0.00	1.36	20.00	0.00
490+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.26	8.97	0.00	14.29	0.00	0.00	0.00	1.36	20.00	0.00
490+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.32	8.97	0.00	16.35	0.00	0.00	0.00	1.36	20.00	0.00
490+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.35	8.97	0.00	18.38	0.00	0.00	0.00	1.36	20.00	0.00
491+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.70	8.97	0.00	21.73	0.00	0.00	0.00	1.36	20.00	0.00
491+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.96	8.97	0.00	26.20	0.79	0.00	0.00	1.36	20.00	0.00
491+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.30	8.97	0.00	26.20	9.13	0.00	0.00	1.36	20.00	0.00
491+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.32	8.97	0.00	26.20	16.15	0.00	0.00	1.36	20.00	0.00
491+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.45	8.97	0.00	26.20	21.28	0.00	0.00	1.36	20.00	0.00
491+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.79	8.97	0.00	26.20	24.62	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
491+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.43	8.97	0.00	26.20	27.26	0.00	0.00	1.36	20.00	0.00
491+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.33	8.97	0.00	26.20	29.16	0.00	0.00	1.36	20.00	0.00
491+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	67.78	8.97	0.00	26.20	32.61	0.00	0.00	1.36	20.00	0.00
491+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.23	8.97	0.00	26.20	40.06	0.00	0.00	1.36	20.00	0.00
491+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	82.66	8.97	0.00	26.20	47.49	0.00	0.00	1.36	20.00	0.00
491+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.39	8.97	0.00	26.20	51.22	0.00	0.00	1.36	20.00	0.00
491+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	90.62	8.97	0.00	26.20	55.45	0.00	0.00	1.36	20.00	0.00
491+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	103.34	8.97	0.00	26.20	68.17	0.00	0.00	1.36	20.00	0.00
491+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	116.20	8.97	0.00	26.20	81.03	0.00	0.00	1.36	20.00	0.00
491+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	133.06	8.97	0.00	26.20	97.89	0.00	0.00	1.36	20.00	0.00
491+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	150.03	8.97	0.00	26.20	114.86	0.00	0.00	1.36	20.00	0.00
491+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	167.39	8.97	0.00	26.20	132.22	0.00	0.00	1.36	20.00	0.00
491+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	175.87	8.97	0.00	26.20	140.70	0.00	0.00	1.36	20.00	0.00
491+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	192.09	8.97	0.00	26.20	156.92	0.00	0.00	1.36	20.00	0.00
491+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	201.04	8.97	0.00	26.20	165.87	0.00	0.00	1.36	20.00	0.00
491+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	182.81	8.97	0.00	26.20	147.64	0.00	0.00	1.36	20.00	0.00
491+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	161.90	8.97	0.00	26.20	126.73	0.00	0.00	1.36	20.00	0.00
491+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	143.99	8.97	0.00	26.20	108.82	0.00	0.00	1.36	20.00	0.00
491+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	130.09	8.97	0.00	26.20	94.92	0.00	0.00	1.36	20.00	0.00
491+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	121.14	8.97	0.00	26.20	85.97	0.00	0.00	1.36	20.00	0.00
491+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	119.45	8.97	0.00	26.20	84.28	0.00	0.00	1.36	20.00	0.00
491+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	114.95	8.97	0.00	26.20	79.78	0.00	0.00	1.36	20.00	0.00
491+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	123.96	8.97	0.00	26.20	88.79	0.00	0.00	1.36	20.00	0.00
491+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	132.06	8.97	0.00	26.20	96.89	0.00	0.00	1.36	20.00	0.00
491+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	134.42	8.97	0.00	26.20	99.25	0.00	0.00	1.36	20.00	0.00
491+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	121.63	8.97	0.00	26.20	86.46	0.00	0.00	1.36	20.00	0.00
491+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.24	8.97	0.00	26.20	71.07	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
491+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	98.37	8.97	0.00	26.20	63.20	0.00	0.00	1.36	20.00	0.00
491+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	92.99	8.97	0.00	26.20	57.82	0.00	0.00	1.36	20.00	0.00
491+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	90.97	8.97	0.00	26.20	55.80	0.00	0.00	1.36	20.00	0.00
491+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.25	8.97	0.00	26.20	54.08	0.00	0.00	1.36	20.00	0.00
491+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	79.89	8.97	0.00	26.20	44.72	0.00	0.00	1.36	20.00	0.00
491+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	72.74	8.97	0.00	26.20	37.57	0.00	0.00	1.36	20.00	0.00
491+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.52	8.97	0.00	26.20	46.35	0.00	0.00	1.36	20.00	0.00
491+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.08	8.97	0.00	26.20	37.91	0.00	0.00	1.36	20.00	0.00
491+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.92	8.97	0.00	26.20	14.75	0.00	0.00	1.36	20.00	0.00
491+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.27	8.97	0.00	26.20	4.10	0.00	0.00	1.36	20.00	0.00
491+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.16	8.97	0.00	25.19	0.00	0.00	0.00	1.36	20.00	0.00
491+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.32	8.97	0.00	23.35	0.00	0.00	0.00	1.36	20.00	0.00
491+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.33	8.97	0.00	23.36	0.00	0.00	0.00	1.36	20.00	0.00
491+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.30	8.97	0.00	23.33	0.00	0.00	0.00	1.36	20.00	0.00
491+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.56	8.97	0.00	25.59	0.00	0.00	0.00	1.36	20.00	0.00
491+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.27	8.97	0.00	26.20	2.10	0.00	0.00	1.36	20.00	0.00
491+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.21	8.97	0.00	18.24	0.00	0.00	0.00	1.36	20.00	0.00
492+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.32	8.97	0.00	14.35	0.00	0.00	0.00	1.36	20.00	0.00
492+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.88	8.97	0.00	13.91	0.00	0.00	0.00	1.36	20.00	0.00
492+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.00	8.97	0.00	23.03	0.00	0.00	0.00	1.36	20.00	0.00
492+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.28	8.97	0.00	6.31	0.00	0.00	0.00	1.36	20.00	0.00
492+080.000	2.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.45	8.97	0.00	0.48	0.00	0.00	0.00	1.36	20.00	24.81
492+100.000	4.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	71.59
492+120.000	7.23	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	119.04
492+140.000	4.43	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	116.56
492+160.000	5.27	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	97.00
492+180.000	7.57	0.03	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	128.40
492+200.000	7.64	0.02	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	152.10

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
492+220.000	6.75	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	143.90
492+240.000	2.52	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	92.70
492+260.000	8.48	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	109.96
492+280.000	3.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	115.24
492+300.000	6.05	3.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	91.00
492+320.000	3.16	2.70	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	92.12
492+340.000	2.04	13.36	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	52.00
492+360.000	2.09	27.03	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	41.30
492+380.000	2.20	41.95	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	42.90
492+400.000	4.51	56.51	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	67.10
492+420.000	9.48	126.58	4.44	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	139.90
492+440.000	11.37	120.02	18.90	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	208.50
492+460.000	11.15	113.49	36.32	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	225.20
492+480.000	12.39	109.22	54.02	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	235.40
492+500.000	14.30	106.63	67.12	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	266.90
492+520.000	19.18	106.50	68.60	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	334.80
492+540.000	18.84	100.59	65.60	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	380.20
492+560.000	21.75	85.75	63.38	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	405.90
492+580.000	22.19	70.56	62.17	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	439.40
492+600.000	21.10	53.65	60.70	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	432.90
492+620.000	21.25	35.48	58.84	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	423.50
492+640.000	19.68	22.37	57.31	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	409.30
492+660.000	19.49	11.50	55.89	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	391.70
492+680.000	17.11	2.60	54.62	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	366.00
492+700.000	19.86	0.30	51.86	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	369.70
492+720.000	15.27	1.94	46.02	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	351.30
492+740.000	10.96	7.04	39.31	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	262.30
492+760.000	6.19	19.58	30.24	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	171.50
492+780.000	3.62	30.92	22.21	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	98.10
492+800.000	1.54	21.12	13.49	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	51.60
492+820.000	1.00	25.18	6.80	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	25.40
492+840.000	1.44	28.66	0.92	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	24.40
492+860.000	2.98	27.98	0.04	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	44.20
492+880.000	3.71	26.41	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	66.90
492+900.000	4.11	24.32	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	78.20
492+920.000	3.16	22.29	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	72.70
492+940.000	5.23	20.57	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	83.90
492+960.000	5.46	19.88	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	106.90
492+980.000	6.53	21.22	0.01	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	119.90
493+000.000	6.46	23.29	0.23	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	129.90
493+020.000	9.30	23.00	0.77	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	157.60
493+040.000	8.70	23.26	0.80	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	180.00
493+060.000	8.20	22.45	0.70	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	169.00
493+080.000	7.94	20.73	0.23	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	161.40

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
493+100.000	5.05	19.60	0.02	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	129.90
493+120.000	7.13	20.60	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	121.80
493+140.000	7.93	25.26	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	150.60
493+160.000	7.71	25.22	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	156.40
493+180.000	8.55	23.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	162.60
493+200.000	6.83	20.17	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	153.80
493+220.000	4.50	16.14	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	113.30
493+240.000	4.65	8.59	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	91.50
493+260.000	5.14	0.07	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	97.90
493+280.000	1.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	63.90
493+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.59	8.97	0.00	8.62	0.00	0.00	0.00	1.36	20.00	12.50
493+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.70	8.97	0.00	4.73	0.00	0.00	0.00	1.36	20.00	0.00
493+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.11	8.97	0.00	3.14	0.00	0.00	0.00	1.36	20.00	0.00
493+360.000	1.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.04	8.97	0.00	2.06	0.00	0.00	0.00	1.36	20.00	13.31
493+380.000	3.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	51.85
493+400.000	13.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	168.95
493+420.000	1.96	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	150.01
493+440.000	2.43	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	43.85
493+460.000	8.65	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	110.75
493+480.000	7.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	163.74
493+500.000	0.86	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	85.80
493+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.81	8.97	0.00	3.84	0.00	0.00	0.00	1.36	20.00	8.56
493+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.15	8.97	0.00	8.18	0.00	0.00	0.00	1.36	20.00	0.00
493+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.41	8.97	0.00	10.44	0.00	0.00	0.00	1.36	20.00	0.00
493+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.83	8.97	0.00	12.86	0.00	0.00	0.00	1.36	20.00	0.00
493+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.58	8.97	0.00	15.61	0.00	0.00	0.00	1.36	20.00	0.00
493+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.48	8.97	0.00	18.51	0.00	0.00	0.00	1.36	20.00	0.00
493+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.00	8.97	0.00	18.03	0.00	0.00	0.00	1.36	20.00	0.00
493+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.50	8.97	0.00	17.53	0.00	0.00	0.00	1.36	20.00	0.00
493+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.99	8.97	0.00	17.02	0.00	0.00	0.00	1.36	20.00	0.00
493+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.50	8.97	0.00	16.53	0.00	0.00	0.00	1.36	20.00	0.00
493+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.00	8.97	0.00	16.03	0.00	0.00	0.00	1.36	20.00	0.00
493+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.38	8.97	0.00	15.41	0.00	0.00	0.00	1.36	20.00	0.00
493+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.38	8.97	0.00	14.41	0.00	0.00	0.00	1.36	20.00	0.00
493+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.43	8.97	0.00	13.46	0.00	0.00	0.00	1.36	20.00	0.00
493+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.36	8.97	0.00	12.39	0.00	0.00	0.00	1.36	20.00	0.00
493+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.49	8.97	0.00	12.52	0.00	0.00	0.00	1.36	20.00	0.00
493+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.78	8.97	0.00	12.81	0.00	0.00	0.00	1.36	20.00	0.00
493+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.17	8.97	0.00	10.20	0.00	0.00	0.00	1.36	20.00	0.00
493+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.39	8.97	0.00	7.42	0.00	0.00	0.00	1.36	20.00	0.00
493+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.82	8.97	0.00	5.85	0.00	0.00	0.00	1.36	20.00	0.00
493+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.84	8.97	0.00	3.87	0.00	0.00	0.00	1.36	20.00	0.00
493+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.81	8.97	0.00	1.84	0.00	0.00	0.00	1.36	20.00	0.00
493+960.000	0.64	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	6.42

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	325,214.73
493+980.000	2.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	32.48
494+000.000	6.23	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	88.37
494+020.000	7.22	3.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	134.54
494+040.000	6.23	0.10	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	134.53
494+060.000	10.14	6.64	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	163.70
494+080.000	20.53	25.90	4.99	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	306.70
494+100.000	23.05	21.00	15.70	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	435.80
494+120.000	39.57	19.83	31.20	21.27	22.62	86.86	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	626.20
494+140.000	26.77	20.19	47.20	27.01	22.91	118.27	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	663.40
494+160.000	15.07	19.69	67.35	14.35	19.57	156.84	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	418.40
494+180.000	8.89	21.03	82.99	8.30	10.48	180.65	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	239.60
494+200.000	6.08	12.41	78.83	5.75	11.94	179.70	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	149.70
494+220.000	5.52	14.16	68.60	5.40	14.11	164.60	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	116.00
494+240.000	6.84	16.26	57.91	6.89	16.78	145.24	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	123.60
494+260.000	9.48	17.28	46.94	9.85	18.52	125.11	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	163.20
494+280.000	12.47	23.73	37.33	13.28	26.65	104.24	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	219.50
494+300.000	9.82	33.17	30.97	10.50	37.73	89.64	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	222.90
494+320.000	13.05	38.13	25.56	14.07	44.44	76.67	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	228.70
494+340.000	13.51	43.39	19.96	14.73	51.43	62.37	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	265.60
494+360.000	15.53	45.52	14.99	17.23	55.28	49.08	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	290.40
494+380.000	9.25	47.10	10.91	10.53	58.41	39.18	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	247.80
494+400.000	8.48	43.45	8.10	9.95	55.38	30.80	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	177.30
494+420.000	7.40	38.53	6.10	8.99	50.58	24.87	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	158.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
494+440.000	9.68	35.40	4.05	11.98	47.41	18.14	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	170.80
494+460.000	7.87	32.49	2.24	10.02	44.53	11.95	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	175.50
494+480.000	3.20	12.14	0.07	8.70	40.01	7.90	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	110.70
494+500.000	4.11	8.55	0.00	11.76	32.64	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	73.10
494+520.000	3.21	7.21	0.00	9.85	24.52	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	73.20
494+540.000	2.91	2.09	0.00	9.56	16.07	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	61.20
494+560.000	0.74	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	36.50
494+580.000	2.28	4.59	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	30.18
494+600.000	2.55	1.55	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	48.29
494+620.000	1.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	41.58
494+640.000	1.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.62	8.97	0.65	0.00	0.00	0.00	0.00	1.36	20.00	26.26
494+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.96	8.97	3.99	0.00	0.00	0.00	0.00	1.36	20.00	10.19
494+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.07	8.97	7.10	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.36	8.97	9.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.71	8.97	12.74	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	27.32	8.97	18.35	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.79	8.97	25.82	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.71	8.97	30.74	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.39	8.97	33.42	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.68	8.97	33.71	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.13	8.97	33.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.84	8.97	32.87	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.64	8.97	32.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.29	8.97	32.32	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.07	8.97	38.10	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.64	8.97	41.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00
494+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.09	8.97	39.12	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
494+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.43	8.97	36.46	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.38	8.97	35.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.92	8.97	35.95	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.12	8.97	36.15	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.07	8.97	38.10	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.64	8.97	39.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.39	8.97	42.42	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.11	8.97	45.14	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.37	8.97	41.40	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46.72	8.97	37.75	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.19	8.97	34.22	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.49	8.97	32.52	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.85	8.97	32.88	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.17	8.97	36.20	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.38	8.97	39.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.48	8.97	42.51	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.89	8.97	42.92	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.48	8.97	42.51	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.89	8.97	41.92	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	52.22	8.97	43.25	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	56.08	8.97	47.11	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.27	8.97	51.30	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	64.83	8.97	55.86	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.16	8.97	60.19	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.92	8.97	74.95	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.30	8.97	79.33	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.44	8.97	79.47	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
495+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.43	8.97	79.46	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.26	8.97	79.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.96	8.97	78.99	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.57	8.97	78.60	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.08	8.97	78.11	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.50	8.97	77.53	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.23	8.97	78.26	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.74	8.97	77.77	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.24	8.97	77.27	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	85.75	8.97	76.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	85.26	8.97	76.29	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.78	8.97	75.81	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	84.29	8.97	75.32	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.80	8.97	74.83	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.41	8.97	74.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.93	8.97	77.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.19	8.97	78.22	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.73	8.97	74.76	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	81.40	8.97	72.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+900.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	80.92	8.97	71.95	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+920.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	80.27	8.97	71.30	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+940.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	67.40	8.97	58.43	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+960.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.29	8.97	45.32	0.00	0.00	0.00	0.00	1.36	20.00	0.00
495+980.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.93	8.97	30.96	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+000.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.36	8.97	24.39	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+020.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.49	8.97	17.52	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.33	8.97	10.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.60	8.97	4.63	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
496+080.000	1.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	12.86
496+100.000	3.40	2.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	46.87
496+120.000	3.90	4.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	73.00
496+140.000	3.97	2.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	78.72
496+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.07	8.97	1.10	0.00	0.00	0.00	0.00	1.36	20.00	39.73
496+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.53	8.97	3.56	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.62	8.97	6.65	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.77	8.97	13.80	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	31.95	8.97	22.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.16	8.97	32.19	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.86	8.97	40.89	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.19	8.97	54.22	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.00	8.97	68.03	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	93.15	8.97	84.18	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	97.77	8.97	88.80	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.11	8.97	90.14	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.08	8.97	91.11	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.13	8.97	92.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.35	8.97	93.38	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	103.67	8.97	94.70	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	106.95	8.97	97.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.33	8.97	101.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.77	8.97	104.80	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.95	8.97	104.98	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.59	8.97	104.62	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.41	8.97	104.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	112.85	8.97	103.88	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	93.75	8.97	84.78	0.00	0.00	0.00	0.00	1.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
496+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.53	8.97	77.56	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.94	8.97	66.97	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	61.37	8.97	52.40	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.78	8.97	40.81	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.53	8.97	31.56	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.88	8.97	21.91	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.25	8.97	10.28	0.00	0.00	0.00	0.00	1.36	20.00	0.00
496+780.000	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	5.95
496+800.000	3.38	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	39.75
496+820.000	9.39	19.97	0.28	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	127.70
496+840.000	15.45	43.31	32.03	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	248.40
496+860.000	10.95	23.01	59.26	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	264.00
496+880.000	8.98	12.33	68.78	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	199.30
496+900.000	12.91	9.43	71.10	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	218.90
496+920.000	11.30	10.42	64.75	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	242.10
496+940.000	14.57	9.15	53.02	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	258.70
496+960.000	19.23	6.98	42.12	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	338.00
496+980.000	10.51	14.20	35.20	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	297.40
497+000.000	10.15	18.15	31.50	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	206.60
497+020.000	10.68	13.32	26.09	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	208.30
497+040.000	10.98	9.17	20.43	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	216.60
497+060.000	7.79	7.66	15.90	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	187.70
497+080.000	6.07	9.33	11.31	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	138.60
497+100.000	6.81	10.30	7.04	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	128.80
497+120.000	6.66	9.10	4.39	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	134.70
497+140.000	2.81	5.29	0.53	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	94.70
497+160.000	3.62	6.03	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	64.30
497+180.000	2.61	2.59	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	62.30
497+200.000	6.65	4.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	92.63
497+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.29	8.97	3.32	0.00	0.00	0.00	0.00	1.36	20.00	66.53
497+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.64	8.97	16.67	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.68	8.97	32.71	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	51.01	8.97	42.04	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	48.85	8.97	39.88	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43.52	8.97	34.55	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.87	8.97	26.90	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.41	8.97	23.44	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.38	8.97	23.41	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.23	8.97	24.26	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.33	8.97	25.36	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.07	8.97	28.10	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.04	8.97	33.07	0.00	0.00	0.00	0.00	1.36	20.00	0.00
497+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.17	15.65	52.52	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	86.56	15.65	70.91	0.00	0.00	0.00	0.00	2.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
497+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.05	15.65	84.40	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	101.03	15.65	85.38	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.45	15.65	75.80	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.83	15.65	62.18	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63.49	15.65	47.84	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	58.99	15.65	43.34	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.97	15.65	45.32	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.92	15.65	51.27	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.28	15.65	53.63	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.09	15.65	54.44	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.78	15.65	56.13	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	78.95	15.65	63.30	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.77	15.65	54.12	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.79	15.65	34.14	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.13	15.65	17.48	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.42	15.65	5.77	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.45	15.65	2.80	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.12	15.65	2.47	0.00	0.00	0.00	0.00	2.36	20.00	0.00
497+880.000	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.81	15.65	1.16	0.00	0.00	0.00	0.00	2.36	20.00	1.28
497+900.000	2.36	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	24.90
497+920.000	9.49	1.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	118.49
497+940.000	7.28	9.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	167.67
497+960.000	2.82	0.00	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	101.00
497+980.000	8.31	0.25	0.00	0.00	0.00	0.00	8.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.10	0.00	2.64	20.00	111.30
498+000.000	9.19	9.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	174.97
498+020.000	7.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	163.96
498+040.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.48	15.65	3.83	0.00	0.00	0.00	0.00	2.36	20.00	72.09
498+060.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.98	15.65	14.33	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+080.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	31.51	15.65	15.86	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+100.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.31	15.65	16.66	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+120.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.24	15.65	12.59	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.23	15.65	10.58	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.72	15.65	11.07	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.43	15.65	13.78	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.43	15.65	6.78	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.65	15.65	6.00	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.45	15.65	4.80	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.00	15.65	2.35	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+280.000	0.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.51	15.65	0.86	0.00	0.00	0.00	0.00	2.36	20.00	5.60
498+300.000	4.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	46.93
498+320.000	6.19	1.43	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	103.19
498+340.000	8.88	3.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	150.64
498+360.000	9.12	4.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	179.97
498+380.000	8.24	5.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	173.56

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
498+400.000	8.55	4.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	167.87
498+420.000	7.72	4.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	162.65
498+440.000	9.07	4.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	167.81
498+460.000	11.16	3.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	202.28
498+480.000	12.09	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	232.52
498+500.000	10.97	0.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	230.64
498+520.000	7.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	187.14
498+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.33	15.65	2.68	0.00	0.00	0.00	0.00	2.36	20.00	77.40
498+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.40	15.65	6.75	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+580.000	4.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	41.20
498+600.000	4.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	83.23
498+620.000	1.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.69	15.65	0.04	0.00	0.00	0.00	0.00	2.36	20.00	60.29
498+640.000	0.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.27	15.65	0.62	0.00	0.00	0.00	0.00	2.36	20.00	24.04
498+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.08	15.65	3.43	0.00	0.00	0.00	0.00	2.36	20.00	5.78
498+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23.76	15.65	8.11	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.02	15.65	14.37	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.95	15.65	21.30	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.13	15.65	9.48	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.04	15.65	4.39	0.00	0.00	0.00	0.00	2.36	20.00	0.00
498+780.000	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.26	15.65	1.61	0.00	0.00	0.00	0.00	2.36	20.00	1.33
498+800.000	2.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	23.67
498+820.000	5.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	78.54
498+840.000	6.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	116.84
498+860.000	5.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	116.68
498+880.000	5.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	109.81
498+900.000	6.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	121.54
498+920.000	7.98	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	147.55
498+940.000	9.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	172.26
498+960.000	10.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	192.85
498+980.000	9.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	194.99
499+000.000	6.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	161.26
499+020.000	5.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	117.32
499+040.000	5.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	102.81
499+060.000	5.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	102.88
499+080.000	4.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	93.98
499+100.000	2.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	71.69
499+120.000	1.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.65	15.65	0.00	0.00	0.00	0.00	0.00	2.36	20.00	46.40
499+140.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.46	15.65	4.81	0.00	0.00	0.00	0.00	2.36	20.00	17.94
499+160.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.86	15.65	19.21	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+180.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.24	15.65	29.59	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+200.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.88	15.65	7.23	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+220.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.23	15.65	6.58	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+240.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.14	15.65	8.49	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+260.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.41	15.65	9.76	0.00	0.00	0.00	0.00	2.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	325,214.73
499+280.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.09	15.65	9.44	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+300.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.44	15.65	6.79	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+320.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.59	15.65	3.94	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+340.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.58	15.65	5.93	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+360.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.26	15.65	9.61	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+380.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.63	15.65	12.98	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+400.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.72	15.65	14.07	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+420.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.10	15.65	14.45	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+440.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	31.43	15.65	15.78	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+460.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.40	15.65	17.75	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+480.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.93	15.65	20.28	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+500.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.79	15.65	23.14	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	41.82	15.65	26.17	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.12	15.65	29.47	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.45	15.65	33.80	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55.50	15.65	39.85	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.42	15.65	46.77	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.19	15.65	52.54	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.83	15.65	56.18	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	69.69	15.65	54.04	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	62.22	15.65	46.57	0.00	0.00	0.00	0.00	2.36	20.00	0.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
499+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.74	15.65	39.09	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	49.30	15.65	33.65	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	45.36	15.65	29.71	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.10	15.65	26.45	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.75	15.65	25.10	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+800.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	39.40	15.65	23.75	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+820.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.45	15.65	20.80	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+840.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	32.53	15.65	16.88	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+860.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.61	15.65	9.96	0.00	0.00	0.00	0.00	2.36	20.00	0.00
499+880.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.13	8.97	2.16	0.00	0.00	0.00	0.00	1.36	20.00	0.00
499+900.000	4.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	44.49
499+920.000	5.63	6.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	100.75
499+940.000	5.68	2.60	0.00	0.00	0.00	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	113.06
499+960.000	15.60	23.16	1.98	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	212.80
499+980.000	15.11	23.80	10.21	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	307.10
500+000.000	12.57	27.34	17.71	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	276.80
500+020.000	12.91	28.07	22.41	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	254.80
500+040.000	12.51	28.43	25.11	0.00	0.00	0.00	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	254.20
500+060.000	12.59	29.96	26.95	13.67	34.43	77.52	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	251.00
500+080.000	10.93	29.05	29.55	11.70	32.75	85.98	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	235.20
500+100.000	10.04	28.33	30.22	10.93	32.42	88.72	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	209.70
500+120.000	12.98	29.34	30.15	14.15	33.75	88.43	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	230.20
500+140.000	12.06	37.33	29.45	12.80	42.18	86.95	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	250.40
500+160.000	13.06	46.14	26.91	13.69	52.15	81.05	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	251.20
500+180.000	11.87	54.62	23.24	12.42	62.26	72.39	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	249.30
500+200.000	11.56	60.42	18.70	12.17	70.06	59.53	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	234.30
500+220.000	14.20	53.16	14.33	15.44	63.56	47.74	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	257.60
500+240.000	11.85	49.78	9.89	13.36	61.67	35.90	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	260.50

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS

TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)

MVP 10 - KM 450+000 A KM 500+789.330

PLANILHA DE CUBAÇÃO

ESTACA	ÁREAS																		DISTÂNCIA	
	ORIGENS										DESTINOS									
	CORTE			EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO					REATERRO		SUBLASTRO		
	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		ATERRO TOTAL	CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO			PRIMEIRA
																			50,789.33	**325,214.73**
500+260.000	9.95	52.37	5.32	11.46	66.59	21.53	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	218.00
500+280.000	7.03	57.17	1.80	8.16	73.50	8.89	0.00	0.00	3.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	1.64	20.00	169.80
500+300.000	5.53	22.43	0.00	12.85	68.12	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	125.60
500+320.000	4.38	14.16	0.00	11.47	52.11	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	99.10
500+340.000	3.89	9.20	0.00	11.02	39.19	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	82.70
500+360.000	2.46	6.60	0.00	7.40	31.03	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	63.50
500+380.000	2.90	9.34	0.00	9.28	20.57	0.00	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	53.60
500+400.000	2.58	0.65	0.00	8.82	11.43	0.00	5.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	0.00	1.64	20.00	54.80
500+420.000	2.90	9.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	54.83
500+440.000	2.24	6.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	51.46
500+460.000	1.87	3.55	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	41.14
500+480.000	2.45	1.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	43.25
500+500.000	3.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.97	8.97	0.00	0.00	0.00	0.00	0.00	1.36	20.00	61.96
500+520.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.96	8.97	0.00	1.99	0.00	0.00	0.00	1.36	20.00	37.42
500+540.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.25	8.97	0.00	6.28	0.00	0.00	0.00	1.36	20.00	0.00
500+560.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.22	8.97	0.00	6.25	0.00	0.00	0.00	1.36	20.00	0.00
500+580.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.68	8.97	0.00	7.71	0.00	0.00	0.00	1.36	20.00	0.00
500+600.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.99	8.97	0.00	10.02	0.00	0.00	0.00	1.36	20.00	0.00
500+620.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.08	8.97	0.00	7.11	0.00	0.00	0.00	1.36	20.00	0.00
500+640.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.63	8.97	0.00	6.66	0.00	0.00	0.00	1.36	20.00	0.00
500+660.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.06	8.97	0.00	6.09	0.00	0.00	0.00	1.36	20.00	0.00
500+680.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.32	8.97	0.00	6.35	0.00	0.00	0.00	1.36	20.00	0.00
500+700.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.61	8.97	0.00	16.64	0.00	0.00	0.00	1.36	20.00	0.00
500+720.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.09	8.97	0.00	19.12	0.00	0.00	0.00	1.36	20.00	0.00
500+740.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.36	8.97	0.00	21.39	0.00	0.00	0.00	1.36	20.00	0.00
500+760.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.72	8.97	0.00	26.20	5.55	0.00	0.00	1.36	20.00	0.00
500+780.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.09	8.97	0.00	26.20	18.92	0.00	0.00	1.36	20.00	0.00
500+789.330	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	60.36	8.97	0.00	26.20	25.19	0.00	0.00	1.36	9.33	0.00

OBSERVAÇÕES:

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	908.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,437.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,725.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,736.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,800.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,877.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,881.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,859.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,902.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,989.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,594.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,440.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,635.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,504.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,278.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,002.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	848.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	755.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	653.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	533.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	427.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	332.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	270.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	235.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	167.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	90.54	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	58.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	44.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	40.94	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	49.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	34.11	0.00	0.00	0.00	0.00	27.20
1.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	8.84	0.00	0.00	0.00	0.00	27.20
49.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
48.43	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
2.90	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
5.30	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
3.10	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
1.20	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.50	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
30.43	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
47.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
24.43	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
12.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
7.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
4.42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
1.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
3.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
20.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
43.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
26.86	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.40	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
15.50	0.00	431.80	293.70	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
48.60	0.00	872.40	667.40	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
85.10	0.00	893.30	833.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
121.20	0.00	923.30	1,057.70	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
157.80	0.00	950.30	1,300.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
195.20	0.00	962.00	1,494.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
233.50	0.00	925.61	1,677.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
278.30	0.00	850.80	1,896.80	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
347.30	0.00	798.99	2,191.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
438.70	0.00	777.70	2,563.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
535.20	0.00	798.50	2,952.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
647.50	0.00	824.90	3,311.60	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
1,047.30	80.80	412.60	1,742.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,287.70	268.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,120.30	496.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
998.60	752.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
928.60	1,032.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
898.60	1,341.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
903.30	1,700.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
954.70	2,141.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,091.50	2,548.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,192.50	2,671.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,195.60	2,671.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,214.00	2,760.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,230.80	2,882.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,221.30	3,032.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,110.50	2,772.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,010.40	2,186.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,248.10	1,258.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
860.00	289.00	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
623.70	58.10	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,276.20	304.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,415.40	552.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,196.10	553.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
992.20	383.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
536.90	138.20	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
267.50	1.30	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
327.50	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
202.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
90.70	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
141.80	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
186.40	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
109.70	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
21.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	65.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	287.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	435.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	298.30	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	267.10	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	443.40	26.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	99.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	149.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	147.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	172.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	235.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	178.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	154.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	356.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	585.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	664.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	640.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	620.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	513.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	340.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	494.00	132.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	494.00	119.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	250.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	223.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	101.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	20.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	479.00	12.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	371.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	248.70	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	139.20	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	44.90	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	3.99	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	29.49	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	25.50	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.84	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.84	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	35.50	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	119.20	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	228.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	354.50	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	471.80	12.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	102.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	269.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	296.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	198.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	180.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	235.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	286.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	226.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	475.80	76.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	432.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	478.10	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	477.10	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	270.40	0.00	0.00	0.00	27.20
33.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	52.80	0.00	0.00	0.00	27.20
96.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
62.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	158.20	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	375.50	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	479.30	120.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	192.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	86.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	26.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	331.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	725.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	825.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	839.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	841.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	840.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	844.30	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	PÁGINA 2 / 5
CONTRATO -	REV. 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	845.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	840.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	840.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	840.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	875.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	987.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,165.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,296.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,330.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,181.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	888.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	607.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	378.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	292.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	359.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	533.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	660.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	710.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	711.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	562.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	350.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	187.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	148.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	249.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	321.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	334.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	427.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	716.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	976.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	891.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	715.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	611.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	478.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	294.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	118.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	44.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	434.40	14.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	185.60	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	13.20	0.00	51.00	0.00	30.00
167.10	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
327.90	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
161.50	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.70	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	108.80	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	480.40	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	838.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	994.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,149.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,372.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,675.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,018.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,282.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,454.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,464.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,356.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,264.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	139.66	1,664.37	0.00	0.00	0.00	0.00	21.17
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	19.90	228.56	0.00	0.00	0.00	0.00	3.02
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.13	0.00	193.09	646.95	0.00	0.00	10.02
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	47.20	0.00	137.81	496.23	0.00	0.00	7.15
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,253.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,506.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,176.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,690.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,578.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,727.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,901.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,854.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,637.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,500.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,341.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	986.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	569.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	451.90	182.30	0.00	0.00	27.20
33.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	189.90	0.00	0.00	0.00	27.20
134.50	0.00	233.10	559.20	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
376.30	0.00	555.60	1,728.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
597.70	0.00	623.50	2,560.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
605.90	0.00	546.17	2,434.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
419.80	0.00	266.88	1,437.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
136.60	0.00	21.71	394.80	0.00	0.00	51.00	0.00	0.00	89.73	11.74	0.00	0.00	51.00	0.00	30.00
0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	477.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,705.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,427.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,920.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,331.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,087.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,951.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,859.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,705.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,541.50	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,366.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,163.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,018.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,955.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,878.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,725.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,387.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	888.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	552.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	523.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	646.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	693.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	471.40	0.00	0.00	0.00	0.00	27.20
41.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	137.30	0.00	0.00	0.00	0.00	27.20
46.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
53.40	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
83.10	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
34.40	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	23.99	0.00	0.00	51.00	0.00	30.00
2.90	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	23.99	0.00	0.00	51.00	0.00	30.00
17.60	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
31.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
19.50	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
4.12	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.51	0.00	0.00	51.00	0.00	30.00
0.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	215.41	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	492.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	323.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	105.60	0.00	0.00	51.00	0.00	0.00	0.00	89.73	46.50	0.00	0.00	51.00	0.00	30.00
152.90	0.00	170.40	296.40	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
406.50	0.30	97.40	708.60	111.70	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
650.20	152.70	85.20	673.10	524.80	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
777.50	477.00	144.40	488.10	1,019.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
908.90	587.60	181.20	553.30	1,137.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
615.70	263.00	356.90	661.70	531.10	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
237.00	0.00	391.60	723.10	199.10	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
738.80	64.00	165.40	803.50	473.30	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,281.40	91.90	147.00	875.70	383.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
945.20	27.90	179.30	861.40	108.80	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
262.50	0.00	182.40	467.80	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
9.20	0.00	108.80	66.10	0.00	51.00	0.00	0.00	0.00	89.73	0.00	2.17	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	238.97	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	498.80	254.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	731.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,119.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,451.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,718.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,929.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,025.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,928.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,007.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,222.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,242.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,162.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,037.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,848.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,560.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,224.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	915.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	577.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	340.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	242.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	409.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	890.30	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,167.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,012.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	893.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,058.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,277.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,447.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,548.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,605.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,647.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,675.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,676.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,633.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,510.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,223.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	807.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	322.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	292.00	30.10	0.00	0.00	27.20
0.90	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	30.00	0.00	51.00	0.00	30.00
189.80	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
457.90	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
329.40	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
60.40	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	181.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	618.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	980.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	641.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	97.50	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	62.20	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	460.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,104.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,448.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,395.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,184.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	775.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	244.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	6.81	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	225.61	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	618.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	813.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	673.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	389.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	323.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	585.40	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	948.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,098.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,087.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	968.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	784.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	612.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	356.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	104.88	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.28	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	17.02	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	149.51	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	386.38	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	254.04	0.00	0.00	0.00	0.00	27.20
3.40	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.15	0.00	0.00	51.00	0.00	30.00
217.60	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
232.00	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
17.80	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	27.37	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	335.07	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	543.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	236.10	0.00	0.00	0.00	0.00	27.20
61.90	0.00	356.70	439.30	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
429.60	0.00	712.16	1,722.70	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
926.70	2.00	589.96	2,920.30	168.40	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
972.60	2.00	553.10	3,155.80	168.40	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
517.00	0.00	683.00	2,110.80	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
103.50	0.00	364.40	591.90	0.00	0.00	51.00	0.00	0.00	89.73	0.00	1.06	0.00	51.00	0.00	30.00
217.50	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	1.06	0.00	51.00	0.00	30.00
904.40	188.00	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
716.40	188.00	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
29.40	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
209.50	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
527.90	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
320.22	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
308.62	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
789.90	123.80	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
483.10	123.80	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	96.60	0.00	51.00	0.00	30.00
14.60	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	96.60	0.00	51.00	0.00	30.00
71.70	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
74.59	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
17.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	83.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	327.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	857.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	930.90	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	425.50	0.00	0.00	0.00	0.00	27.20
13.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	107.10	0.00	0.00	0.00	0.00	27.20
27.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
20.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
30.40	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
174.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
173.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
23.40	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	16.89	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	44.71	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	145.12	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	386.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	703.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,059.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,547.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,105.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,155.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,779.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,403.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,169.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,189.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,241.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,324.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,519.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,486.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,190.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	863.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	591.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	441.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	375.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	429.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	721.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,280.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,454.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	815.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	162.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	166.80	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	296.20	0.00	0.00	0.00	27.20
51.20	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	129.40	0.00	51.00	0.00	30.00
422.80	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
1,215.10	257.80	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,904.10	788.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,737.40	1,247.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
4,372.90	1,611.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
5,792.60	1,844.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
5,383.40	1,909.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
3,796.70	1,839.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,357.50	1,670.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,027.40	2,032.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,310.00	2,917.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,205.30	2,419.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,660.10	1,121.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,464.30	543.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,169.50	165.50	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
290.30	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.43	343.01	0.00	0.00	0.00	0.00	27.20

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.43	786.89	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,164.95	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,412.62	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,306.88	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	703.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	129.10	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	67.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	280.69	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	460.33	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	214.38	0.00	0.00	0.00	0.00	27.20
116.30	0.00	75.39	84.00	0.00	0.00	51.00	0.00	0.00	89.73	7.63	0.00	0.00	51.00	0.00	30.00
662.30	211.20	95.49	286.00	214.30	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
911.10	272.10	106.70	426.90	351.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
365.10	60.90	86.60	224.90	137.10	0.00	0.00	34.00	0.00	89.73	0.00	0.00	0.00	0.00	34.00	30.00
62.90	0.00	107.80	236.60	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
137.20	0.00	410.72	665.20	160.20	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
150.20	532.80	408.36	493.50	908.60	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
503.30	1,385.70	197.14	319.60	1,783.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
994.70	1,724.70	191.40	554.70	1,967.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,705.80	1,793.30	173.80	848.00	1,731.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,234.10	1,626.60	174.10	614.90	1,795.90	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
232.40	1,509.90	199.30	159.30	1,981.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
280.50	1,104.40	157.10	224.00	1,636.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
180.50	300.20	106.20	277.40	836.40	0.00	0.00	34.00	0.00	89.73	0.00	0.00	0.00	0.00	34.00	30.00
41.87	0.60	48.40	145.80	185.00	0.00	0.00	0.00	0.00	179.46	0.00	39.95	0.00	0.00	0.00	27.20
5.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	39.95	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
208.40	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
208.40	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
20.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	246.23	0.00	0.00	0.00	0.00	0.00	37.20
25.89	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	0.00	0.00	0.00	81.00	0.00	50.00
5.50	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	110.20	0.00	0.00	81.00	0.00	50.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	879.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,638.90	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,194.70	0.00	0.00	0.00	0.00	47.20
37.70	0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	156.50	325.40	0.00	0.00	81.00	0.00	50.00
284.40	0.00	0.00	0.00	0.00	0.00	162.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
582.60	0.00	0.00	0.00	0.00	0.00	162.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
1,127.50	8.40	0.00	0.00	0.00	0.00	162.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
1,990.20	179.40	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
4,734.00	587.70	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
5,426.30	993.90	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
2,966.80	1,077.00	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,857.90	985.10	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,566.70	837.90	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,577.90	492.90	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,066.60	140.30	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
320.40	0.00	0.00	0.00	0.00	0.00	162.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
197.70	0.00	0.00	0.00	0.00	0.00	162.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
837.20	14.70	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
1,796.10	73.40	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,495.00	72.80	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
498.51	23.30	0.00	0.00	0.00	0.00	0.00	54.00	0.00	156.50	0.00	0.00	0.00	0.00	54.00	50.00
145.49	9.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
32.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	9.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	459.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	806.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	557.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	200.60	0.00	0.00	81.00	0.00	50.00
358.80	468.70	0.00	0.00	0.00	81.00	0.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
665.30	770.70	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
306.50	302.00	0.00	0.00	0.00	81.00	0.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	0.00	57.00	0.00	81.00	0.00	50.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	57.00	0.00	0.00	0.00	47.20
138.80	0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	156.50	0.00	0.00	0.00	81.00	0.00	50.00
494.00	435.30	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
981.90	1,074.50	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,260.30	921.70	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
656.20	282.50	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
22.60	0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	156.50	0.00	96.50	0.00	81.00	0.00	50.00
11.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	112.15	0.00	0.00	0.00	47.20
20.05	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	0.00	15.65	0.00	81.00	0.00	50.00
8.20	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	0.00	105.22	0.00	81.00	0.00	50.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	140.14	0.00	0.00	0.00	47.20
222.30	0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	156.50	0.00	34.92	0.00	81.00	0.00	50.00
788.40	387.40	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
1,259.80	1,294.60	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,413.50	2,096.60	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,558.50	2,063.10	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,323.00	1,586.30	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,590.70	1,458.50	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
2,065.50	1,673.70	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,747.60	2,024.50	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,628.70	2,226.20	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,736.90	2,194.80	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,690.70	1,987.40	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,535.70	1,506.70	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,240.20	661.90	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
498.50	77.30	0.00	0.00	0.00	81.00	0.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	140.17	0.00	0.00	81.00	0.00	50.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	795.97	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,663.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,114.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,430.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,621.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,603.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,672.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,531.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,350.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,424.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,758.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	3,234.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	3,443.10	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	3,488.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	3,468.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	3,344.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	3,042.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,556.80	0.00	0.00	0.00	0.00	47.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,313.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,459.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,805.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,908.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	2,242.10	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,232.44	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	603.52	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	617.58	0.00	0.00	0.00	0.00	47.20
3.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	403.80	0.00	0.00	0.00	0.00	47.20
66.03	0.00	147.20	63.60	0.00	0.00	81.00	0.00	0.00	156.50	0.00	0.00	0.00	81.00	0.00	50.00
322.00	0.00	381.70	585.50	0.00	0.00	162.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
1,086.10	165.80	457.60	1,311.70	351.80	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
1,971.60	381.90	268.50	1,820.70	800.40	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
2,159.30	264.90	136.60	2,026.10	549.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,227.80	48.80	281.20	1,431.20	100.40	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
218.70	0.00	190.00	436.00	0.00	81.00	81.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
5.10	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	0.00	0.00	0.00	81.00	0.00	50.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	304.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	503.38	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	198.98	0.00	0.00	81.00	0.00	50.00
288.80	0.00	0.00	0.00	0.00	81.00	81.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
1,040.80	208.30	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
1,429.70	370.30	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
823.40	162.00	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
156.50	0.00	0.00	0.00	0.00	81.00	81.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
714.70	247.90	0.00	0.00	0.00	81.00	0.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
1,455.80	878.60	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,733.20	1,359.50	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
2,010.00	1,486.60	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,916.60	1,540.00	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,523.30	1,580.50	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,190.50	1,599.00	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,170.60	1,612.30	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,371.10	1,530.80	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,709.60	1,220.70	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,785.10	858.00	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
1,318.20	542.00	0.00	0.00	0.00	0.00	0.00	108.00	0.00	0.00	0.00	0.00	0.00	0.00	108.00	52.80
543.90	185.50	0.00	0.00	0.00	0.00	81.00	54.00	0.00	0.00	0.00	0.00	0.00	81.00	54.00	52.80
56.80	0.00	0.00	0.00	0.00	0.00	81.00	0.00	0.00	156.50	0.00	112.89	0.00	81.00	0.00	50.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	748.06	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	1,570.90	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	2,095.26	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	246.23	0.00	1,421.53	330.99	0.00	0.00	37.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	385.10	330.99	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	129.75	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	48.85	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	294.60	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	304.80	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	52.40	0.00	51.00	0.00	30.00
95.40	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
233.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
175.20	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
41.00	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
55.40	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
292.80	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
855.00	68.00	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
730.00	90.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
120.50	22.30	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
79.60	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
189.10	369.30	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
210.60	735.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
245.50	569.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
156.00	202.60	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
7.50	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	262.00	79.60	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	159.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	475.40	80.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	475.40	102.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	349.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	418.70	246.30	0.00	0.00	27.20
14.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	181.62	0.00	0.00	0.00	27.20
14.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	24.92	0.00	0.00	0.00	27.20
10.70	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
10.40	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
2.40	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
8.20	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
117.60	9.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
315.70	86.90	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
464.70	223.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
535.40	365.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
542.60	379.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
341.00	165.30	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
104.20	5.90	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
56.07	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
25.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	262.00	10.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	455.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	885.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	671.50	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	PÁGINA 2 / 5
CONTRATO -	REV. 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	429.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	441.80	198.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	179.80	0.00	51.00	0.00	30.00
2.80	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
2.80	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	13.79	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	13.79	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	194.90	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	456.90	424.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	779.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	522.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	429.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	297.90	0.00	0.00	27.20
2.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	262.00	35.40	0.00	0.00	27.20
148.40	14.00	0.00	0.00	0.00	0.00	0.00	34.00	0.00	89.73	0.00	0.00	0.00	0.00	34.00	30.00
497.40	132.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
712.80	264.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
743.20	368.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
753.70	436.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
663.20	228.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
291.40	14.00	0.00	0.00	0.00	0.00	0.00	34.00	0.00	89.73	0.00	0.00	0.00	0.00	34.00	30.00
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
413.60	414.60	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
872.90	1,023.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
850.60	1,163.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
864.90	980.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
741.00	727.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
712.90	908.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
973.70	1,558.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,071.20	2,006.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,053.40	1,949.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
968.20	1,437.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
923.30	1,053.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
960.40	1,276.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,016.60	1,659.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,020.10	1,676.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
959.70	1,356.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
865.50	875.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
781.40	434.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
790.00	456.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
878.50	901.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
963.20	1,333.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
991.30	1,494.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
968.80	1,364.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
881.50	990.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
790.70	568.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
798.30	543.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
888.70	969.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
984.70	1,516.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,057.70	1,985.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,187.10	2,268.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,143.30	2,228.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,045.90	2,188.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,308.00	2,786.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,583.20	3,848.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,691.50	5,040.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,681.80	6,444.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,612.80	8,174.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,437.40	10,209.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,244.40	11,567.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,208.60	11,921.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,275.80	11,781.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,580.00	10,675.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,668.10	8,505.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,415.80	5,487.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,063.40	2,695.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
962.80	2,247.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,223.30	4,361.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,214.80	7,246.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
958.50	9,063.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
725.50	9,529.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
740.20	8,831.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,084.00	6,983.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,275.20	4,555.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,180.00	2,715.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,080.90	2,318.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,191.00	3,159.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,319.20	4,770.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,037.60	6,911.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
641.70	9,150.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
593.10	10,034.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
755.60	8,708.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
980.10	6,133.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,039.30	4,040.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,201.90	4,424.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,575.90	6,608.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,768.40	8,829.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,852.90	10,267.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,816.60	10,254.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,466.90	9,074.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
996.60	7,174.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
926.90	5,073.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,023.20	3,243.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
944.00	1,996.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
861.20	1,535.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
837.40	1,523.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
824.00	1,450.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
789.50	1,237.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
730.00	887.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
662.10	521.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
584.50	197.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
270.70	21.30	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	20.69	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	20.69	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	74.10	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	259.20	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	447.10	52.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	223.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	472.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	692.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	935.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,151.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,324.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,586.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,856.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,030.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,133.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,108.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,099.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,357.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,639.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,734.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,409.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.54	0.00	173.84	632.68	0.00	0.00	9.02
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59.96	0.00	175.08	580.71	0.00	0.00	9.09
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33.20	0.00	96.93	218.41	0.00	0.00	5.03
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.07	0.00	330.15	667.34	0.00	0.00	17.14
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	634.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	443.10	165.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	237.80	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	122.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	159.50	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	253.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	405.50	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	507.90	104.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	248.90	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	352.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	499.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	678.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	813.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	810.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	807.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	944.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,154.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,317.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,205.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	912.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	770.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	908.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,136.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,168.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,090.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,397.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,748.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,793.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,843.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,903.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,007.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,162.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,352.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,363.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,217.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,060.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,884.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,769.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,758.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,819.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,911.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,979.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,012.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,070.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,096.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,917.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,601.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,247.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	825.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	359.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	350.40	55.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	88.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.50	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
11.90	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
12.20	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.80	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
119.90	30.30	257.50	91.90	63.90	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
223.50	229.40	530.74	160.60	394.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
256.10	545.00	589.07	159.40	842.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
401.90	883.60	635.84	226.60	1,233.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
578.40	1,202.90	638.69	310.20	1,552.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
578.60	1,353.80	580.13	305.70	1,684.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
449.70	1,209.50	532.15	243.70	1,537.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
452.00	859.10	524.94	274.70	1,186.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
392.70	518.10	550.94	257.40	772.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
294.30	349.70	583.20	190.60	516.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
323.60	326.90	516.80	208.70	501.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
311.40	257.80	464.80	213.50	419.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
246.60	125.80	475.40	180.90	198.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
105.30	25.10	347.65	80.50	34.30	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
0.00	0.00	106.75	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
115.40	0.00	253.65	195.50	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
354.50	107.70	492.35	359.20	154.60	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
391.90	281.70	447.20	264.50	410.90	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
295.10	398.40	390.40	193.10	582.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
279.70	520.50	346.00	176.10	746.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
245.40	579.40	326.80	157.10	812.90	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
295.60	395.10	304.15	203.90	559.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
193.60	111.80	294.80	163.40	167.20	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
6.00	0.00	437.98	32.80	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
178.70	99.40	531.03	121.20	152.30	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
302.00	104.20	407.94	306.90	181.50	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
123.30	4.80	261.94	185.70	29.20	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
33.00	0.00	413.99	39.50	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
286.90	91.50	608.09	199.10	131.50	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
454.40	259.80	601.90	293.20	384.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
417.50	399.00	561.80	271.40	611.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
489.20	421.80	476.10	312.20	664.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
554.20	288.80	482.10	365.30	481.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
523.90	121.90	576.00	367.10	225.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
424.00	45.30	602.20	309.10	76.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
197.50	21.10	543.40	166.70	27.30	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
15.40	0.00	308.39	33.80	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	51.59	0.00	0.00	51.00	0.00	0.00	0.00	89.73	132.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	249.50	0.00	0.00	51.00	0.00	0.00	0.00	89.73	132.00	0.00	0.00	51.00	0.00	30.00
188.80	142.40	306.20	121.10	293.90	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
433.90	300.70	378.10	288.00	561.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
606.00	544.50	533.50	378.70	825.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
643.20	1,007.60	446.60	364.80	1,366.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
569.20	1,483.50	477.60	293.90	1,832.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
631.30	1,850.10	534.70	305.60	2,146.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
798.80	1,993.80	779.59	455.30	2,264.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,081.70	1,657.60	1,002.40	619.30	1,940.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,232.40	847.50	1,129.28	687.60	1,110.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
605.10	195.70	1,007.37	358.90	313.20	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
0.00	0.00	462.40	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
444.20	66.50	316.30	179.20	71.50	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
675.00	138.60	517.80	287.10	166.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
230.80	72.10	399.00	107.90	95.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
0.00	0.00	127.60	0.00	0.00	51.00	0.00	0.00	0.00	89.73	206.20	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	375.80	0.00	0.00	0.00	0.00	27.20
7.90	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	169.60	0.00	0.00	51.00	0.00	30.00
128.40	340.20	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
331.20	673.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
210.70	333.30	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	98.39	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	101.59	0.00	0.00	0.00	0.00	27.20
253.30	52.00	0.00	0.00	0.00	0.00	0.00	34.00	0.00	89.73	3.20	0.00	0.00	0.00	34.00	30.00
284.90	52.00	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
31.60	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	262.00	137.80	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,495.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,399.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	7,198.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	6,631.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,846.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	262.00	372.50	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	262.00	225.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	267.18	225.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	5.18	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	262.00	233.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,897.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	5,000.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	7,866.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	9,267.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	9,131.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	8,523.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	7,157.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,579.20	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,533.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,393.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	467.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	247.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	941.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,793.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,055.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,834.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,304.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	978.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,825.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,051.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,244.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,203.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,496.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	6,194.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	6,348.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	6,499.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	8,361.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	10,768.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	11,837.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	10,467.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	6,776.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,808.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,475.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,758.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,693.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	974.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	248.33	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	334.14	86.03	0.00	0.00	27.20
122.80	0.40	178.81	351.60	225.10	0.00	51.00	0.00	0.00	89.73	0.00	72.14	0.00	51.00	0.00	30.00
431.60	419.60	407.84	532.10	855.90	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
773.30	679.40	495.57	457.80	1,060.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
464.50	260.20	635.63	277.30	429.90	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
537.70	374.90	943.09	309.40	631.10	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,063.80	1,179.10	961.88	628.70	1,771.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
867.70	1,256.20	750.11	564.90	2,009.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
501.10	452.00	1,065.73	555.60	869.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
234.50	0.00	1,341.80	448.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
462.30	303.50	1,546.40	435.10	499.60	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
819.00	806.30	1,625.74	615.30	1,345.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
722.50	845.30	847.94	623.60	1,324.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
290.80	342.50	130.30	304.50	478.70	0.00	0.00	34.00	0.00	89.73	0.00	88.66	0.00	0.00	34.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	350.66	559.60	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	268.31	559.60	0.00	0.00	27.20
68.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	6.31	0.00	0.00	0.00	27.20
68.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	262.00	1,722.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,982.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,033.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,242.25	0.00	0.00	27.20
44.50	53.90	0.00	0.00	0.00	0.00	0.00	34.00	0.00	89.73	0.00	262.00	468.65	0.00	34.00	30.00
182.80	281.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
177.80	227.70	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
39.50	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	262.00	277.60	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,954.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	463.01	1,677.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	463.01	98.50	0.00	0.00	27.20
140.29	0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	272.98	98.50	0.00	0.00	27.20
1,696.69	220.80	0.00	0.00	0.00	0.00	0.00	34.00	0.00	89.73	0.00	10.98	0.00	0.00	34.00	30.00
5,465.70	578.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
7,360.50	804.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
4,772.60	1,005.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,850.20	1,102.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
852.80	1,042.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
599.20	792.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
349.80	292.20	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
74.60	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	262.00	360.80	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	634.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,713.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,874.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	773.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,095.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,414.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,511.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	6,777.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	8,546.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	9,632.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	10,035.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	10,225.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	9,894.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	8,566.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	8,260.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	10,223.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	13,553.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	15,831.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	16,265.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	16,223.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	13,933.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	10,939.70	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	9,035.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	6,882.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	5,736.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,442.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,909.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,890.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,828.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	5,329.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	7,044.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	7,078.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	4,043.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	279.44	898.60	0.00	0.00	27.20
150.70	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	17.44	0.00	51.00	0.00	30.00
200.20	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
49.50	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	139.40	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	602.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	648.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	385.90	0.00	0.00	0.00	0.00	27.20
87.60	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	200.70	0.00	0.00	51.00	0.00	30.00
125.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
38.00	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	102.41	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	364.41	461.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,000.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,882.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,784.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,645.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	391.10	1,204.60	0.00	0.00	27.20
78.70	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	129.10	0.00	51.00	0.00	30.00
203.80	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
228.20	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
153.70	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
136.26	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	1.28	0.00	51.00	0.00	30.00
86.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	1.28	0.00	0.00	0.00	27.20
298.44	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
1,061.70	150.70	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,224.90	706.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,129.50	1,481.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,702.10	2,229.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,137.00	2,783.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,435.60	2,942.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,531.00	2,774.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,112.40	2,377.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,522.80	2,012.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,198.90	2,013.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,293.80	2,278.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,104.80	2,331.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
928.50	1,425.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
749.30	426.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
444.40	229.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
393.60	171.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
704.00	127.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
843.50	554.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
545.60	1,099.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,040.70	1,434.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,765.90	2,094.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,361.90	4,373.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,677.00	6,686.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,338.90	5,749.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,690.70	3,229.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,188.70	1,541.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
941.80	525.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
384.90	46.40	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
4.30	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	18.60	0.00	51.00	0.00	30.00
0.30	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	18.60	0.00	51.00	0.00	30.00
5.23	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
4.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	175.10	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	437.10	313.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	539.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	470.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	448.20	244.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	305.90	0.00	0.00	0.00	27.20
6.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	119.70	0.00	0.00	0.00	27.20
6.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	95.20	0.00	0.00	0.00	27.20
46.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	95.20	0.00	0.00	0.00	27.20
86.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
39.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	39.90	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	301.90	27.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	162.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	244.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	160.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	106.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	126.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	327.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	511.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	715.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,092.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,308.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,307.40	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,350.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,748.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,045.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,235.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	399.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	791.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,560.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,571.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,019.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	597.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	411.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	435.50	211.00	0.00	0.00	27.20
12.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	173.50	0.00	0.00	0.00	27.20
34.20	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
22.00	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	29.00	0.00	51.00	0.00	30.00
18.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	34.99	0.00	0.00	0.00	27.20
27.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	7.59	0.00	0.00	0.00	27.20
9.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	89.80	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	350.20	74.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	346.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	283.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	343.90	11.30	0.00	0.00	27.20
38.90	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	81.90	0.00	51.00	0.00	30.00
198.20	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
172.20	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
12.90	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	92.60	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	816.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,523.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,594.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	5,133.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,657.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,834.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,827.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,073.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,387.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,001.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,904.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,951.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,814.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,791.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,684.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,315.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,587.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,571.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,415.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	558.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	238.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	147.20	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	109.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	64.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	26.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	5.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	1.18	0.00	0.00	0.00	27.20
3.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	1.18	0.00	0.00	0.00	27.20
47.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
43.94	0.00	271.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
24.00	0.00	580.30	524.90	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
98.10	0.00	639.00	1,322.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
201.50	0.00	563.90	1,933.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
295.90	0.00	386.40	2,526.40	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
320.10	0.00	212.80	2,814.80	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
230.50	0.00	378.60	2,359.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
81.50	0.00	440.10	1,369.20	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
2.60	0.00	122.10	433.10	0.00	51.00	0.00	0.00	0.00	89.73	0.00	8.20	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	242.90	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	496.70	254.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	512.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	414.90	258.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	366.80	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	475.90	58.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	118.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	519.50	60.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	519.50	46.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	154.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	353.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	906.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,893.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,108.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,983.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,589.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,973.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	632.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	368.40	141.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	231.70	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	387.30	195.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	793.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	854.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	546.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	568.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	312.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	320.25	33.30	0.00	0.00	27.20
239.10	0.00	44.12	507.50	0.00	0.00	51.00	0.00	0.00	89.73	0.00	58.25	0.00	51.00	0.00	30.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
939.90	228.90	75.94	1,046.00	453.30	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,051.80	853.50	169.02	762.70	1,405.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
431.50	1,488.50	249.20	279.20	2,143.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
87.40	1,816.40	244.70	61.40	2,469.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
72.90	1,873.70	277.80	61.70	2,616.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
147.70	1,680.80	278.10	113.80	2,422.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
180.50	1,283.10	189.20	133.20	1,897.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
288.30	755.70	199.70	219.20	1,259.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
300.10	232.20	356.16	403.40	446.50	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
110.60	0.00	212.66	258.90	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	87.50	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	349.50	123.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	511.20	123.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	442.40	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	455.20	110.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	232.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	427.30	121.60	0.00	0.00	27.20
10.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	165.30	0.00	0.00	0.00	27.20
30.71	0.00	294.00	486.20	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
116.30	0.00	586.60	1,297.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
212.00	0.00	555.00	1,709.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
220.50	0.00	668.72	1,691.60	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
150.30	0.00	830.38	1,324.70	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
57.48	0.00	424.06	530.50	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
11.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	203.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	593.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	503.40	0.00	0.00	0.00	0.00	27.20
17.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	113.80	0.00	0.00	0.00	0.00	27.20
120.08	0.00	174.00	684.20	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
462.00	0.00	331.20	1,796.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
676.70	0.00	292.30	2,063.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
626.20	0.00	280.50	1,855.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
629.50	0.00	348.30	1,819.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
513.70	0.00	394.00	1,535.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
257.39	0.00	191.10	619.80	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
92.29	0.00	192.90	110.20	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
294.50	0.00	268.40	819.30	51.10	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
602.80	211.50	187.70	1,091.80	650.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
679.60	448.40	211.40	762.10	1,280.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
640.70	431.70	208.40	725.90	1,284.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
571.90	337.50	224.20	690.10	1,072.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
556.50	242.40	247.80	686.20	811.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
636.20	157.30	254.90	774.00	555.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
736.70	68.80	263.10	829.30	243.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
524.10	11.20	223.70	843.60	58.20	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
222.50	0.00	282.30	717.10	28.10	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
103.60	0.00	344.00	435.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
26.30	0.00	227.10	265.80	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
3.30	0.00	82.70	102.20	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
65.98	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
112.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
94.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
89.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
81.91	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
39.92	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
52.79	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
52.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	20.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	133.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	245.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	253.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	209.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	173.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	230.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	276.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	247.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	215.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	197.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	207.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	236.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	249.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	248.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	289.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	384.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	439.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	527.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	630.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	613.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	605.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	657.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	715.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	739.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	685.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	511.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	211.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	18.17	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.61	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.48	0.00	0.00	0.00	0.00	27.20
26.77	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
39.87	0.00	192.50	189.30	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
92.70	0.00	334.50	657.80	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
221.20	0.00	229.80	1,179.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
317.70	0.00	199.30	1,516.40	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
340.50	0.00	223.20	1,596.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
283.40	0.00	286.90	1,443.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
194.00	0.00	331.10	1,128.60	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
101.20	0.00	305.70	745.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
26.20	0.00	149.80	269.60	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
34.69	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
34.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	18.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	85.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	187.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	300.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	393.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	488.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	627.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	774.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	918.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,030.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	988.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	842.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	720.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	666.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	682.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	730.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	804.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	847.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	865.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	928.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,062.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,227.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,317.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,360.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,401.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,374.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,304.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,237.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,177.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,070.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	728.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	385.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	308.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	267.00	0.00	0.00	0.00	0.00	27.20

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	170.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	69.88	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	11.78	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
12.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
35.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
52.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
36.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
7.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	59.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	194.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	281.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	296.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	275.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	313.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	225.40	0.00	0.00	0.00	0.00	27.20
60.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	38.10	0.00	0.00	0.00	0.00	27.20
154.61	0.00	120.36	333.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
304.70	0.00	228.40	960.60	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
475.40	0.00	174.14	1,461.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
930.80	52.60	175.70	1,658.60	224.80	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,440.20	176.50	195.30	1,702.50	633.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,513.50	335.80	85.70	1,662.06	1,059.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,448.40	445.40	20.20	1,536.16	1,334.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,328.00	427.60	88.20	1,435.90	1,257.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,088.70	322.40	232.80	1,232.20	987.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
914.70	196.60	338.80	1,100.70	656.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
942.60	84.40	312.30	1,075.80	330.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
682.10	16.10	312.00	1,152.70	86.40	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
271.50	0.00	343.20	1,035.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
121.10	0.00	291.90	659.80	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
34.10	0.00	207.59	253.60	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
23.12	0.00	85.19	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
23.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	9.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	51.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	55.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	14.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	46.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	203.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	379.70	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	440.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	401.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	286.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	208.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	216.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	278.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	424.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	606.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	771.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	780.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	719.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	718.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	717.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	723.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	752.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	807.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	865.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	895.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	909.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	947.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,019.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,109.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,196.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,286.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,378.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,446.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,486.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,517.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,537.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,526.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,499.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,416.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,249.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,058.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	865.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	756.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	745.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	804.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	981.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,148.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,225.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,276.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,327.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,383.60	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,435.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,459.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,454.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,445.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,499.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,595.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,708.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,739.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,651.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,561.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,473.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,403.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,383.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,393.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,382.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,328.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,165.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,144.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,185.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,053.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	986.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	985.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	985.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	985.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	919.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	803.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	709.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	640.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	616.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	616.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	616.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	616.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	559.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	408.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	292.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	213.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	101.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	23.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	372.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	946.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	1,138.90	205.50	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	1,117.50	402.60	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	999.70	197.10	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	775.40	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
36.28	0.00	328.60	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
36.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	51.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	184.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	287.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	308.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	281.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	183.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	62.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	12.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	5.50	0.00	0.00	0.00	0.00	27.20
24.55	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
72.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
83.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
37.77	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
12.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
37.65	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
47.71	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
45.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
46.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
49.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
56.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
111.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
144.30	0.00	436.30	733.60	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
184.80	0.00	976.10	1,733.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
252.20	0.00	784.80	2,306.70	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
253.70	0.00	510.80	2,393.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
223.80	0.00	528.90	2,058.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
176.60	0.00	521.10	1,791.40	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
125.50	0.00	544.70	1,470.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
71.70	0.00	615.10	1,186.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
40.70	0.00	703.40	1,198.50	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
166.20	0.00	758.90	1,842.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
412.70	0.00	573.90	2,759.80	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
461.30	0.00	422.50	2,805.40	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
305.60	0.00	478.10	2,096.80	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
152.20	0.00	358.00	1,459.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
43.10	0.00	265.50	788.70	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
29.37	0.00	153.10	201.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
29.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	29.60	0.00	0.00	0.00	0.00	27.20
44.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	29.60	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
57.75	0.00	258.73	361.60	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
94.40	0.00	472.03	1,178.80	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
215.20	0.00	420.60	1,954.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
273.30	0.00	383.90	2,360.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
275.30	0.00	374.80	2,444.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
228.30	0.00	541.30	2,216.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
171.10	0.00	520.90	1,858.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
192.10	0.00	434.60	1,632.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
266.60	0.00	358.30	1,835.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
331.60	0.00	108.40	2,274.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
353.00	0.00	20.80	2,497.10	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
316.30	0.00	13.90	2,477.17	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
228.40	0.00	19.00	2,181.57	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
125.70	0.00	170.93	1,613.60	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
39.20	0.00	151.93	618.70	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	15.50	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	78.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	211.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	167.00	0.00	0.00	0.00	0.00	27.20
32.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	18.50	0.00	0.00	0.00	0.00	27.20
46.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
14.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	5.35	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	5.35	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	15.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	45.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	82.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	140.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	211.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	285.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	368.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	492.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	664.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	878.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,112.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,264.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,373.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,542.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,665.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,766.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,877.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,929.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,940.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,009.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,190.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,299.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,405.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,545.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,608.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,732.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,023.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,994.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,542.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,975.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,366.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,171.60	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,299.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,414.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,496.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,430.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,262.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,092.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	972.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	877.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	819.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	831.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	958.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,184.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,377.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,505.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,527.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,422.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,250.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,114.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,024.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	911.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	786.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	648.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	485.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	321.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	230.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	255.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	365.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	494.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	596.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	632.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	631.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	664.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	684.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	670.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	657.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	670.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	701.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	738.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	812.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	892.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	955.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,009.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,046.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,117.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,213.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,286.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,304.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,421.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,605.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,449.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,089.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	754.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	378.10	0.00	0.00	0.00	0.00	27.20
13.10	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	94.30	0.00	0.00	51.00	0.00	30.00
422.10	66.10	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
685.60	272.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
477.90	386.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
396.10	201.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
194.80	21.60	0.00	0.00	0.00	51.00	0.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	93.60	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	118.08	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	121.78	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	330.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	459.60	0.00	0.00	0.00	0.00	27.20
2.97	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	226.60	0.00	0.00	0.00	0.00	27.20
66.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
63.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	127.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	557.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	633.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	280.20	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	76.00	0.00	0.00	0.00	0.00	27.20
4.47	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
122.30	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
425.10	8.30	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
935.10	170.10	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,216.50	468.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,063.60	799.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
852.80	1,158.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
721.00	1,452.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
717.30	1,654.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,414.00	1,695.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,699.50	1,716.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
3,492.40	1,923.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
3,736.70	2,192.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
3,058.60	2,328.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,683.80	2,330.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,329.30	2,355.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,462.10	2,304.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,174.80	2,135.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
868.30	1,871.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
682.60	1,406.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
564.00	833.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
410.70	471.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
451.80	432.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
588.10	464.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
629.90	384.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
621.40	212.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
400.60	58.30	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
126.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
28.20	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
46.28	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
56.97	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
10.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	28.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	48.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	31.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	81.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	282.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	618.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,009.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,345.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,535.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,538.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,310.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	900.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	472.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	160.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	22.70	0.00	0.00	0.00	0.00	27.20
0.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
23.97	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
72.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
57.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
9.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	25.30	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	103.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	159.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	167.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	187.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	201.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	209.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	237.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	287.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	362.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	452.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	562.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	689.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	765.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	767.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	760.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	772.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	783.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	790.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	815.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	885.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,088.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,286.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,312.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,329.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,420.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,461.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,365.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,189.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	942.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	709.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	523.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	320.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	179.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	101.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	40.60	0.00	0.00	0.00	0.00	27.20
1.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	7.80	0.00	0.00	0.00	0.00	27.20
1.54	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
20.60	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
90.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
144.70	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
124.30	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
77.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
42.10	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
17.10	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
2.40	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
2.27	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
2.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	64.60	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	191.50	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	222.60	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	95.70	0.00	0.00	0.00	27.20
46.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
51.05	0.00	386.10	334.50	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
24.50	0.00	749.20	831.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
19.50	0.00	609.40	914.50	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
53.60	0.00	473.60	1,150.50	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
173.00	0.00	398.10	1,765.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
220.40	0.00	330.90	1,942.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
157.10	0.00	367.13	1,531.40	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
80.40	0.00	494.83	1,067.50	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
24.30	0.00	802.77	445.60	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	514.97	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	72.80	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	281.20	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	470.40	17.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	104.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	260.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	412.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	525.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	632.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	757.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	890.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,010.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,117.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,221.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,259.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,280.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,347.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,419.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,494.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,547.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,545.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,515.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,477.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,453.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,447.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,434.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,423.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,415.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,410.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,409.50	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,410.70	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,404.80	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,388.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,371.40	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,365.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,387.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,436.60	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,491.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,541.10	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,586.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,629.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.73	0.00	262.00	830.10	0.00	0.00	13.60
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	44.87	0.00	131.00	417.70	0.00	0.00	6.80
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.94	0.00	256.76	819.61	0.00	0.00	13.33
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.59	0.00	10.48	33.43	0.00	0.00	0.54
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,612.58	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,498.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,397.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,272.60	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,022.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	626.90	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	215.20	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	376.00	14.30	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	114.00	0.00	0.00	0.00	27.20
7.70	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
179.10	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
623.50	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
1,391.20	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
3,153.50	131.70	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
4,711.50	516.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
5,113.90	1,031.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
5,250.80	1,452.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
5,095.40	1,705.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
4,912.30	1,831.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
5,007.50	1,851.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
5,211.60	1,766.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
5,165.40	1,545.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
4,963.50	1,207.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
4,604.20	840.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
3,954.50	496.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,705.90	166.10	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,553.60	3.10	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
1,119.70	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
810.10	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
568.50	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
344.30	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
136.20	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
89.95	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
89.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
18.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	46.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	160.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	292.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	341.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	298.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	242.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	180.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	104.70	0.00	0.00	0.00	0.00	27.20
9.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	30.80	0.00	0.00	0.00	0.00	27.20
9.70	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
59.80	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
168.20	0.30	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
389.90	17.40	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
411.50	18.70	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
221.50	1.60	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
100.50	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
76.75	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
67.25	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
85.47	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
85.47	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
34.76	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
34.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	12.90	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	87.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	106.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	52.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	69.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	291.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	533.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	390.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	127.90	0.00	0.00	0.00	0.00	27.20
0.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	27.40	0.00	0.00	0.00	0.00	27.20
0.49	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
40.70	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
105.40	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
112.70	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
61.40	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
13.40	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.70	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	9.86	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	46.86	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	137.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	260.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	307.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	183.00	0.00	0.00	0.00	0.00	27.20
3.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	36.40	0.00	0.00	0.00	0.00	27.20
6.57	0.00	196.20	296.90	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
62.40	0.00	531.70	923.90	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
190.00	0.00	700.70	1,627.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
354.70	0.00	659.40	2,286.80	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
511.30	0.00	563.90	2,631.00	184.40	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
759.80	29.50	484.60	2,467.10	511.60	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
1,119.40	73.80	419.40	2,153.90	667.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,366.30	109.10	396.80	2,149.10	780.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,484.30	147.10	384.00	2,275.80	984.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,523.60	166.90	371.70	2,311.70	1,132.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,497.40	175.10	464.50	2,260.60	1,220.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,418.10	180.10	483.35	2,150.30	1,292.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,237.20	158.10	364.85	1,915.90	1,191.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
807.10	72.00	411.00	1,770.50	918.40	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
394.60	3.50	385.70	1,616.70	386.70	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
151.80	0.00	275.40	1,082.20	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
10.77	0.00	134.70	363.80	0.00	51.00	0.00	0.00	0.00	89.73	7.67	0.00	0.00	51.00	0.00	30.00
4.57	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	87.47	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	261.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	477.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	756.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,084.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,446.40	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,889.30	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,422.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,996.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,472.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,815.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,086.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,213.10	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,285.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,632.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,748.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,528.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	4,109.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,553.70	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	3,176.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,901.50	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,500.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,870.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,329.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	912.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	511.20	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	155.47	0.00	0.00	0.00	0.00	27.20
0.20	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	4.57	0.00	0.00	51.00	0.00	30.00
80.70	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
193.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
271.10	2.10	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
374.40	156.30	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
318.10	496.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
186.20	744.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
161.60	796.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
206.60	774.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
261.50	757.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
285.00	758.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
325.20	764.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
446.40	781.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
647.50	950.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,000.10	1,309.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,344.00	1,730.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,445.70	2,153.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,524.90	2,524.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,639.10	2,720.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,681.00	2,696.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,638.80	2,415.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,750.40	1,914.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,936.00	1,293.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,771.90	716.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,391.10	299.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
739.40	57.80	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
141.80	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
28.20	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
36.50	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
56.15	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
40.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	103.80	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	549.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	751.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	543.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	422.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	317.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	201.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	68.60	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
1.60	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
4.50	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
11.57	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
8.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	112.90	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	367.40	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	343.60	0.00	0.00	0.00	0.00	27.20
0.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	89.10	0.00	0.00	0.00	0.00	27.20
4.66	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
98.80	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
333.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
576.60	65.80	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
698.30	278.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
767.10	604.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
914.30	996.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,180.50	1,440.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,472.60	1,919.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,640.50	2,486.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,666.70	3,022.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,639.00	3,271.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,566.20	3,206.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,362.80	3,015.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
989.40	2,836.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
625.70	2,676.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
482.30	2,574.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
530.70	2,608.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
718.20	2,805.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
949.60	3,107.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,146.90	3,403.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,156.00	3,622.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,040.10	3,673.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,032.40	3,558.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
970.80	3,416.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
811.00	3,378.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
799.30	3,671.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
758.80	4,469.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
738.90	5,474.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
881.20	6,358.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,086.20	7,160.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,268.70	7,842.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,141.90	8,147.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
815.70	8,049.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
558.60	7,778.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
449.60	7,407.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
435.30	6,873.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
446.10	6,247.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
467.30	5,676.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
512.00	5,158.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
609.10	4,684.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
741.10	4,294.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
782.30	3,989.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
661.00	3,739.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
520.40	3,526.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
475.00	3,307.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
475.70	3,048.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
455.60	2,688.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
394.80	2,253.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
429.20	1,805.60	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
539.00	1,381.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
580.00	1,065.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
567.30	845.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
528.00	666.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
452.50	522.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
322.50	419.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
193.60	331.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
116.50	258.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
86.50	201.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
77.10	145.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
71.10	88.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
67.30	35.90	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
50.00	6.20	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
18.10	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
50.93	0.89	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
60.90	0.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
10.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	1.88	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	22.25	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	52.54	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	88.24	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	122.84	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	184.04	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	260.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	306.34	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	347.24	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	401.04	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	479.27	7.87	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	99.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	252.74	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	374.24	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	458.94	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	518.74	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	564.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	617.64	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	726.64	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	875.44	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	987.04	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,066.64	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,236.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,491.94	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,789.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,127.44	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,470.74	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,729.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,976.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,227.84	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	3,135.04	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,743.64	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,355.44	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	2,037.34	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,808.84	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,702.44	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,640.54	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,685.64	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,856.74	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,961.34	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,857.04	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,575.24	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,342.64	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,210.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,136.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	1,098.74	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	987.94	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	822.84	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	839.14	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	842.54	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	526.54	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	188.44	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	513.87	40.97	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	485.34	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	467.04	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	466.84	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	489.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	517.87	20.97	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	444.37	20.97	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	325.84	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	282.54	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	369.34	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	293.34	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	67.88	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	4.81	0.00	0.00	0.00	27.20
9.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
9.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.30	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.50	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.20	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
4.70	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
4.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
33.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
60.88	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
160.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
403.90	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
689.80	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
984.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
1,830.90	44.40	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
2,466.00	233.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,335.10	552.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,227.10	903.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,158.50	1,211.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,131.30	1,357.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
2,070.90	1,342.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,863.40	1,289.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,563.10	1,255.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,242.10	1,228.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
891.30	1,195.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
578.50	1,161.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
338.70	1,132.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
141.00	1,105.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
29.00	1,064.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
22.40	978.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
89.80	853.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
266.20	695.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
505.00	524.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
520.40	357.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
463.00	202.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
538.40	77.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
566.40	9.60	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
543.90	0.40	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
507.30	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
466.10	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
428.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
404.50	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
411.00	0.10	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
445.10	2.40	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
462.90	10.00	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
462.60	15.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
457.10	15.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
431.80	9.30	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
403.30	2.50	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
402.00	0.20	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
458.60	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
504.80	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
482.20	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
431.70	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
363.10	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
247.30	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
86.60	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.70	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	86.17	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	133.44	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	78.64	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	51.99	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	20.62	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	38.37	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	120.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	186.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	232.94	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	284.64	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	341.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	365.34	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	355.54	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	345.44	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	335.44	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	325.54	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	314.34	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	298.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	278.64	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	258.44	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	249.04	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	253.24	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	230.04	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	176.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	132.64	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	97.14	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	57.04	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	18.37	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
1.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
33.42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
33.31	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
67.40	0.00	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
325.40	49.90	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
469.00	206.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
408.30	469.00	212.70	226.20	868.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
400.20	784.00	482.80	455.30	2,051.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
398.80	1,145.50	413.60	424.80	2,751.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
407.20	1,503.40	226.50	300.50	3,374.90	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
334.40	1,618.20	140.50	224.20	3,603.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
265.70	1,474.30	111.50	260.50	3,443.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
304.20	1,265.10	122.90	308.90	3,098.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
335.40	1,048.50	167.40	353.00	2,703.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
410.10	842.70	231.30	451.70	2,293.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
569.00	683.00	237.80	643.80	1,938.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
713.00	565.30	245.70	821.70	1,663.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
815.20	455.20	288.00	958.70	1,390.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
889.10	349.50	319.60	1,067.10	1,114.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
926.20	259.00	277.60	1,136.90	882.60	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
905.50	190.10	204.80	1,137.90	699.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
819.80	142.00	189.40	1,059.60	556.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
739.30	101.50	209.70	979.90	430.10	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
678.90	62.90	220.00	919.40	300.90	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
446.30	23.10	187.20	845.40	198.50	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
206.90	0.70	204.60	726.50	79.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
157.60	0.00	216.10	571.60	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
93.00	0.00	194.10	405.90	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
20.90	0.00	95.60	160.70	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
45.86	4.01	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
61.31	4.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
15.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	6.46	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	46.33	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	110.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	164.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	221.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	310.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	441.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	565.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	641.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	671.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	668.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	660.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	655.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	649.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	704.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	797.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	807.84	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	755.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	718.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	713.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	720.94	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	742.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	777.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	820.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	875.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	865.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	791.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	719.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	667.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	653.94	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	690.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	756.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	819.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	854.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	854.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	844.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	851.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	903.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	984.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,071.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,160.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,351.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,542.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,587.94	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,589.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,587.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,582.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,575.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,567.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,556.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,557.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,560.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,550.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,540.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,530.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,520.94	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,511.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,501.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,492.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,523.94	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,561.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,529.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,471.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,443.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,432.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,297.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,037.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	762.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	553.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	419.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	278.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	149.84	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	46.27	0.00	0.00	0.00	0.00	27.20
25.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
70.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
68.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
23.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	10.97	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	46.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	102.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	204.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	367.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	551.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	730.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	951.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,222.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,522.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,729.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,789.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,812.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,832.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,855.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,880.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,926.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,993.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,061.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,097.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,095.94	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,090.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	2,083.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,886.54	0.00	0.00	0.00	0.00	27.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,623.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,445.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	1,193.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	932.04	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	723.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	534.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	321.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	102.77	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
199.70	2.80	0.00	0.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
632.80	323.10	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
663.20	912.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
353.40	1,280.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
217.60	1,398.80	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
198.50	1,358.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
195.70	1,177.70	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
161.30	951.40	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
211.80	773.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
323.50	667.00	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
314.70	575.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
224.90	465.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
168.30	363.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
169.90	272.10	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
196.30	183.50	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
194.00	114.30	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
143.90	49.20	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
113.20	5.30	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
86.20	0.00	0.00	0.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
68.95	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
43.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	33.17	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	199.84	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	493.74	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	747.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	819.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	744.24	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	614.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	503.34	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	468.44	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	476.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	496.14	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	534.54	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	611.64	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	246.23	855.87	0.00	0.00	0.00	0.00	37.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,234.30	0.00	0.00	0.00	0.00	47.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,553.10	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,697.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,611.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,379.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,100.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	911.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	886.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	965.90	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,049.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,080.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,105.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,194.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,174.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	882.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	516.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	232.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	85.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	52.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	36.32	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	11.62	0.00	0.00	0.00	0.00	47.20
16.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
107.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
90.83	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	0.00	0.00	0.00	81.00	0.00	50.00
2.50	0.00	0.00	0.00	0.00	162.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	162.00	0.00	52.80
94.96	0.00	0.00	0.00	0.00	81.00	0.00	0.00	0.00	156.50	0.00	0.00	0.00	81.00	0.00	50.00
92.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	38.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	181.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	301.90	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	325.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	292.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	231.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	216.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	248.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	205.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	127.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	108.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	71.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	32.11	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	8.61	0.00	0.00	0.00	0.00	47.20
14.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
45.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
71.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
92.37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
101.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
98.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
93.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
75.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
51.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
21.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
1.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	26.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	94.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	67.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	6.64	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	40.54	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	115.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	224.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	356.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	307.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	138.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	59.95	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	16.05	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	0.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	48.10	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	240.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	488.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	368.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	138.10	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	150.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	182.50	0.00	0.00	0.00	0.00	47.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	192.00	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	162.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	107.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	98.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	155.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	225.90	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	270.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	285.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	302.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	335.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	380.30	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	434.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	493.10	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	556.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	632.70	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	736.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	866.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	993.10	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,087.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,102.20	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	1,006.10	0.00	0.00	0.00	0.00	47.20

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	856.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	727.40	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	633.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	561.60	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	515.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	488.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	445.50	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	376.80	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	313.00	268.37	0.00	0.00	0.00	0.00	47.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	246.23	121.14	0.00	0.00	0.00	0.00	37.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	21.57	0.00	0.00	0.00	0.00	27.20
61.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
87.06	0.00	0.00	0.00	0.00	0.00	51.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
257.60	19.80	0.00	0.00	0.00	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
469.60	121.90	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
511.40	279.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
554.10	401.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
565.00	475.20	0.00	0.00	0.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
583.90	520.60	136.70	344.30	775.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
590.10	565.00	253.70	671.80	1,635.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
573.80	597.70	226.30	651.70	1,747.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
576.70	603.70	250.80	661.70	1,771.50	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
666.70	596.00	269.50	759.30	1,753.80	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
834.70	563.60	264.90	943.30	1,680.00	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,007.60	501.50	261.10	1,144.10	1,534.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,150.40	419.40	245.90	1,323.20	1,319.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,135.80	330.30	276.10	1,336.20	1,072.70	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,029.40	242.20	288.00	1,252.30	836.40	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
NUMERAÇÃO DA TLSA FTN-MVP10-NV4D-0643-001	**PÁGINA** 2 / 5
CONTRATO -	**REV.** 00

VOLUME GEOMÉTRICO															
ORIGENS															
CORTE		EMPRÉSTIMOS			REBAIXO			SUBSTITUIÇÃO	CORPO DE ATERRO				REATERRO		SUBLASTRO
SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA	PRIMEIRA	SEGUNDA	TERCEIRA		CAMADA FINAL	CORPO DE ATERRO	TRANSIÇÃO	ATERRO EM ROCHA	EM SOLO	EM RACHÃO	
655,922.86	**858,109.00**	**119,572.58**	**276,804.46**	**147,666.80**	**15,492.00**	**26,256.00**	**35,936.00**	**0.00**	**316,787.87**	**788,438.57**	**273,320.03**	**824,566.86**	**41,748.00**	**35,936.00**	**78,782.20**
1,021.50	152.10	248.20	1,282.60	574.30	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
1,095.40	71.20	196.20	1,400.90	304.20	0.00	0.00	68.00	0.00	0.00	0.00	0.00	0.00	0.00	68.00	32.80
796.00	18.00	210.13	1,416.20	88.90	0.00	51.00	34.00	0.00	0.00	0.00	0.00	0.00	51.00	34.00	32.80
365.90	0.00	243.23	1,202.30	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
233.60	0.00	224.90	913.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
158.00	0.00	184.20	702.20	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
159.40	0.00	166.80	516.00	0.00	0.00	102.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
99.90	0.00	181.00	320.00	0.00	51.00	51.00	0.00	0.00	0.00	0.00	0.00	0.00	102.00	0.00	32.80
97.85	0.00	88.20	114.30	0.00	51.00	0.00	0.00	0.00	89.73	0.00	0.00	0.00	51.00	0.00	30.00
152.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
96.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
48.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
12.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	0.00	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	19.87	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	82.64	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	125.24	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	139.54	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	177.24	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	171.24	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	137.64	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	127.44	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	124.34	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	229.84	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	357.54	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	405.04	0.00	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	475.87	55.47	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	179.46	0.00	524.00	244.64	0.00	0.00	27.20
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	83.72	0.00	244.45	205.75	0.00	0.00	12.69

IDENTIFICAÇÃO	BOTA - FORA			0.00	OBSERVAÇÕES
DESTINO	ESTACA INICIAL	ESTACA FINAL	CENTRO DE MASSA	VOLUME COMPACTADO	

OBSERVAÇÕES: